정상상태 유도 모음

『동태 확률 일반균형 모형』부록 자료

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이 문서는 교재『동태 확률 일반균형 모형』의 각 장에서 정상상태(steady state)를 균형조건으로부터 단계별로 유도하는 과정 전체를 수록한 것이다. 본문에는 각 장의 유도 순서(어떤 타깃이 어떤 모수를 고정하고, 어떤 순서로 변수들이 풀리는지)와 결과만 요약되어 있으며, 이 문서가 그 대수적 전개의 전문이다. 각 절의 식 번호는 해당 장의 균형 조건식 번호를 그대로 따른다. 기본 RBC 모형(제2장)의 정상상태 유도는 교육적 원형으로서 교재 본문에 전 과정이 수록되어 있고, 제8장(두국가 모형)과 제14장(금융 가속기)의 유도는 본문 자체가 축차적 유도 전문(닫힌 형태)이므로 이 문서에는 별도로 수록하지 않았다. 각 장의 정상상태를 수치적으로 계산하는 코드는 다운로드 코드 묶음의 `*_steadystate.m` 파일들에 구현되어 있다.

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1 제5장 가격 경직성

물가상승률 π와 노동시간 n은 주어졌다고 가정한다. π는 중앙은행의 물가안정목표로 설정한다. 일반적으로 분기모형에서는 연율이 2%가 되도록 $\pi = (1+0.02)^{\frac{1}{4}}$로 설정한다. 노동 n은 하루의 1/3(24시간 중 8시간)을 노동에 사용한다는 의미에서 모형의 파라미터 χ 값을 $n = 1/3$이 되도록 설정한다. 또한 기술수준 z_t의 정상상태 값 z는 1로 정규화한다.

$$\begin{aligned} c_t^{-\sigma} &= \lambda_t \\ \Rightarrow \lambda &= c^{-\sigma} \end{aligned} \tag{2}$$

$$\begin{aligned} c_t^{-\sigma} &= \beta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \frac{R_t}{\pi_{t+1}} \right] \\ \Rightarrow c^{-\sigma} &= \beta c^{-\sigma} \frac{R}{\pi} \\ \Rightarrow 1 &= \beta \frac{R}{\pi} \\ \Rightarrow R &= \frac{\pi}{\beta} \end{aligned} \tag{4}$$

$$\begin{aligned} \lambda_t = \xi_t &\left[\left(1 - \frac{\psi}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 \right) - \psi \left(\frac{i_t}{i_{t-1}} - 1 \right) \left(\frac{i_t}{i_{t-1}} \right) \right] \\ &+ \beta \mathbb{E}_t \left[\xi_{t+1} \psi \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \\ \Rightarrow \lambda = \xi &\left[\left(1 - \frac{\psi}{2} \left(\frac{i}{i} - 1 \right)^2 \right) - \psi \left(\frac{i}{i} - 1 \right) \left(\frac{i}{i} \right) \right] \\ &+ \beta \xi \psi \left(\frac{i}{i} - 1 \right) \left(\frac{i}{i} \right)^2 \\ \Rightarrow \lambda &= \xi \end{aligned} \tag{6}$$

$$\begin{aligned} \xi_t &= \beta \mathbb{E}_t \left[\lambda_{t+1} r_{t+1}^k + \xi_{t+1}(1-\delta) \right] \\ \Rightarrow \xi &= \beta \lambda r^k + \beta \xi (1-\delta) \\ \Rightarrow [1 - \beta(1-\delta)] \xi &= \beta \lambda r^k \\ \Rightarrow \xi &= \frac{\beta}{1-\beta(1-\delta)} \lambda r^k \\ \Rightarrow r^k &= \frac{1-\beta(1-\delta)}{\beta} \end{aligned} \tag{5}$$

$$1 = (1-\theta)(p_t^{opt})^{1-\varepsilon_p} + \theta \left(\frac{(\pi_{t-1}^{\iota} \pi^{1-\iota})^{\gamma}}{\pi_t} \right)^{1-\varepsilon_p} \tag{10}$$

$$
\begin{aligned}
&\Rightarrow 1 = (1-\theta)(p^{opt})^{1-\varepsilon_p} + \theta\pi^{(1-\gamma)(\varepsilon_p-1)} \\
&\Rightarrow (1-\theta)(p^{opt})^{1-\varepsilon_p} = 1 - \theta\pi^{(1-\gamma)(\varepsilon_p-1)} \\
&\Rightarrow (p^{opt})^{1-\varepsilon_p} = \frac{1-\theta\pi^{(1-\gamma)(\varepsilon_p-1)}}{1-\theta} \\
&\Rightarrow p^{opt} = \left[\frac{1-\theta\pi^{(1-\gamma)(\varepsilon_p-1)}}{1-\theta}\right]^{\frac{1}{1-\varepsilon_p}}
\end{aligned}
$$

$$
\Delta_t = (1-\theta)(p_t^{opt})^{-\varepsilon_p} + \theta\left(\frac{\pi_t}{(\pi_{t-1}^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p}\Delta_{t-1} \tag{12}
$$

$$
\begin{aligned}
&\Rightarrow s = (1-\theta)(p^{opt})^{-\varepsilon_p} + \theta\left(\frac{\pi}{(\pi^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p} s \\
&\Rightarrow s = (1-\theta)(p^{opt})^{-\varepsilon_p} + \theta\pi^{\varepsilon_p(1-\gamma)} s \\
&\Rightarrow \left(1-\theta\pi^{\varepsilon_p(1-\gamma)}\right) s = (1-\theta)(p^{opt})^{-\varepsilon_p} \\
&\Rightarrow s = \frac{(1-\theta)(p^{opt})^{-\varepsilon_p}}{1-\theta\pi^{\varepsilon_p(1-\gamma)}}
\end{aligned}
$$

$$
x_{1,t} = mc_t y_t + \theta\mathbb{E}_t\left[\beta\frac{\lambda_{t+1}}{\lambda_t}\left(\frac{\pi_{t+1}}{(\pi_t^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p} x_{1,t+1}\right] \tag{8}
$$

$$
\begin{aligned}
&\Rightarrow x_1 = mcy + \theta\beta\frac{\lambda}{\lambda}\left(\frac{\pi}{(\pi^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p} x_1 \\
&\Rightarrow x_1 = mcy + \theta\beta\pi^{\varepsilon_p(1-\gamma)} x_1 \\
&\Rightarrow \left[1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}\right] x_1 = mcy \\
&\Rightarrow x_1 = \frac{mcy}{1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}}
\end{aligned}
$$

$$
x_{2,t} = y_t + \theta\mathbb{E}_t\left[\beta\frac{\lambda_{t+1}}{\lambda_t}\left(\frac{\pi_{t+1}}{(\pi_t^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p-1} x_{2,t+1}\right] \tag{9}
$$

$$
\begin{aligned}
&\Rightarrow x_2 = y + \theta\beta\frac{\lambda}{\lambda}\left(\frac{\pi}{(\pi^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p-1} x_2 \\
&\Rightarrow x_2 = y + \theta\beta\pi^{(1-\gamma)(\varepsilon_p-1)} x_2 \\
&\Rightarrow \left[1-\theta\beta\pi^{(1-\gamma)(\varepsilon_p-1)}\right] x_2 = y \\
&\Rightarrow x_2 = \frac{y}{1-\theta\beta\pi^{(1-\gamma)(\varepsilon_p-1)}}
\end{aligned}
$$

$$
\begin{aligned}
p_t^{opt} &= \frac{\varepsilon_p}{\varepsilon_p-1}\frac{x_{1,t}}{x_{2,t}} \\
\Rightarrow p^{opt} &= \frac{\varepsilon_p}{\varepsilon_p-1}\frac{x_1}{x_2}
\end{aligned} \tag{7}
$$

$$\Rightarrow p^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\frac{mcy}{1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}}}{\frac{y}{1-\theta\beta\pi^{(1-\gamma)(\varepsilon_p-1)}}}$$

$$\Rightarrow p^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} mc \frac{1-\theta\beta\pi^{(1-\gamma)(\varepsilon_p-1)}}{1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}}$$

$$\Rightarrow mc = \left(\frac{\varepsilon_p - 1}{\varepsilon_p}\right)\left(\frac{1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}}{1-\theta\beta\pi^{(1-\gamma)(\varepsilon_p-1)}}\right) p^{opt}$$

$$d_t = y_t\left(1 - mc_t\Delta_t\right) \tag{13}$$
$$\Rightarrow d = y\left(1 - mcs\right)$$

$$r_t^k = mc_t\alpha\frac{y_t\Delta_t}{k_t} \tag{14}$$
$$\Rightarrow r^k = mc\alpha\frac{ys}{k}$$
$$\Rightarrow k = \frac{mc\alpha s}{r^k}y$$

$$y_t = \frac{z_t k_t^\alpha n_t^{1-\alpha}}{\Delta_t} \tag{11}$$
$$\Rightarrow y = \frac{zk^\alpha n^{1-\alpha}}{s}$$
$$\Rightarrow y = \frac{z}{s}n^{1-\alpha}\left(\frac{mc\alpha s}{r^k}y\right)^\alpha$$
$$\Rightarrow y = \frac{z}{s}n^{1-\alpha}\left(\frac{mc\alpha s}{r^k}\right)^\alpha y^\alpha$$
$$\Rightarrow y^{1-\alpha} = \frac{z}{s}n^{1-\alpha}\left(\frac{mc\alpha s}{r^k}\right)^\alpha$$
$$\Rightarrow y = \left[\frac{z}{s}n^{1-\alpha}\left(\frac{mc\alpha s}{r^k}\right)^\alpha\right]^{\frac{1}{1-\alpha}}$$

$$w_t = mc_t(1-\alpha)\frac{y_t\Delta_t}{n_t} \tag{15}$$
$$\Rightarrow w = mc(1-\alpha)\frac{ys}{n}$$

$$k_{t+1} = (1-\delta)k_t + i_t\left[1 - \frac{\psi}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2\right] \tag{1}$$
$$\Rightarrow k = (1-\delta)k + i\left[1 - \frac{\psi}{2}\left(\frac{i}{i} - 1\right)^2\right]$$
$$\Rightarrow k = (1-\delta)k + i$$

$$\Rightarrow i = \delta k$$

$$\begin{aligned} y_t &= c_t + i_t \\ \Rightarrow y &= c + i \\ \Rightarrow y &= c + \delta k \\ \Rightarrow c &= y - \delta k \end{aligned} \tag{16}$$

마지막으로 $n = 1/3$ 이 되도록 χ를 설정한다.

$$\begin{aligned} \chi n_t^{\phi} &= c_t^{-\sigma} w_t \\ \Rightarrow \chi &= c^{-\sigma} w \\ \Rightarrow \chi &= \frac{c^{-\sigma} w}{n^{\phi}} \end{aligned} \tag{3}$$

정상상태 값 요약

정상상태 해의 도출 결과를 요약하면 다음과 같다.

$$\begin{aligned} n &= \frac{1}{3} \\ \pi &= (1.02)^{\frac{1}{4}} \\ R &= \frac{\pi}{\beta} \\ r^k &= \frac{1 - \beta(1-\delta)}{\beta} \\ p^{opt} &= \left[\frac{1 - \theta \pi^{(1-\gamma)(\varepsilon_p - 1)}}{1 - \theta} \right]^{\frac{1}{1-\varepsilon_p}} \\ s &= \frac{(1-\theta)(p^{opt})^{-\varepsilon_p}}{1 - \theta \pi^{\varepsilon_p (1-\gamma)}} \\ mc &= \left(\frac{\varepsilon_p - 1}{\varepsilon_p} \right) \left(\frac{1 - \theta \beta \pi^{\varepsilon_p (1-\gamma)}}{1 - \theta \beta \pi^{(1-\gamma)(\varepsilon_p - 1)}} \right) p^{opt} \\ y &= \left[\frac{z}{s} n^{1-\alpha} \left(\frac{mc \alpha s}{r^k} \right)^{\alpha} \right]^{\frac{1}{1-\alpha}} \\ k &= \frac{mc \alpha s}{r^k} y \\ x_1 &= \frac{mcy}{1 - \theta \beta \pi^{\varepsilon_p (1-\gamma)}} \\ x_2 &= \frac{y}{1 - \theta \beta \pi^{(1-\gamma)(\varepsilon_p - 1)}} \\ w &= mc(1-\alpha) \frac{ys}{n} \\ c &= y - \delta k \\ d &= y\,(1 - mcs) \end{aligned}$$

$$\chi = \frac{c^{-\sigma} w}{n^{\phi}}$$

2 제9장 소규모 개방경제 모형

단순화를 위해 π, π_d, π_f, π^*, R^*는 주어진 것으로 가정한다. 그리고 $rer = 1$, $b^f = 0$으로 가정한다. 또한 $n = 1/3$으로 가정하고 χ는 $n = 1/3$이 되도록 설정한다.

$$
\begin{aligned}
& p^*_{f,t} = 1 \\
& \Rightarrow p^*_f = 1
\end{aligned}
\tag{29}
$$

$$
\begin{aligned}
& \pi_{d,t} = \frac{p_{d,t}}{p_{d,t-1}}\pi_t \\
& \Rightarrow \pi_d = \frac{p_d}{p_d}\pi \\
& \Rightarrow \pi_d = \pi
\end{aligned}
\tag{27}
$$

$$
\begin{aligned}
& \pi_{f,t} = \frac{p_{f,t}}{p_{f,t-1}}\pi_t \\
& \Rightarrow \pi_f = \frac{p_f}{p_f}\pi \\
& \Rightarrow \pi_f = \pi
\end{aligned}
\tag{28}
$$

$$
\begin{aligned}
& \Phi_t = \Phi \exp\left(-\eta\left[\frac{b^f_t}{p_{d,t}y_{d,t}} - \frac{b^f}{p_d y_d}\right]\right) v_t \\
& \Rightarrow \Phi = \Phi \exp\left(-\eta\left[\frac{b^f}{p_d y_d} - \frac{b^f}{p_d y_d}\right]\right) v \\
& \Rightarrow \Phi = \Phi
\end{aligned}
\tag{4}
$$

$$
\begin{aligned}
& c_t^{-\sigma} = \beta \mathbb{E}_t\left[c_{t+1}^{-\sigma}\frac{R_t}{\pi_{t+1}}\right] \\
& \Rightarrow c^{-\sigma} = \beta c^{-\sigma}\frac{R}{\pi} \\
& \Rightarrow 1 = \beta\frac{R}{\pi} \\
& \Rightarrow R = \frac{\pi}{\beta}
\end{aligned}
\tag{8}
$$

$$
c_t^{-\sigma} = \xi_t\left[1 - \frac{\psi}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2 - \psi\left(\frac{i_t}{i_{t-1}} - 1\right)\frac{i_t}{i_{t-1}}\right] + \beta\mathbb{E}_t\left[\xi_{t+1}\psi\left(\frac{i_{t+1}}{i_t} - 1\right)\left(\frac{i_{t+1}}{i_t}\right)^2\right] \tag{6}
$$

$$\Rightarrow c^{-\sigma} = \xi \left[1 - \frac{\psi}{2} \left(\frac{i}{i} - 1 \right)^2 - \psi \left(\frac{i}{i} - 1 \right) \frac{i}{i} \right] + \beta \xi \psi \left(\frac{i}{i} - 1 \right) \left(\frac{i}{i} \right)^2$$
$$\Rightarrow \xi = c^{-\sigma}$$

$$\begin{aligned} \xi_t &= \beta \mathbb{E}_t \left[\xi_{t+1} (1-\delta) \right] + \beta \mathbb{E}_t \left[c_{t+1}^{-\sigma} r_{t+1}^k \right] \qquad (7) \\ \Rightarrow \xi &= \beta \xi (1-\delta) + \beta c^{-\sigma} r^k \\ \Rightarrow \xi &= \beta \xi (1-\delta) + \beta \xi r^k \\ \Rightarrow 1 &= \beta (1-\delta) + \beta r^k \\ \Rightarrow \beta r^k &= 1 - \beta (1-\delta) \\ \Rightarrow r^k &= \frac{1 - \beta (1-\delta)}{\beta} \end{aligned}$$

$$\begin{aligned} c_t^{-\sigma} &= \beta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \frac{\tilde{e}_{t+1} \Phi_t R_t^*}{\pi_{t+1}} \right] \qquad (9) \\ \Rightarrow c^{-\sigma} &= \beta c^{-\sigma} \frac{\tilde{e} \Phi R^*}{\pi} \\ \Rightarrow 1 &= \beta \frac{\tilde{e} \Phi R^*}{\pi} \\ \Rightarrow \tilde{e} &= \frac{\pi}{\beta} \frac{1}{\Phi R^*} \qquad (9^*) \end{aligned}$$

$$\begin{aligned} rer_t &= \tilde{e}_t \frac{\pi_t^*}{\pi_t} rer_{t-1} \qquad (25) \\ \Rightarrow rer &= \tilde{e} \frac{\pi^*}{\pi} rer \\ \Rightarrow 1 &= \tilde{e} \frac{\pi^*}{\pi} \\ \Rightarrow \pi^* &= \frac{\pi}{\tilde{e}} \end{aligned}$$

일반적으로 물가목표제를 채택하고 있는 국가들은 연율 2%의 물가목표를 가지고 있으므로 $\pi = \pi^*$로 설정하는 것이 타당하다. 이를 위해서는 $\tilde{e} = 1$이 되어야 하며, Φ 값을 이에 맞게 설정해야 한다. 즉, 식 (9*)로부터Φ를 다음과 같이 설정한다.

$$\begin{aligned} \tilde{e} &= \frac{\pi}{\beta} \frac{1}{\Phi R^*} \\ \Rightarrow 1 &= \frac{\pi}{\beta} \frac{1}{\Phi R^*} \\ \Rightarrow \Phi &= \frac{\pi}{\beta} \frac{1}{R^*} \end{aligned}$$

$$p_{f,t} = rer_t p_{f,t}^* \qquad (23)$$

$$\Rightarrow p_f = rerp_f^*$$
$$\Rightarrow p_f = 1$$

$$\begin{aligned} 1 &= \omega p_{d,t}^{1-\rho} + (1-\omega) p_{f,t}^{1-\rho} \\ \Rightarrow 1 &= \omega p_d^{1-\rho} + (1-\omega) p_f^{1-\rho} \\ \Rightarrow p_d^{1-\rho} &= 1 - (1-\omega) p_f^{1-\rho} \\ \Rightarrow p_d^{1-\rho} &= \frac{1-(1-\omega)p_f^{1-\rho}}{\omega} \\ \Rightarrow p_d &= \left(\frac{1-(1-\omega)p_f^{1-\rho}}{\omega} \right)^{\frac{1}{1-\rho}} \end{aligned} \tag{1}$$

$$\begin{aligned} tot_t &= \frac{p_{d,t}}{p_{f,t}} \\ \Rightarrow tot &= \frac{p_d}{p_f} \end{aligned} \tag{26}$$

$$\begin{aligned} p_{d,t} &= rer_t p_{d,t}^* \\ \Rightarrow p_d &= rerp_d^* \\ \Rightarrow p_d^* &= \frac{p_d}{rer} \end{aligned} \tag{24}$$

$$\begin{aligned} 1 &= (1-\theta)(p_{d,t}^{opt})^{1-\varepsilon_p} + \theta \left[\frac{(\pi_{d,t-1}^{\iota} \pi_d^{1-\iota})^{\gamma}}{\pi_{d,t}} \right]^{1-\varepsilon_p} \\ \Rightarrow 1 &= (1-\theta)(p_d^{opt})^{1-\varepsilon_p} + \theta \left[\frac{(\pi_d^{\iota} \pi_d^{1-\iota})^{\gamma}}{\pi_d} \right]^{1-\varepsilon_p} \\ \Rightarrow (1-\theta)(p_d^{opt})^{1-\varepsilon_p} &= 1 - \theta \pi_d^{(\gamma-1)(1-\varepsilon_p)} \\ \Rightarrow (p_d^{opt})^{1-\varepsilon_p} &= \frac{1-\theta\pi_d^{(\gamma-1)(1-\varepsilon_p)}}{1-\theta} \\ \Rightarrow p_d^{opt} &= \left[\frac{1-\theta\pi_d^{(\gamma-1)(1-\varepsilon_p)}}{1-\theta} \right]^{\frac{1}{1-\varepsilon_p}} \end{aligned} \tag{13}$$

$$\begin{aligned} \Delta_t &= (1-\theta)(p_{d,t}^{opt})^{-\varepsilon_p} + \theta \left(\frac{(\pi_{d,t-1}^{\iota} \pi_d^{1-\iota})^{\gamma}}{\pi_{d,t}} \right)^{-\varepsilon_p} \Delta_{t-1} \\ \Rightarrow s &= (1-\theta)(p_d^{opt})^{-\varepsilon_p} + \theta \left(\frac{(\pi_d^{\iota} \pi_d^{1-\iota})^{\gamma}}{\pi_d} \right)^{-\varepsilon_p} s \end{aligned} \tag{15}$$

$$
\begin{aligned}
&\Rightarrow s = (1-\theta)(p_d^{opt})^{-\varepsilon_p} + \theta \pi_d^{\varepsilon_p(1-\gamma)} s \\
&\Rightarrow \left[1 - \theta \pi_d^{\varepsilon_p(1-\gamma)}\right] s = (1-\theta)(p_d^{opt})^{-\varepsilon_p} \\
&\Rightarrow s = \frac{(1-\theta)(p_d^{opt})^{-\varepsilon_p}}{1 - \theta \pi_d^{\varepsilon_p(1-\gamma)}}
\end{aligned}
$$

$$
x_{d,n,t} = y_{d,t} mc_t + \theta \mathbb{E}_t \left[Q_{d,t,t+1} \left(\frac{\pi_{d,t+1}}{(\pi_{d,t}^{\iota} \pi_d^{1-\iota})^{\gamma}} \right)^{\varepsilon_p} x_{d,n,t+1} \right] \tag{11}
$$

$$
\begin{aligned}
&\Rightarrow x_{d,n} = y_d mc + \theta \underbrace{Q_d}_{=\beta} \left(\frac{\pi_d}{(\pi_d^{\iota} \pi_d^{1-\iota})^{\gamma}} \right)^{\varepsilon_p} x_{d,n} \\
&\Rightarrow x_{d,n} = y_d mc + \theta \beta \pi_d^{\varepsilon_p(1-\gamma)} x_{d,n} \\
&\Rightarrow \left[1 - \theta \beta \pi_d^{\varepsilon_p(1-\gamma)}\right] x_{d,n} = y_d mc \\
&\Rightarrow x_{d,n} = \frac{y_d mc}{1 - \theta \beta \pi_d^{\varepsilon_p(1-\gamma)}}
\end{aligned}
$$

$$
x_{d,d,t} = y_{d,t} + \theta \mathbb{E}_t \left[Q_{d,t,t+1} \left(\frac{\pi_{d,t+1}}{(\pi_{d,t}^{\iota} \pi_d^{1-\iota})^{\gamma}} \right)^{\varepsilon_p - 1} x_{d,d,t+1} \right] \tag{12}
$$

$$
\begin{aligned}
&\Rightarrow x_{d,d} = y_d + \theta Q_d \left(\frac{\pi_d}{(\pi_d^{\iota} \pi_d^{1-\iota})^{\gamma}} \right)^{\varepsilon_p - 1} x_{d,d} \\
&\Rightarrow x_{d,d} = y_d + \theta \beta \pi_d^{(1-\gamma)(\varepsilon_p - 1)} x_{d,d} \\
&\Rightarrow \left[1 - \theta \beta \pi_d^{(1-\gamma)(\varepsilon_p - 1)}\right] x_{d,d} = y_d \\
&\Rightarrow x_{d,d} = \frac{y_d}{1 - \theta \beta \pi_d^{(1-\gamma)(\varepsilon_p - 1)}}
\end{aligned}
$$

$$
\begin{aligned}
p_{d,t}^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_{d,n,t}}{x_{d,d,t}} \qquad (10) \\
\Rightarrow p_d^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_{d,n}}{x_{d,d}} \\
\Rightarrow p_d^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\frac{y_d mc}{1 - \theta \beta \pi_d^{\varepsilon_p(1-\gamma)}}}{\frac{y_d}{1 - \theta \beta \pi_d^{(1-\gamma)(\varepsilon_p - 1)}}} \\
\Rightarrow p_d^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{mc \left[1 - \theta \beta \pi_d^{(1-\gamma)(\varepsilon_p - 1)}\right]}{1 - \theta \beta \pi_d^{\varepsilon_p(1-\gamma)}} \\
\Rightarrow mc &= \frac{\varepsilon_p - 1}{\varepsilon_p} \frac{1 - \theta \beta \pi_d^{\varepsilon_p(1-\gamma)}}{\left[1 - \theta \beta \pi_d^{(1-\gamma)(\varepsilon_p - 1)}\right]} p_d^{opt}
\end{aligned}
$$

만일 $\pi_d = 1$ 이라면, 다음이 성립한다.

$$mc = \frac{\varepsilon_p - 1}{\varepsilon_p} p_d^{opt}$$

$$\begin{aligned} r_t^k &= p_{d,t} mc_t \alpha \frac{y_{d,t} \Delta_t}{k_t} \\ \Rightarrow r^k &= p_d mc \alpha \frac{y_d s}{k} \\ \Rightarrow k &= \frac{p_d mc \alpha s}{r^k} y_d \end{aligned} \tag{17}$$

$$\begin{aligned} y_{d,t} &= \frac{z_t k_t^\alpha n_t^{1-\alpha}}{\Delta_t} \\ \Rightarrow y_d &= \frac{z k^\alpha n^{1-\alpha}}{s} \\ \Rightarrow y_d &= \frac{1}{s} k^\alpha n^{1-\alpha} \\ \Rightarrow y_d &= \frac{1}{s} \left(\underbrace{\frac{p_d mc \alpha y_d s}{r^k}}_{=k} \right)^\alpha n^{1-\alpha} \\ \Rightarrow y_d^{1-\alpha} &= \frac{1}{s} \left(\frac{p_d mc \alpha s}{r^k} \right)^\alpha n^{1-\alpha} \\ \Rightarrow y_d &= \left[\frac{1}{s} \left(\frac{p_d mc \alpha s}{r^k} \right)^\alpha n^{1-\alpha} \right]^{\frac{1}{1-\alpha}} \end{aligned} \tag{14}$$

$$\begin{aligned} d_t &= y_{d,t} \left(1 - mc_t \Delta_t\right) \\ \Rightarrow d &= y_d \left(1 - mcs\right) \end{aligned} \tag{16}$$

$$\begin{aligned} w_t &= p_{d,t} mc_t (1-\alpha) \frac{y_{d,t} \Delta_t}{n_t} \\ \Rightarrow w &= p_d mc (1-\alpha) \frac{y_d s}{n} \end{aligned} \tag{18}$$

$$\begin{aligned} k_{t+1} &= (1-\delta) k_t + i_t - \frac{\psi}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 i_t \\ \Rightarrow k &= (1-\delta) k + i - \frac{\psi}{2} \left(\frac{i}{i} - 1 \right)^2 i \\ \Rightarrow i &= \delta k \end{aligned} \tag{3}$$

$b = b^f = 0$ 이라고 가정했으므로 가계의 예산제약식으로부터 다음이 성립한다.

$$
\begin{aligned}
& c_t + i_t + b_t + b_t^f \\
& \qquad = w_t n_t + r_t^k k_t + \frac{R_{t-1}}{\pi_t} b_{t-1} + \frac{\tilde{e}_t \Phi_{t-1} R_{t-1}^*}{\pi_t} b_{t-1}^f + p_{d,t} d_t \qquad (2) \\
\Rightarrow & c + i + b + b^f = wn + r^k k + \frac{R}{\pi} b + \frac{\tilde{e} \Phi R^*}{\pi} b^f + p_d d \\
\Rightarrow & c + i = wn + r^k k + p_d d \\
\Rightarrow & c = wn + r^k k + p_d d - \underbrace{\delta k}_{=i}
\end{aligned}
$$

$$
\begin{aligned}
exp_t &= c_t + i_t \qquad (19) \\
\Rightarrow exp &= c + i \\
\Rightarrow exp &= c + \underbrace{\delta k}_{=i}
\end{aligned}
$$

$$
\begin{aligned}
y_{d,t} &= \omega p_{d,t}^{-\rho} exp_t + \omega^* {p_{d,t}^*}^{-\rho} exp_t^* \qquad (20) \\
\Rightarrow y_d &= \omega p_d^{-\rho} exp + \omega^* {p_d^*}^{-\rho} exp_t^* \\
\Rightarrow \omega^* {p_d^*}^{-\rho} exp_t^* &= y_d - \omega p_d^{-\rho} exp \\
\Rightarrow exp^* &= \frac{y_d - \omega p_d^{-\rho} exp}{\omega^* {p_d^*}^{-\rho}}
\end{aligned}
$$

$$
\begin{aligned}
y_{f,t} &= (1-\omega) p_{f,t}^{-\rho} exp_t \qquad (21) \\
\Rightarrow y_f &= (1-\omega) p_f^{-\rho} exp
\end{aligned}
$$

$$
\begin{aligned}
y_t &\equiv p_{d,t} y_{d,t} \qquad (36) \\
\Rightarrow y &= p_d y_d
\end{aligned}
$$

$$
\begin{aligned}
x_t &\equiv p_{d,t} \left[\omega^* (p_{d,t}^*)^{-\rho} exp_t^* \right] \qquad (37) \\
\Rightarrow x &\equiv p_d \left[\omega^* (p_d^*)^{-\rho} exp^* \right]
\end{aligned}
$$

$$
\begin{aligned}
m_t &\equiv p_{f,t} y_{f,t} \qquad (38) \\
\Rightarrow m &\equiv p_f y_f
\end{aligned}
$$

$$
tb_t = x_t - m_t \qquad (39)
$$

$$\Rightarrow tb = x - m$$

마지막으로, $n = 1/3$이 되도록 하는 χ는 다음과 같이 구할 수 있다.

$$\begin{aligned} \chi n_t^{\phi} &= c_t^{-\sigma} w_t \\ \Rightarrow \chi n^{\phi} &= c^{-\sigma} w \\ \Rightarrow \chi &= \frac{c^{-\sigma} w}{n^{\phi}} \end{aligned} \tag{5}$$

3 제10장 임금 경직성과 소비 습관

물가상승률 π와 임금상승률 π_w, 노동시간 n은 주어졌다고 가정한다. π와 π_w는 물가안정목표로 설정한다. 일반적으로 분기모형에서는 연율이 2%가 되도록 $(1+0.02)^{\frac{1}{4}}$로 설정한다. 노동 n은 하루의 1/3(24시간 중 8시간)을 노동에 사용한다는 의미에서 모형의 파라미터 χ 값을 $n = 1/3$이 되도록 설정한다. 또한 기술수준 z_t의 정상상태 값 z는 1로 정규화한다.

$$\begin{aligned} \lambda_t &= \beta \mathbb{E}_t \left[\lambda_{t+1} \frac{R_t}{\pi_{t+1}} \right] \\ \Rightarrow \lambda &= \beta \lambda \frac{R}{\pi} \\ \Rightarrow R &= \frac{\pi}{\beta} \end{aligned} \tag{3}$$

$$\begin{aligned} Q_{t,t+1} &= \beta \frac{\lambda_{t+1}}{\lambda_t} \\ \Rightarrow Q &= \beta \frac{\lambda}{\lambda} = \beta \end{aligned} \tag{24}$$

$$\begin{aligned} \lambda_t = \xi_t &\left[1 - \frac{\psi}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 - \psi \left(\frac{i_t}{i_{t-1}} - 1 \right) \frac{i_t}{i_{t-1}} \right] \\ &+ \beta \mathbb{E}_t \left[\xi_{t+1} \psi \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \\ \Rightarrow \lambda = \xi &\left[1 - \frac{\psi}{2} \left(\frac{i}{i} - 1 \right)^2 - \psi \left(\frac{i}{i} - 1 \right) \frac{i}{i} \right] + \beta \xi \psi \left(\frac{i}{i} - 1 \right) \left(\frac{i}{i} \right)^2 \\ \Rightarrow \lambda &= \xi \end{aligned} \tag{4}$$

$$\begin{aligned} q_t &= \frac{\xi_t}{\lambda_t} \\ \Rightarrow q &= \frac{\lambda}{\lambda} = 1 \end{aligned} \tag{23}$$

$$\begin{aligned} \xi_t &= \beta \mathbb{E}_t \left[\lambda_{t+1} r^k_{t+1} + \xi_{t+1}(1-\delta) \right] \\ \Rightarrow \xi &= \beta [\lambda r^k + \xi(1-\delta)] \\ \Rightarrow \lambda &= \beta \lambda r^k + \beta \lambda (1-\delta) \\ \Rightarrow 1 &= \beta r^k + \beta (1-\delta) \\ \Rightarrow \beta r^k &= 1 - \beta (1-\delta) \\ \Rightarrow r^k &= \frac{1}{\beta} - (1-\delta) \end{aligned} \tag{5}$$

$$
\begin{aligned}
1 &= (1-\theta)\left(\tilde{p}_t^{opt}\right)^{1-\varepsilon_p} + \theta\left(\frac{(\pi_{t-1}^{\iota}\pi^{1-\iota})^{\gamma}}{\pi_t}\right)^{1-\varepsilon_p} \qquad (12)\\
\Rightarrow 1 &= (1-\theta)\left(\tilde{p}^{opt}\right)^{1-\varepsilon_p} + \theta\left(\frac{(\pi^{\iota}\pi^{1-\iota})^{\gamma}}{\pi}\right)^{1-\varepsilon_p}\\
\Rightarrow 1 &= (1-\theta)\left(\tilde{p}^{opt}\right)^{1-\varepsilon_p} + \theta\left(\frac{\pi^{\gamma}}{\pi}\right)^{1-\varepsilon_p}\\
\Rightarrow (1-\theta)\left(\tilde{p}^{opt}\right)^{1-\varepsilon_p} &= 1-\theta\left(\pi^{\gamma-1}\right)^{1-\varepsilon_p}\\
\Rightarrow \left(\tilde{p}^{opt}\right)^{1-\varepsilon_p} &= \frac{1-\theta\pi^{(\gamma-1)(1-\varepsilon_p)}}{1-\theta}\\
\Rightarrow \tilde{p}^{opt} &= \left[\frac{1-\theta\pi^{(\gamma-1)(1-\varepsilon_p)}}{1-\theta}\right]^{\frac{1}{1-\varepsilon_p}}
\end{aligned}
$$

참고로 $\pi = 1$ 이면, $\tilde{p}^{opt} = 1$ 이다.

$$
\begin{aligned}
\Delta_t &= (1-\theta)(\tilde{p}_t^{opt})^{-\varepsilon_p} + \theta\left(\frac{(\pi_{t-1}^{\iota}\pi^{1-\iota})^{\gamma}}{\pi_t}\right)^{-\varepsilon_p}\Delta_{t-1} \qquad (15)\\
\Rightarrow s &= (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta\left(\frac{(\pi^{\iota}\pi^{1-\iota})^{\gamma}}{\pi}\right)^{-\varepsilon_p} s\\
\Rightarrow s &= (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta\left(\frac{\pi^{\gamma}}{\pi}\right)^{-\varepsilon_p} s\\
\Rightarrow s &= (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta\left(\pi^{\gamma-1}\right)^{-\varepsilon_p} s\\
\Rightarrow s &= (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta\pi^{-\varepsilon_p(\gamma-1)} s\\
\Rightarrow \left[1-\theta\pi^{-\varepsilon_p(\gamma-1)}\right] s &= (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p}\\
\Rightarrow s &= \frac{(1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p}}{1-\theta\pi^{-\varepsilon_p(\gamma-1)}}
\end{aligned}
$$

참고로 $\pi = 1$ 이면 $\tilde{p}^{opt} = 1$ 이므로, $s = 1$ 이다.

$$
\begin{aligned}
x_{n,t} &= y_t mc_t + \theta\mathbb{E}_t\left[Q_{t,t+1}\left(\frac{\pi_{t+1}}{(\pi_t^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p} x_{n,t+1}\right] \qquad (10)\\
\Rightarrow x_n &= ymc + \theta q\left(\frac{\pi}{(\pi^{\iota}\pi^{1-\iota})^{\gamma}}\right)^{\varepsilon_p} x_n\\
\Rightarrow x_n &= ymc + \theta\beta\left(\frac{\pi}{\pi^{\gamma}}\right)^{\varepsilon_p} x_n\\
\Rightarrow x_n &= ymc + \theta\beta\pi^{\varepsilon_p(1-\gamma)} x_n\\
\Rightarrow \left[1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}\right] x_n &= ymc\\
\Rightarrow x_n &= \frac{ymc}{1-\theta\beta\pi^{\varepsilon_p(1-\gamma)}}
\end{aligned}
$$

$$
\begin{aligned}
x_{d,t} &= y_t + \theta \mathbb{E}_t \left[Q_{t,t+1} \left(\frac{\pi_{t+1}}{(\pi_t^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p - 1} x_{d,t+1} \right] \qquad (11) \\
\Rightarrow x_d &= y + \theta q \left(\frac{\pi}{(\pi^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p - 1} x_d \\
\Rightarrow x_d &= y + \theta \beta \left(\frac{\pi}{\pi^{\gamma}} \right)^{\varepsilon_p - 1} x_d \\
\Rightarrow x_d &= y + \theta \beta \left(\pi^{1-\gamma} \right)^{\varepsilon_p - 1} x_d \\
\Rightarrow x_d &= y + \theta \beta \pi^{(\varepsilon_p - 1)(1-\gamma)} x_d \\
\Rightarrow & \left[1 - \theta \beta \pi^{(\varepsilon_p - 1)(1-\gamma)} \right] x_d = y \\
\Rightarrow x_d &= \frac{y}{1 - \theta \beta \pi^{(\varepsilon_p - 1)(1-\gamma)}}
\end{aligned}
$$

$$
\begin{aligned}
\tilde{p}_t^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_{n,t}}{x_{d,t}} \qquad (9) \\
\Rightarrow \tilde{p}^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_n}{x_d} \\
\Rightarrow \tilde{p}^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\frac{ymc}{1 - \theta \beta \pi^{\varepsilon_p (1-\gamma)}}}{\frac{y}{1 - \theta \beta \pi^{(\varepsilon_p - 1)(1-\gamma)}}} \\
\Rightarrow \tilde{p}^{opt} &= \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{1 - \theta \beta \pi^{(\varepsilon_p - 1)(1-\gamma)}}{1 - \theta \beta \pi^{\varepsilon_p (1-\gamma)}} mc \\
\Rightarrow mc &= \frac{\varepsilon_p - 1}{\varepsilon_p} \frac{1 - \theta \beta \pi^{\varepsilon_p (1-\gamma)}}{1 - \theta \beta \pi^{(\varepsilon_p - 1)(1-\gamma)}} \tilde{p}^{opt}
\end{aligned}
$$

참고로 $\pi = 1$ 이면 $\tilde{p}^{opt} = 1$ 이므로, $mc = \frac{\varepsilon_p - 1}{\varepsilon_p}$ 이다.

$$
\begin{aligned}
1 &= (1 - \theta_w) \left(\tilde{w}_t^{opt} \right)^{1-\varepsilon_w} + \theta_w \left(\frac{(\pi_{w,t-1}^{\iota_w} \pi^{1-\iota_w})^{\gamma_w}}{\pi_{w,t}} \right)^{1-\varepsilon_w} \qquad (13) \\
\Rightarrow 1 &= (1 - \theta_w) \left(\tilde{w}^{opt} \right)^{1-\varepsilon_w} + \theta_w \left(\frac{(\pi_w^{\iota_w} \pi^{1-\iota_w})^{\gamma_w}}{\pi_w} \right)^{1-\varepsilon_w} \\
\Rightarrow 1 &= (1 - \theta_w) \left(\tilde{w}^{opt} \right)^{1-\varepsilon_w} + \theta_w \left(\frac{\pi_w^{\gamma_w}}{\pi_w} \right)^{1-\varepsilon_w} \\
\Rightarrow (1 - \theta_w) & \left(\tilde{w}^{opt} \right)^{1-\varepsilon_w} = 1 - \theta_w \left(\pi_w^{\gamma_w - 1} \right)^{1-\varepsilon_w} \\
\Rightarrow & \left(\tilde{w}^{opt} \right)^{1-\varepsilon_w} = \frac{1 - \theta_w \pi_w^{(\gamma_w - 1)(1-\varepsilon_w)}}{1 - \theta_w} \\
\Rightarrow \tilde{w}^{opt} &= \left[\frac{1 - \theta_w \pi_w^{(\gamma_w - 1)(1-\varepsilon_w)}}{1 - \theta_w} \right]^{\frac{1}{1-\varepsilon_w}}
\end{aligned}
$$

참고로 $\pi_w = 1$ 이면, $\tilde{w}^{opt} = 1$ 이다.

$$\Delta_{w,t} = (1-\theta_w)(\tilde{w}_t^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_{w,t-1}^{\iota_w} \pi^{1-\iota_w})^{\gamma_w}}{\pi_{w,t}} \right)^{-\varepsilon_w} \Delta_{w,t-1} \tag{17}$$

$$\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_w^{\iota_w} \pi^{1-\iota_w})^{\gamma_w}}{\pi_w} \right)^{-\varepsilon_w} \Delta_w$$

$$\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{\pi_w^{\gamma_w}}{\pi_w} \right)^{-\varepsilon_w} \Delta_w$$

$$\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \left(\pi_w^{\gamma_w - 1} \right)^{-\varepsilon_w} \Delta_w$$

$$\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \pi_w^{-\varepsilon_w(\gamma_w - 1)} \Delta_w$$

$$\Rightarrow \left[1 - \theta_w \pi_w^{-\varepsilon_w(\gamma_w - 1)} \right] \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w}$$

$$\Rightarrow \Delta_w = \frac{(1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w}}{1 - \theta_w \pi_w^{-\varepsilon_w(\gamma_w - 1)}}$$

참고로 $\pi_w = 1$ 이면 $\tilde{w}^{opt} = 1$ 이므로, $\Delta_w = 1$ 이다.

$$n_t^{eff} = \frac{n_t}{\Delta_{w,t}} \tag{16}$$

$$\Rightarrow n^{eff} = \frac{n}{\Delta_w}$$

$$r_t^k = mc_t \alpha \frac{y_t \Delta_t}{k_t} \tag{19}$$

$$\Rightarrow r^k = mc\alpha \frac{ys}{k}$$

$$\Rightarrow \frac{k}{y} = mc\alpha \frac{s}{r^k}$$

$$\Rightarrow k = \frac{k}{y} y$$

$$y_t = \frac{z_t k_t^{\alpha} (n_t^{eff})^{1-\alpha}}{\Delta_t} \tag{14}$$

$$\Rightarrow y = \frac{z k^{\alpha} (n^{eff})^{1-\alpha}}{s}$$

$$\Rightarrow 1 = \frac{z}{s} \left(\frac{k}{y} \right)^{\alpha} \left(\frac{n^{eff}}{y} \right)^{1-\alpha}$$

$$\Rightarrow 1 = \frac{z}{s} \left(mc\alpha \frac{s}{r^k} \right)^{\alpha} \left(\frac{n^{eff}}{y} \right)^{1-\alpha}$$

$$\Rightarrow y^{1-\alpha} = \frac{z}{s} \left(mc\alpha \frac{s}{r^k} \right)^{\alpha} \left(n^{eff} \right)^{1-\alpha}$$

$$\Rightarrow y = \left[\frac{z}{s} \left(mc\alpha \frac{s}{r^k} \right)^{\alpha} \right]^{\frac{1}{1-\alpha}} n^{eff}$$

$$
\begin{aligned}
w_t &= mc_t(1-\alpha)\frac{y_t\Delta_t}{n_t^{eff}} \\
\Rightarrow w &= mc(1-\alpha)\frac{ys}{n^{eff}}
\end{aligned}
\tag{20}
$$

$$
\begin{aligned}
d_t &= y_t\left(1-mc_t\Delta_t\right) \\
\Rightarrow d &= y\left(1-mcs\right)
\end{aligned}
\tag{18}
$$

$$
\begin{aligned}
k_{t+1} &= (1-\delta)k_t + \left[1-\frac{\psi}{2}\left(\frac{i_t}{i_{t-1}}-1\right)^2\right]i_t \\
\Rightarrow k &= (1-\delta)k + \left[1-\frac{\psi}{2}\left(\frac{i}{i}-1\right)^2\right]i \\
\Rightarrow k &= (1-\delta)k + i \\
\Rightarrow i &= \delta k
\end{aligned}
\tag{1}
$$

$$
\begin{aligned}
y_t &= c_t + i_t \\
\Rightarrow y &= c + i \\
\Rightarrow y &= c + \delta k \\
\Rightarrow c &= y - \delta k
\end{aligned}
\tag{21}
$$

$$
\begin{aligned}
\lambda_t &= (c_t-\hbar c_{t-1})^{-\sigma} - \beta\hbar\mathbb{E}_t\left[(c_{t+1}-\hbar c_t)^{-\sigma}\right] \\
\Rightarrow \lambda &= (c-\hbar c)^{-\sigma} - \beta\hbar\left[(c-\hbar c)^{-\sigma}\right] \\
\Rightarrow \lambda &= (1-\beta\hbar)(c-\hbar c)^{-\sigma}
\end{aligned}
\tag{2}
$$

$$
\begin{aligned}
x_{w,d,t} &= \lambda_t w_t n_t^{eff} + \beta\theta_w\mathbb{E}_t\left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w-1} x_{w,d,t+1}\right] \\
\Rightarrow x_{w,d} &= \lambda w n^{eff} + \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w-1} x_{w,d} \\
\Rightarrow &\left[1-\beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w-1}\right] x_{w,d} = \lambda w n^{eff} \\
\Rightarrow x_{w,d} &= \frac{\lambda w n^{eff}}{1-\beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w-1}}
\end{aligned}
\tag{8}
$$

$$\tilde{w}_t^{opt}(h)^{1+\varepsilon_w\phi} = \frac{\varepsilon_w}{\varepsilon_w - 1}\frac{x_{w,n,t}}{x_{w,d,t}} \tag{6}$$
$$\Rightarrow \tilde{w}^{opt}(h)^{1+\varepsilon_w\phi} = \frac{\varepsilon_w}{\varepsilon_w - 1}\frac{x_{w,n}}{x_{w,d}}$$
$$\Rightarrow x_{w,n} = \frac{\varepsilon_w - 1}{\varepsilon_w} x_{w,d}\tilde{w}^{opt}(h)^{1+\varepsilon_w\phi}$$

마지막으로 $n = 1/3$이 되도록 하는 χ는 다음과 같다.

$$x_{w,n,t} = \chi\left(n_t^{eff}\right)^{1+\phi} + \beta\theta_w\mathbb{E}_t\left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w(1+\phi)} x_{w,n,t+1}\right] \tag{7}$$
$$\Rightarrow x_{w,n} = \chi\left(n^{eff}\right)^{1+\phi} + \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w(1+\phi)} x_{w,n}$$
$$\Rightarrow \chi\left(n^{eff}\right)^{1+\phi} = x_{w,n} - \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w(1+\phi)} x_{w,n}$$
$$\Rightarrow \chi = \left[1 - \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w(1+\phi)}\right]\frac{x_{w,n}}{(n^{eff})^{1+\phi}}$$

4 제12장 탐색-매칭 마찰과 실업

소비재 인플레이션율 π, 취업률 n, 긴장도 θ는 주어진 것으로 가정한다. 한편, 정상상태에서 $b = 0$이다.

$$
\begin{aligned}
1 &= \mathbb{E}_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} \frac{R_t}{\pi_{t+1}} \right] \\
\Rightarrow 1 &= \beta \frac{\lambda}{\lambda} \frac{R}{\pi} \\
\Rightarrow R &= \frac{\pi}{\beta}
\end{aligned}
\tag{3}
$$

$$
\begin{aligned}
1 = Q_t &\left[1 - \frac{\psi}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 - \psi \left(\frac{i_t}{i_{t-1}} - 1 \right) \frac{i_t}{i_{t-1}} \right] + \mathbb{E}_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} Q_{t+1} \psi \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \\
\Rightarrow 1 = Q &\left[1 - \frac{\psi}{2} \left(\frac{i}{i} - 1 \right)^2 - \psi \left(\frac{i}{i} - 1 \right) \frac{i}{i} \right] + \beta \frac{\lambda}{\lambda} Q \psi \left(\frac{i}{i} - 1 \right) \left(\frac{i}{i} \right)^2 \\
\Rightarrow Q &= 1
\end{aligned}
\tag{4}
$$

$$
\begin{aligned}
Q_t &= \mathbb{E}_t \left\{ \beta \frac{\lambda_{t+1}}{\lambda_t} \left[r^k_{t+1} + Q_{t+1}(1-\delta) \right] \right\} \\
\Rightarrow Q &= \beta \frac{\lambda}{\lambda} \left[r^k + Q(1-\delta) \right] \\
\Rightarrow Q &= \beta r^k + \beta Q (1-\delta) \\
\Rightarrow \beta r^k &= Q \left[1 - \beta(1-\delta) \right] \\
\Rightarrow r^k &= Q \frac{1 - \beta(1-\delta)}{\beta}
\end{aligned}
\tag{5}
$$

$$
\begin{aligned}
u_t &= 1 - (1-\rho) n_t \\
\Rightarrow u &= 1 - (1-\rho) n
\end{aligned}
\tag{9}
$$

$$
\begin{aligned}
u^s_t &= \rho n_t \\
\Rightarrow u^s &= \rho n
\end{aligned}
\tag{10}
$$

$$
\begin{aligned}
n_{t+1} &= (1-\rho) n_t + m_t \\
\Rightarrow n &= (1-\rho) n + m
\end{aligned}
\tag{8}
$$

$$\Rightarrow m = \rho n$$

$$f_t = \frac{m_t}{u_t} \tag{12}$$
$$\Rightarrow f = \frac{m}{u}$$

$$\theta_t = \frac{v_t}{u_t} \tag{11}$$
$$\Rightarrow \theta = \frac{v}{u}$$
$$\Rightarrow v = \theta u$$

$$q_t = \frac{m_t}{v_t} \tag{13}$$
$$\Rightarrow q = \frac{m}{v}$$

$$m_t = \mu_t u_t^{\xi} v_t^{1-\xi} \tag{6}$$
$$\Rightarrow m = \mu u^{\xi} v^{1-\xi}$$
$$\Rightarrow \mu = \frac{m}{u^{\xi} v^{1-\xi}}$$

$$\mu_t = \bar{\mu} \zeta_t^{\mu} \left(\frac{u_t^s}{u_t} \right)^{\eta} \tag{7}$$
$$\Rightarrow \mu = \bar{\mu} \zeta^{\mu} \left(\frac{u^s}{u} \right)^{\eta}$$
$$\Rightarrow \bar{\mu} = \frac{\mu}{\zeta^{\mu} \left(\frac{u^s}{u} \right)^{\eta}}$$

$$1 = (1-\theta_p)(\tilde{p}_t^{opt})^{1-\varepsilon_p} + \theta_p \left(\frac{(\pi_{t-1}^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}}{\pi_t} \right)^{1-\varepsilon_p} \tag{25}$$
$$\Rightarrow 1 = (1-\theta_p)(\tilde{p}^{opt})^{1-\varepsilon_p} + \theta_p \left(\frac{(\pi^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}}{\pi} \right)^{1-\varepsilon_p}$$
$$\Rightarrow 1 = (1-\theta_p)(\tilde{p}^{opt})^{1-\varepsilon_p} + \theta_p \pi^{(1-\gamma_p)(\varepsilon_p - 1)}$$
$$\Rightarrow (1-\theta_p)(\tilde{p}^{opt})^{1-\varepsilon_p} = 1 - \theta_p \pi^{(1-\gamma_p)(\varepsilon_p - 1)}$$
$$\Rightarrow (\tilde{p}^{opt})^{1-\varepsilon_p} = \frac{1 - \theta_p \pi^{(1-\gamma_p)(\varepsilon_p - 1)}}{1-\theta_p}$$

$$\Rightarrow \tilde{p}^{opt} = \left[\frac{1 - \theta_p \pi^{(1-\gamma_p)(\varepsilon_p - 1)}}{1 - \theta_p} \right]^{\frac{1}{1-\varepsilon_p}}$$

$$\Delta_t = (1-\theta_p)(\tilde{p}_t^{opt})^{-\varepsilon_p} + \theta_p \left(\frac{\pi_t}{(\pi_{t-1}^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}} \right)^{\varepsilon_p} \Delta_{t-1} \tag{27}$$

$$\Rightarrow \Delta = (1-\theta_p)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta_p \left(\frac{\pi}{(\pi^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}} \right)^{\varepsilon_p} \Delta$$

$$\Rightarrow \Delta = (1-\theta_p)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta_p \pi^{\varepsilon_p (1-\gamma_p)} \Delta$$

$$\Rightarrow \left(1 - \theta_p \pi^{\varepsilon_p (1-\gamma_p)} \right) \Delta = (1-\theta_p)(\tilde{p}^{opt})^{-\varepsilon_p}$$

$$\Rightarrow \Delta = \frac{(1-\theta_p)}{1 - \theta_p \pi^{\varepsilon_p (1-\gamma_p)}} (\tilde{p}^{opt})^{-\varepsilon_p}$$

$$x_{n,t} = \varphi_t y_t + \theta_p \mathbb{E}_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} \left(\frac{\pi_{t+1}}{(\pi_t^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}} \right)^{\varepsilon_p} x_{n,t+1} \right] \tag{23}$$

$$\Rightarrow x_n = \varphi y + \theta_p \beta \frac{\lambda}{\lambda} \left(\frac{\pi}{(\pi^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}} \right)^{\varepsilon_p} x_n$$

$$\Rightarrow x_n = \varphi y + \theta_p \beta \pi^{\varepsilon_p (1-\gamma_p)} x_n$$

$$\Rightarrow \left[1 - \theta_p \beta \pi^{\varepsilon_p (1-\gamma_p)} \right] x_n = \varphi y$$

$$\Rightarrow x_n = \frac{\varphi y}{1 - \theta_p \beta \pi^{\varepsilon_p (1-\gamma_p)}}$$

$$x_{d,t} = y_t + \theta_p \mathbb{E}_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} \left(\frac{\pi_{t+1}}{(\pi_t^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}} \right)^{\varepsilon_p - 1} x_{d,t+1} \right] \tag{24}$$

$$\Rightarrow x_d = y + \theta_p \beta \frac{\lambda}{\lambda} \left(\frac{\pi}{(\pi^{\iota_p} \pi^{1-\iota_p})^{\gamma_p}} \right)^{\varepsilon_p - 1} x_d$$

$$\Rightarrow x_d = y + \theta_p \beta \pi^{(\varepsilon_p - 1)(1-\gamma_p)} x_d$$

$$\Rightarrow \left[1 - \theta_p \beta \pi^{(\varepsilon_p - 1)(1-\gamma_p)} \right] x_d = y$$

$$\Rightarrow x_d = \frac{y}{1 - \theta_p \beta \pi^{(\varepsilon_p - 1)(1-\gamma_p)}}$$

$$\tilde{p}_t^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_{n,t}}{x_{d,t}} \tag{22}$$

$$\Rightarrow \tilde{p}^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_n}{x_d}$$

$$\Rightarrow \tilde{p}^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\frac{\varphi y}{1 - \theta_p \beta \pi^{\varepsilon_p (1-\gamma_p)}}}{\frac{y}{1 - \theta_p \beta \pi^{(\varepsilon_p - 1)(1-\gamma_p)}}}$$

$$\Rightarrow \tilde{p}^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{1 - \theta_p \beta \pi^{(\varepsilon_p - 1)(1-\gamma_p)}}{1 - \theta_p \beta \pi^{\varepsilon_p (1-\gamma_p)}} \varphi$$

$$\Rightarrow \varphi = \frac{\varepsilon_p - 1}{\varepsilon_p} \frac{1 - \theta_p \beta \pi^{\varepsilon_p (1-\gamma_p)}}{1 - \theta_p \beta \pi^{(\varepsilon_p - 1)(1-\gamma_p)}} \tilde{p}^{opt}$$

$$\begin{aligned} r_t^k &= \varphi_t \alpha \frac{y_t^w}{k_t} \\ \Rightarrow r^k &= \varphi \alpha \frac{y^w}{k} \\ \Rightarrow r^k &= \varphi \alpha \frac{z k^\alpha n^{1-\alpha}}{k} \\ \Rightarrow r^k &= \varphi \alpha z \left(\frac{n}{k}\right)^{1-\alpha} \\ \Rightarrow \left(\frac{n}{k}\right)^{1-\alpha} &= \frac{r^k}{\varphi \alpha z} \\ \Rightarrow \frac{n}{k} &= \left(\frac{r^k}{\varphi \alpha z}\right)^{\frac{1}{1-\alpha}} \\ \Rightarrow k &= \left(\frac{r^k}{\varphi \alpha z}\right)^{-\frac{1}{1-\alpha}} n \end{aligned} \tag{16}$$

$$\begin{aligned} y_t^w &= z_t k_t^\alpha n_t^{1-\alpha} \\ \Rightarrow y^w &= z k^\alpha n^{1-\alpha} \end{aligned} \tag{15}$$

$$\begin{aligned} y_t^w &= \Delta_t y_t \\ \Rightarrow y^w &= s y \\ \Rightarrow y &= \frac{y^w}{s} \end{aligned} \tag{26}$$

$$\begin{aligned} k_{t+1} &= (1-\delta) k_t + \left[1 - \frac{\psi}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2\right] i_t \\ \Rightarrow k &= (1-\delta) k + \left[1 - \frac{\psi}{2}\left(\frac{i}{i} - 1\right)^2\right] i \\ \Rightarrow k &= (1-\delta) k + i \\ \Rightarrow i &= \delta k \end{aligned} \tag{1}$$

$$\begin{aligned} y_t &= c_t + i_t + \kappa v_t \\ \Rightarrow y &= c + i + \kappa v \\ \Rightarrow c &= y - i - \kappa v \end{aligned} \tag{36}$$

$$\lambda_t = (c_t - \hbar c_{t-1})^{-\sigma} - \beta\hbar\mathbb{E}_t\left[(c_{t+1} - \hbar c_t)^{-\sigma}\right] \tag{2}$$
$$\Rightarrow \lambda = (c - \hbar c)^{-\sigma} - \beta\hbar(c - \hbar c)^{-\sigma}$$
$$\Rightarrow \lambda = (1 - \beta\hbar)(c - \hbar c)^{-\sigma}$$

$$\frac{\kappa}{q_t} = \mathbb{E}_t\left[\beta\frac{\lambda_{t+1}}{\lambda_t}J_{t+1}\right] \tag{18}$$
$$\Rightarrow \frac{\kappa}{q} = \beta\frac{\lambda}{\lambda}J$$
$$\Rightarrow \frac{\kappa}{q} = \beta J$$
$$\Rightarrow J = \frac{\kappa}{q\beta}$$

$$J_t = \varphi_t(1-\alpha)\frac{y_t^w}{n_t} - w_t + \mathbb{E}_t\left[\beta\frac{\lambda_{t+1}}{\lambda_t}(1-\rho)J_{t+1}\right] \tag{17}$$
$$\Rightarrow J = \varphi(1-\alpha)\frac{y^w}{n} - w + \beta\frac{\lambda}{\lambda}(1-\rho)J$$
$$\Rightarrow J = \varphi(1-\alpha)\frac{y^w}{n} - w + \beta(1-\rho)J$$
$$\Rightarrow [1 - \beta(1-\rho)]\, J = \varphi(1-\alpha)\frac{y^w}{n} - w$$
$$\Rightarrow w = \varphi(1-\alpha)\frac{y^w}{n} - [1 - \beta(1-\rho)]\, J$$

$$w_t = (1-\theta_w)w_t^{opt} + \theta_w w_{t-1}\frac{\left(\pi_{t-1}^{\iota_w}\pi^{1-\iota_w}\right)^{\gamma_w}}{\pi_t} \tag{28}$$
$$\Rightarrow w = (1-\theta_w)w^{opt} + \theta_w w\frac{\left(\pi^{\iota_w}\pi^{1-\iota_w}\right)^{\gamma_w}}{\pi}$$
$$\Rightarrow w = (1-\theta_w)w^{opt} + \theta_w w\pi^{(\gamma_w-1)}$$
$$\Rightarrow (1-\theta_w)w^{opt} = \left(1 - \theta_w\pi^{(\gamma_w-1)}\right)w$$
$$\Rightarrow w^{opt} = \frac{1 - \theta_w\pi^{(\gamma_w-1)}}{1-\theta_w}w$$

$$D_t = 1 + \mathbb{E}_t\left[\beta\frac{\lambda_{t+1}}{\lambda_t}(1-\rho)\theta_w\frac{\left(\pi_t^{\iota_w}\pi^{1-\iota_w}\right)^{\gamma_w}}{\pi_{t+1}}D_{t+1}\right] \tag{19}$$
$$\Rightarrow D = 1 + \beta\frac{\lambda}{\lambda}(1-\rho)\theta_w\frac{\left(\pi^{\iota_w}\pi^{1-\iota_w}\right)^{\gamma_w}}{\pi}D$$
$$\Rightarrow D = 1 + \beta(1-\rho)\theta_w\pi^{(\gamma_w-1)}D$$
$$\Rightarrow \left[1 - \beta(1-\rho)\theta_w\pi^{(\gamma_w-1)}\right]D = 1$$

$$\Rightarrow D = \frac{1}{1-\beta(1-\rho)\theta_w \pi^{(\gamma_w - 1)}}$$

$$w_t^{opt} D_t = w_t^o + \mathbb{E}_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} (1-\rho)\theta_w D_{t+1} w_{t+1}^{opt} \right] \tag{21}$$
$$\Rightarrow w^{opt} D = w^o + \beta \frac{\lambda}{\lambda} (1-\rho)\theta_w D w^{opt}$$
$$\Rightarrow w^{opt} D = w^o + \beta (1-\rho)\theta_w D w^{opt}$$
$$\Rightarrow w^o = \left[1 - \beta(1-\rho)\theta_w\right] D w^{opt}$$

$$w_t^o = (1-\eta_w) \left(\underbrace{\frac{\chi n_t^\phi}{\lambda_t}}_{=MRS_t} + b^u \right) + \eta_w \left[\kappa\theta_t + \varphi_t \underbrace{(1-\alpha)\frac{y_t^w}{n_t}}_{=MPL_t} \right] \tag{20}$$
$$\Rightarrow w^o = (1-\eta_w) \left(\frac{\chi n^\phi}{\lambda} + b^u \right) + \eta_w \left[\kappa\theta + \varphi(1-\alpha)\frac{y_t^w}{n_t} \right]$$
$$\Rightarrow (1-\eta_w) \left(\frac{\chi n^\phi}{\lambda} + b^u \right) = w^o - \eta_w \left[\kappa\theta + \varphi(1-\alpha)\frac{y_t^w}{n_t} \right]$$
$$\Rightarrow \frac{\chi n^\phi}{\lambda} + b^u = \frac{w^o - \eta_w \left[\kappa\theta + \varphi(1-\alpha)\frac{y_t^w}{n_t} \right]}{1-\eta_w}$$
$$\Rightarrow \frac{\chi n^\phi}{\lambda} = \frac{w^o - \eta_w \left[\kappa\theta + \varphi(1-\alpha)\frac{y_t^w}{n_t} \right]}{1-\eta_w} - b^u$$
$$\Rightarrow \chi = \left\{ \frac{w^o - \eta_w \left[\kappa\theta + \varphi(1-\alpha)\frac{y_t^w}{n_t} \right]}{1-\eta_w} - b^u \right\} \frac{\lambda}{n^\phi}$$

$$H_t = \left(w_t - \underbrace{\frac{\chi n_t^\phi}{\lambda_t}}_{=MRS_t} - b^u \right) + \mathbb{E}_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} (1-\rho-f_t) H_{t+1} \right] \tag{14}$$
$$\Rightarrow H = \left(w - \frac{\chi n^\phi}{\lambda} - b^u \right) + \beta \frac{\lambda}{\lambda} (1-\rho-f) H$$
$$\Rightarrow H = \left(w - \frac{\chi n^\phi}{\lambda} - b^u \right) + \beta (1-\rho-f) H$$
$$\Rightarrow \left[1 - \beta(1-\rho-f)\right] H = w - \frac{\chi n^\phi}{\lambda} - b^u$$
$$\Rightarrow H = \frac{w - \frac{\chi n^\phi}{\lambda} - b^u}{1-\beta(1-\rho-f)}$$

$$d_t^w = \varphi_t y_t^w - w_t n_t - r_t^k k_t - \kappa v_t \tag{29}$$

$$\Rightarrow d^w = \varphi y^w - wn - r^k k - \kappa v$$

$$\begin{aligned} d_t^r &= y_t - \varphi_t y_t^w \\ \Rightarrow d^r &= y - \varphi y^w \end{aligned} \tag{30}$$

$$\begin{aligned} d_t &= d_t^w + d_t^r \\ \Rightarrow d &= d^w + d^r \end{aligned} \tag{31}$$

$$\begin{aligned} & b^u(1-n_t) + \frac{R_{t-1}}{\pi_t} b_{t-1} = t_t + b_t \\ \Rightarrow & b^u(1-n) + \frac{R}{\pi} \underbrace{b}_{=0} = t + \underbrace{b}_{=0} \\ \Rightarrow & t = b^u(1-n) \end{aligned} \tag{32}$$

5 제13장 주택시장과 담보제약

본 모형의 정상상태는 모든 외생 충격이 0이고, 변수들이 시간에 따라 변하지 않는 상태($x_t = x_{t-1} = x$)로 정의된다. 정상상태의 인플레이션율(π), 가계의 노동 공급(n^s, n^b), 그리고 총 주택 공급($\bar{h}$)은 데이터와 일치하도록 외생적으로 주어진다고 가정한다. 이에 따라, 주어진 노동 공급 목표치를 달성할 수 있도록 가계의 노동 비효용 파라미터 χ_s와 χ_b를 역산(Reverse engineering)하여 내생적으로 결정한다.

또한, 본 모형의 정상상태에서는 은행 부문이 규제 당국이 제시한 목표 자기자본 비율을 완벽하게 준수한다고 가정한다.

$$\frac{k_b}{b} = \nu$$

이는 거시경제적 불확실성이 제거된 결정론적(Deterministic) 장기 균형 상태에서, 비용 최소화를 추구하는 은행이 불필요한 자기자본 조정 비용(Capital Adjustment Costs)을 지불할 유인이 없음을 반영한 것이다. 이 목표 자본 비율이 은행의 자본 축적 방정식(식 31) 및 이윤 구조와 정합성을 갖도록 하기 위해, 은행 자본 소멸률 파라미터 δ_b를 역산하여 결정한다.

주요 내생변수의 정상상태 값은 다음과 같은 재귀적(Recursive) 순서로 도출된다.

$$\begin{aligned}
\lambda_t^s &= \beta^s \mathbb{E}_t \left[\lambda_{t+1}^s \frac{R_{d,t}^D}{\pi_{t+1}} \right] \\
\Rightarrow \lambda^s &= \beta^s \lambda^s \frac{R_d^D}{\pi} \\
\Rightarrow 1 &= \beta^s \frac{R_d^D}{\pi} \\
\Rightarrow R_d^D &= \frac{\pi}{\beta^s}
\end{aligned} \tag{3}$$

$$\begin{aligned}
R_{b,t} &= R_{d,t} - \kappa_b \left(\frac{k_{b,t}}{b_t} - \nu \zeta_t^{\nu} \right) \left(\frac{k_{b,t}}{b_t} \right)^2 \\
\Rightarrow R_b &= R_d - \kappa_b \left(\frac{k_b}{b} - \nu \zeta^{\nu} \right) \left(\frac{k_b}{b} \right)^2 \\
\Rightarrow R_b &= R_d
\end{aligned} \tag{26}$$

$$\begin{aligned}
&1 - \varepsilon_{Lb} + \varepsilon_{Lb} \frac{R_{b,t}}{R_{b,t}^b} - \kappa_{Lb} \left(\frac{R_{b,t}^b}{R_{b,t-1}^b} - 1 \right) \frac{R_{b,t}^b}{R_{b,t-1}^b} - \frac{\kappa_{Lb}}{2} \left(\frac{R_{b,t}^b}{R_{b,t-1}^b} - 1 \right)^2 (1 - \varepsilon_{Lb}) \\
&+ \mathbb{E}_t \left[\Lambda_{t,t+1}^s \kappa_{Lb} \left(\frac{R_{b,t+1}^b}{R_{b,t}^b} - 1 \right) \left(\frac{R_{b,t+1}^b}{R_{b,t}^b} \right)^2 \frac{\omega_b b_{t+1}^b}{\omega_b b_t^b} \right] = 0 \\
\Rightarrow &1 - \varepsilon_{Lb} + \varepsilon_{Lb} \frac{R_b}{R_b^b} - \kappa_{Lb} \left(\frac{R_b^b}{R_b^b} - 1 \right) \frac{R_b^b}{R_b^b} - \frac{\kappa_{Lb}}{2} \left(\frac{R_b^b}{R_b^b} - 1 \right)^2 (1 - \varepsilon_{Lb})
\end{aligned} \tag{27}$$

$$
\begin{aligned}
&+\Lambda^s\kappa_{Lb}\left(\frac{R_b^b}{R_b^b}-1\right)\left(\frac{R_b^b}{R_b^b}\right)^2\frac{\omega_b b^b}{\omega_b b^b}=0\\
\Rightarrow&1-\varepsilon_{Lb}+\varepsilon_{Lb}\frac{R_b}{R_b^b}=0\\
\Rightarrow&\varepsilon_{Lb}\frac{R_b}{R_b^b}=\varepsilon_{Lb}-1\\
\Rightarrow&R_b^b=\frac{\varepsilon_{Lb}}{\varepsilon_{Lb}-1}R_b
\end{aligned}
$$

$$
\begin{aligned}
&1-\varepsilon_{Le}+\varepsilon_{Le}\frac{R_{b,t}}{R_{b,t}^e}-\kappa_{Le}\left(\frac{R_{b,t}^e}{R_{b,t-1}^e}-1\right)\frac{R_{b,t}^e}{R_{b,t-1}^e}-\frac{\kappa_{Le}}{2}\left(\frac{R_{b,t}^e}{R_{b,t-1}^e}-1\right)^2(1-\varepsilon_{Le})\\
&+\mathbb{E}_t\left[\Lambda_{t,t+1}^s\kappa_{Le}\left(\frac{R_{b,t+1}^e}{R_{b,t}^e}-1\right)\left(\frac{R_{b,t+1}^e}{R_{b,t}^e}\right)^2\frac{b_{t+1}^e}{b_t^e}\right]=0 \qquad (28)\\
\Rightarrow&1-\varepsilon_{Le}+\varepsilon_{Le}\frac{R_b}{R_b^e}-\kappa_{Le}\left(\frac{R_b^e}{R_b^e}-1\right)\frac{R_b^e}{R_b^e}-\frac{\kappa_{Le}}{2}\left(\frac{R_b^e}{R_b^e}-1\right)^2(1-\varepsilon_{Le})\\
&+\Lambda^s\kappa_{Le}\left(\frac{R_b^e}{R_b^e}-1\right)\left(\frac{R_b^e}{R_b^e}\right)^2\frac{b^e}{b^e}=0\\
\Rightarrow&1-\varepsilon_{Le}+\varepsilon_{Le}\frac{R_b}{R_b^e}=0\\
\Rightarrow&\varepsilon_{Le}\frac{R_b}{R_b^e}=\varepsilon_{Le}-1\\
\Rightarrow&R_b^e=\frac{\varepsilon_{Le}}{\varepsilon_{Le}-1}R_b
\end{aligned}
$$

$$
\begin{aligned}
&-1-\varepsilon_D+\varepsilon_D\frac{R_{d,t}}{R_{d,t}^D}-\kappa_D\left(\frac{R_{d,t}^D}{R_{d,t-1}^D}-1\right)\frac{R_{d,t}^D}{R_{d,t-1}^D}-\frac{\kappa_D}{2}\left(\frac{R_{d,t}^D}{R_{d,t-1}^D}-1\right)^2(1+\varepsilon_D)\\
&+\mathbb{E}_t\left[\Lambda_{t,t+1}^s\kappa_D\left(\frac{R_{d,t+1}^D}{R_{d,t}^D}-1\right)\left(\frac{R_{d,t+1}^D}{R_{d,t}^D}\right)^2\frac{d_{t+1}}{d_t}\right]=0 \qquad (29)\\
\Rightarrow&-1-\varepsilon_D+\varepsilon_D\frac{R_d}{R_d^D}-\kappa_D\left(\frac{R_d^D}{R_d^D}-1\right)\frac{R_d^D}{R_d^D}-\frac{\kappa_D}{2}\left(\frac{R_d^D}{R_d^D}-1\right)^2(1+\varepsilon_D)\\
&+\Lambda^s\kappa_D\left(\frac{R_d^D}{R_d^D}-1\right)\left(\frac{R_d^D}{R_d^D}\right)^2\frac{d}{d}=0\\
\Rightarrow&-1-\varepsilon_D+\varepsilon_D\frac{R_d}{R_d^D}=0\\
\Rightarrow&\varepsilon_D\frac{R_d}{R_d^D}=\varepsilon_D+1\\
\Rightarrow&R_d=\frac{\varepsilon_D+1}{\varepsilon_D}R_d^D
\end{aligned}
$$

여기서 ε_D는 예금 수요의 가격 탄력성으로 양(+)의 값을 가지므로, 우변의 계수는 1보다 큰 값이다 ($\frac{\varepsilon_D+1}{\varepsilon_D} = 1 + \frac{1}{\varepsilon_D} > 1$).

이는 정상상태에서 도매 금리(정책 금리, R_d)가 가계에게 지급하는 소매 예금 금리(R_d^D)보다 항상 높게 형성됨($R_d > R_d^D$)을 의미한다. 경제학적으로 이는 소매 은행이 예금 시장에서 보유한 독점적 경쟁력(Market Power)을 이용하여, 자금 조달 비용을 낮추기 위해 시장 기준 금리보다 낮은 이자를 지급하는 예금 마크다운(Deposit Markdown) 행태를 반영한 것이다. 결과적으로 은행은 이 금리차 ($R_d - R_d^D$)만큼의 확정적인 예금 마진을 확보하게 된다.

이와 대조적으로, 대출 시장에서는 은행이 자금의 기회비용인 도매 금리에 일정한 마진을 가산하는 대출 마크업(Loan Markup) 행태를 보인다($R_b^b, R_b^e > R_d$). 따라서 본 모형의 정상상태 금리 체계는 자금의 조달에서 운용에 이르기까지 다음과 같은 명확한 위계 구조를 갖는다.

$$\underbrace{R_d^D}_{\text{소매 예금 금리}} \quad < \quad \underbrace{R_d}_{\text{정책(도매) 금리}} \quad < \quad \underbrace{R_b^b, R_b^e}_{\text{소매 대출 금리}}$$

이는 은행이 예금 시장에서는 기준 금리보다 낮은 비용으로 자금을 조달하여 1차 마진($R_d - R_d^D$)을 확보하고, 대출 시장에서는 기준 금리보다 높은 수익률로 자금을 운용하여 2차 마진($R_b^L - R_d$)을 남김으로써, 양방향에서 중개 이윤(Spread)을 창출하는 금융 기관의 본질적인 수익 구조를 보여준다.

$$\begin{aligned}
1 &= q_t^k \zeta_t^i \left[1 - \frac{\psi}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 - \psi \left(\frac{i_t}{i_{t-1}} - 1 \right) \frac{i_t}{i_{t-1}} \right] \\
&\quad + \mathbb{E}_t \left[\Lambda_{t,t+1}^s q_{t+1}^k \zeta_{t+1}^i \psi \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \qquad (25) \\
\Rightarrow 1 &= q^k \zeta^i \left[1 - \frac{\psi}{2} \left(\frac{i}{i} - 1 \right)^2 - \psi \left(\frac{i}{i} - 1 \right) \frac{i}{i} \right] \\
&\quad + \Lambda^s q^k \zeta^i \psi \left(\frac{i}{i} - 1 \right) \left(\frac{i}{i} \right)^2 \\
\Rightarrow 1 &= q^k \underbrace{\zeta^i}_{=1} \\
\Rightarrow q^k &= 1
\end{aligned}$$

$$\begin{aligned}
1 &= (1-\theta)(\tilde{p}_t^{opt})^{1-\varepsilon_p} + \theta \left(\frac{(\pi_{t-1}^{\iota} \pi^{1-\iota})^{\gamma}}{\pi_t} \right)^{1-\varepsilon_p} \qquad (37) \\
\Rightarrow 1 &= (1-\theta)(\tilde{p}^{opt})^{1-\varepsilon_p} + \theta \left(\frac{(\pi^{\iota} \pi^{1-\iota})^{\gamma}}{\pi} \right)^{1-\varepsilon_p} \\
\Rightarrow 1 &= (1-\theta)(\tilde{p}^{opt})^{1-\varepsilon_p} + \theta \pi^{(\gamma-1)(1-\varepsilon_p)} \\
\Rightarrow (1-\theta)(\tilde{p}^{opt})^{1-\varepsilon_p} &= 1 - \theta \pi^{(\gamma-1)(1-\varepsilon_p)} \\
\Rightarrow (\tilde{p}^{opt})^{1-\varepsilon_p} &= \frac{1 - \theta \pi^{(\gamma-1)(1-\varepsilon_p)}}{1-\theta}
\end{aligned}$$

$$\Rightarrow \tilde{p}^{opt} = \left[\frac{1 - \theta \pi^{(\gamma-1)(1-\varepsilon_p)}}{1-\theta} \right]^{\frac{1}{1-\varepsilon_p}}$$

$$\Delta_t = (1-\theta)(\tilde{p}_t^{opt})^{-\varepsilon_p} + \theta \left(\frac{\pi_t}{(\pi_{t-1}^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p} \Delta_{t-1} \tag{38}$$

$$\Rightarrow \Delta = (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta \left(\frac{\pi}{(\pi^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p} \Delta$$

$$\Rightarrow \Delta = (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p} + \theta \pi^{(1-\gamma)\varepsilon_p} \Delta$$

$$\Rightarrow \left[1 - \theta \pi^{(1-\gamma)\varepsilon_p} \right] \Delta = (1-\theta)(\tilde{p}^{opt})^{-\varepsilon_p}$$

$$\Rightarrow \Delta = \frac{1-\theta}{1-\theta \pi^{(1-\gamma)\varepsilon_p}} (\tilde{p}^{opt})^{-\varepsilon_p}$$

$$x_{n,t} = mc_t^r y_t + \theta \mathbb{E}_t \left[\Lambda_{t,t+1}^s \left(\frac{\pi_{t+1}}{(\pi_t^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p} x_{n,t+1} \right] \tag{22}$$

$$\Rightarrow x_n = mc^r y + \theta \underbrace{\Lambda^s}_{=\beta^s} \left(\frac{\pi}{(\pi^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p} x_n$$

$$\Rightarrow x_n = mc^r y + \theta \beta^s \pi^{(1-\gamma)\varepsilon_p} x_n$$

$$\Rightarrow \left[1 - \theta \beta^s \pi^{(1-\gamma)\varepsilon_p} \right] x_n = mc^r y$$

$$\Rightarrow x_n = \frac{mc^r y}{1 - \theta \beta^s \pi^{(1-\gamma)\varepsilon_p}}$$

$$x_{d,t} = y_t + \theta \mathbb{E}_t \left[\Lambda_{t,t+1}^s \left(\frac{\pi_{t+1}}{(\pi_t^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p - 1} x_{d,t+1} \right] \tag{23}$$

$$\Rightarrow x_d = y + \theta \underbrace{\Lambda^s}_{=\beta^s} \left(\frac{\pi}{(\pi^{\iota} \pi^{1-\iota})^{\gamma}} \right)^{\varepsilon_p - 1} x_d$$

$$\Rightarrow x_d = y + \theta \beta^s \pi^{(1-\gamma)(\varepsilon_p - 1)} x_d$$

$$\Rightarrow \left[1 - \theta \beta^s \pi^{(1-\gamma)(\varepsilon_p - 1)} \right] x_d = y$$

$$\Rightarrow x_d = \frac{y}{1 - \theta \beta^s \pi^{(1-\gamma)(\varepsilon_p - 1)}}$$

$$\tilde{p}_t^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_{n,t}}{x_{d,t}} \tag{21}$$

$$\Rightarrow \tilde{p}^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{x_n}{x_d}$$

$$\Rightarrow \tilde{p}^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\frac{mc^r y}{1 - \theta \beta^s \pi^{(1-\gamma)\varepsilon_p}}}{\frac{y}{1 - \theta \beta^s \pi^{(1-\gamma)(\varepsilon_p - 1)}}}$$

$$\Rightarrow \tilde{p}^{opt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{1 - \theta\beta^s \pi^{(1-\gamma)(\varepsilon_p - 1)}}{1 - \theta\beta^s \pi^{(1-\gamma)\varepsilon_p}} mc^r$$
$$\Rightarrow mc^r = \frac{\varepsilon_p - 1}{\varepsilon_p} \frac{1 - \theta\beta^s \pi^{(1-\gamma)\varepsilon_p}}{1 - \theta\beta^s \pi^{(1-\gamma)(\varepsilon_p - 1)}} \tilde{p}^{opt}$$

$$mc_t^r = p_t^w \tag{20}$$
$$\Rightarrow p^w = mc^r$$

$$\lambda^e - \mu^e = \beta^e \lambda^e \frac{R_b^e}{\pi} \tag{16}$$
$$\Rightarrow 1 - \frac{\mu^e}{\lambda^e} = \beta^e \frac{R_b^e}{\pi}$$
$$\Rightarrow \frac{\mu^e}{\lambda^e} = 1 - \beta^e \frac{R_b^e}{\pi}$$

$$\lambda^e q^k = \beta^e \lambda^e \left(MPK + q^k(1-\delta) \right) + \mu^e m_E \frac{q^k (1-\delta)\pi}{R_b^e} \tag{17}$$
$$\Rightarrow q^k = \beta^e \left(MPK + q^k(1-\delta) \right) + \frac{\mu^e}{\lambda^e} m_E \frac{q^k (1-\delta)\pi}{R_b^e}$$
$$\Rightarrow q^k \left[1 - \beta^e (1-\delta) - \left(1 - \beta^e \frac{R_b^e}{\pi} \right) m_E \frac{(1-\delta)\pi}{R_b^e} \right] = \beta^e MPK$$
$$\Rightarrow MPK = \frac{q^k}{\beta^e} \left[1 - (1-\delta) \left(\beta^e + m_E \left(\frac{\pi}{R_b^e} - \beta^e \right) \right) \right]$$

$$MPK_t = p_t^w \alpha \frac{y_t^w}{k_{t-1}}$$
$$\Rightarrow MPK = p^w \alpha \frac{y^w}{k}$$
$$\Rightarrow \frac{k}{y^w} = \frac{p^w \alpha}{MPK}$$

$$y^w = z k^\alpha \left[(\omega_s n^s)^{1-\mu} (\omega_b n^b)^\mu \right]^{1-\alpha} \tag{12}$$
$$\Rightarrow y^w = z \left(\frac{k}{y^w} y^w \right)^\alpha \left[(\omega_s n^s)^{1-\mu} (\omega_b n^b)^\mu \right]^{1-\alpha}$$
$$\Rightarrow (y^w)^{1-\alpha} = z \left(\frac{k}{y^w} \right)^\alpha \left[(\omega_s n^s)^{1-\mu} (\omega_b n^b)^\mu \right]^{1-\alpha}$$
$$\Rightarrow y^w = \left[z \left(\frac{k}{y^w} \right)^\alpha \left[(\omega_s n^s)^{1-\mu} (\omega_b n^b)^\mu \right]^{1-\alpha} \right]^{\frac{1}{1-\alpha}}$$

n^s, n^b, z가 주어졌고, k/y^w는 위에서 결정되어 있으므로 y^w도 결정된다. (y^w가 결정되면 k도 결정된다.)

$$
\begin{aligned}
y_t &= \frac{z_t k_{t-1}^{\alpha}\left[(\omega_s n_t^s)^{1-\mu}(\omega_b n_t^b)^{\mu}\right]^{1-\alpha}}{\Delta_t} \\
\Rightarrow y &= \frac{y^w}{\Delta}
\end{aligned} \tag{36}
$$

$$
\begin{aligned}
k_t &= (1-\delta)k_{t-1} + \zeta_t^i\left[1-\frac{\psi}{2}\left(\frac{i_t}{i_{t-1}}-1\right)^2\right]i_t \\
\Rightarrow k &= (1-\delta)k + \underbrace{\zeta^i}_{=1}\left[1-\frac{\psi}{2}\left(\frac{i}{i}-1\right)^2\right]i \\
\Rightarrow k &= (1-\delta)k + i \\
\Rightarrow i &= \delta k
\end{aligned} \tag{24}
$$

$$
\begin{aligned}
b_t^e &= m_{E,t}\mathbb{E}_t\left[\frac{q_{t+1}^k(1-\delta)k_t\pi_{t+1}}{R_{b,t}^e}\right] \\
\Rightarrow b^e &= m_E\frac{q^k(1-\delta)\pi}{R_b^e}k
\end{aligned} \tag{14}
$$

$$
\begin{aligned}
w_t^s &= p_t^w(1-\alpha)(1-\mu)\frac{y_t^w}{\omega_s n_t^s} \\
\Rightarrow w^s &= p^w(1-\alpha)(1-\mu)\frac{y^w}{\omega_s n^s}
\end{aligned} \tag{18}
$$

$$
\begin{aligned}
w_t^b &= p_t^w(1-\alpha)\mu\frac{y_t^w}{\omega_b n_t^b} \\
\Rightarrow w^b &= p^w(1-\alpha)\mu\frac{y^w}{\omega_b n^b}
\end{aligned} \tag{19}
$$

$$
\begin{aligned}
c_t^e + \omega_s w_t^s n_t^s + \omega_b w_t^b n_t^b + \frac{R_{b,t-1}^e b_{t-1}^e}{\pi_t} + q_t^k\left[k_t-(1-\delta)k_{t-1}\right] &= p_t^w y_t^w + b_t^e \\
\Rightarrow c^e + \omega_s w^s n^s + \omega_b w^b n^b + \frac{R_b^e b^e}{\pi} + q^k\left[k-(1-\delta)k\right] &= p^w y^w + b^e \\
\Rightarrow c^e + \omega_s w^s n^s + \omega_b w^b n^b + \frac{R_b^e b^e}{\pi} + q^k\delta k &= p^w y^w + b^e
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
&\Rightarrow c^e = p^w y^w + b^e - \omega_s w^s n^s - \omega_b w^b n^b - \frac{R_b^e b^e}{\pi} - q^k \delta k \\
&\Rightarrow c^e = p^w y^w - \omega_s w^s n^s - \omega_b w^b n^b + \left(1 - \frac{R_b^e}{\pi}\right) b^e - q^k \delta k \\
&\Rightarrow c^e = p^w y^w - \underbrace{p^w (1-\alpha)(1-\mu) y^w}_{=\omega_s w^s n^s} - \underbrace{p^w (1-\alpha)\mu y^w}_{=\omega_b w^b n^b} + \left(1 - \frac{R_b^e}{\pi}\right) b^e - q^k \delta k
\end{aligned}
$$

b^b 결정

기업가 부문에서 b^e와 y^w가 결정되었으나, 차입가계 대출 b^b, 저축가계 소비 c^s, 실질 주택 가격 q는 상호 의존적이므로 연립하여 도출한다.

먼저, 은행의 총대출(b), 자기자본(k_b), 예금(d), 그리고 총배당금(div), 저축가계의 소비(c^s), 저축가계의 한계효용(λ^s), 주택가격(q), 저축가계의 주택보유량(h^s), 차입가계의 주택보유량(h^b), 차입가계의 소비(c^b), 차입가계의 한계효용(λ^b), 차입가계 담보제약의 그림자 가격(μ^b) 등을 미지수 b^b에 대한 1차 선형 함수로 표현한다.

$$
\begin{aligned}
b_t &= \omega_b b_t^b + b_t^e \\
\Rightarrow b(b^b) &= \omega_b b^b + b^e
\end{aligned}
\tag{34}
$$

$$
\begin{aligned}
\frac{k_b}{b} &= \nu \\
\Rightarrow k_b(b^b) &= \nu b(b^b)
\end{aligned}
$$

$$
\begin{aligned}
d_t &= b_t - k_{b,t} \\
\Rightarrow d &= b - k_b \\
\Rightarrow d &= b - \nu b \\
\Rightarrow d(b^b) &= (1-\nu) b(b^b)
\end{aligned}
\tag{42}
$$

$$
\begin{aligned}
\Pi_{m,t} &= y_t - p_t^w (\Delta_t y_t) \\
\Rightarrow \Pi_m &= \frac{y^w}{\Delta} - p^w \Delta \frac{y^w}{\Delta} \\
\Rightarrow \Pi_m &= y^w \left(\frac{1}{\Delta} - p^w\right)
\end{aligned}
\tag{40}
$$

$$
\Pi_{k,t} = q_t^k \zeta_t^i \left[1 - \frac{\psi}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2\right] i_t - i_t \tag{41}
$$

$$
\begin{aligned}
&\Rightarrow \Pi_k = \underbrace{q^k}_{=1}\underbrace{\zeta^i}_{=1}\left[1-\frac{\psi}{2}\left(\frac{i}{i}-1\right)^2\right]i - i \\
&\Rightarrow \Pi_k = i - i \\
&\Rightarrow \Pi_k = 0
\end{aligned}
$$

$$
\begin{aligned}
div_t &= \Pi_{m,t} + \Pi_{k,t} + \delta_b \frac{k_{b,t-1}}{\pi_t} \qquad (39)\\
\Rightarrow div &= \Pi_m + \underbrace{\Pi_k}_{=0} + \delta_b \frac{k_b}{\pi} \\
\Rightarrow div(b^b) &= y^w\left(\frac{1}{\Delta} - p^w\right) + \delta_b \frac{k_b(b^b)}{\pi}
\end{aligned}
$$

$$
\begin{aligned}
c_t^s + q_t(h_t^s - h_{t-1}^s) + d_t^s &= w_t^s n_t^s + \frac{R_{d,t-1}^D d_{t-1}^s}{\pi_t} + \frac{div_t}{\omega_s} \qquad (1)\\
\Rightarrow c^s + q(h^s - h^s) + \frac{d}{\omega_s} &= w^s n^s + \frac{R_d^D}{\pi}\frac{d}{\omega_s} + \frac{div}{\omega_s} \\
\Rightarrow c^s &= w^s n^s - \left(1 - \frac{R_d^D}{\pi}\right)\frac{d}{\omega_s} + \frac{div}{\omega_s} \\
\Rightarrow c^s(b^b) &= w^s n^s - \left(1 - \frac{R_d^D}{\pi}\right)\frac{d(b^b)}{\omega_s} + \frac{div(b^b)}{\omega_s}
\end{aligned}
$$

$$
\begin{aligned}
\lambda_t^s &= \zeta_t^c (c_t^s - \hbar c_{t-1}^s)^{-\sigma} - \beta^s \hbar \mathbb{E}_t\left[\zeta_{t+1}^c (c_{t+1}^s - \hbar c_t^s)^{-\sigma}\right] \qquad (2)\\
\Rightarrow \lambda^s &= \underbrace{\zeta^c}_{=1}(c^s - \hbar c^s)^{-\sigma} - \beta^s \hbar \underbrace{\zeta^c}_{=1}(c^s - \hbar c^s)^{-\sigma} \\
\Rightarrow \lambda^s &= (1 - \beta^s \hbar)(c^s - \hbar c^s)^{-\sigma} \\
\Rightarrow \lambda^s(b^b) &= (1 - \beta^s \hbar)(1 - \hbar)^{-\sigma} c^s(b^b)^{-\sigma}
\end{aligned}
$$

$$
\begin{aligned}
h_t &= \omega_s h_t^s + \omega_b h_t^b = \bar{h} \qquad (33)\\
\Rightarrow &\omega_s h^s + \omega_b h^b = \bar{h} \\
\Rightarrow &\omega_s \underbrace{\frac{\xi}{(1-\beta^s)\lambda^s q}}_{=h^s} + \omega_b \underbrace{\frac{b^b}{m_H q \pi} R_b^b}_{=h^b} = \bar{h} \\
\Rightarrow &q\bar{h} = \omega_s \frac{\xi}{(1-\beta^s)\lambda^s} + \omega_b \frac{b^b}{m_H \pi} R_b^b \\
\Rightarrow &q(b^b) = \left(\omega_s \frac{\xi}{(1-\beta^s)\lambda^s(b^b)} + \omega_b \frac{b^b}{m_H \pi} R_b^b\right)\frac{1}{\bar{h}}
\end{aligned}
$$

여기서 식 (4), (7)로 부터 h^s, h^b를 도출하였다.

$$\lambda_t^s q_t = \frac{\zeta_t^h \xi}{h_t^s} + \beta^s \mathbb{E}_t \left[\lambda_{t+1}^s q_{t+1}\right] \tag{4}$$

$$\Rightarrow \lambda^s q = \frac{\zeta^h \xi}{h^s} + \beta^s \lambda^s q$$

$$\Rightarrow \lambda^s q - \beta^s \lambda^s q = \frac{\xi}{h^s}$$

$$\Rightarrow (1-\beta^s)\lambda^s q = \frac{\xi}{h^s}$$

$$\Rightarrow h^s(b^b) = \frac{\xi}{(1-\beta^s)\lambda^s(b^b)q(b^b)}$$

$$b_t^b = m_{H,t}\mathbb{E}_t \left[\frac{q_{t+1} h_t^b \pi_{t+1}}{R_{b,t}^b}\right] \tag{7}$$

$$\Rightarrow b^b = m_H \frac{q h^b \pi}{R_b^b}$$

$$\Rightarrow h^b(b^b) = \frac{b^b}{m_H q(b^b)\pi} R_b^b$$

$$c_t^b + q_t(h_t^b - h_{t-1}^b) + \frac{R_{b,t-1}^b b_{t-1}^b}{\pi_t} = w_t^b n_t^b + b_t^b \tag{6}$$

$$\Rightarrow c^b + q(h^b - h^b) + \frac{R_b^b b^b}{\pi} = w^b n^b + b^b$$

$$\Rightarrow c^b + \frac{R_b^b b^b}{\pi} = w^b n^b + b^b$$

$$\Rightarrow c^b = w^b n^b + b^b - \frac{R_b^b b^b}{\pi}$$

$$\Rightarrow c^b(b^b) = w^b n^b + \left(1 - \frac{R_b^b}{\pi}\right) b^b$$

$$\lambda_t^b = \zeta_t^c (c_t^b - \hbar c_{t-1}^b)^{-\sigma} - \beta^b \hbar \mathbb{E}_t \left[\zeta_{t+1}^c (c_{t+1}^b - \hbar c_t^b)^{-\sigma}\right] \tag{8}$$

$$\Rightarrow \lambda^b = \underbrace{\zeta^c}_{=1}(c^b - \hbar c^b)^{-\sigma} - \beta^b \hbar \underbrace{\zeta^c}_{=1}(c^b - \hbar c^b)^{-\sigma}$$

$$\Rightarrow \lambda^b = (c^b - \hbar c^b)^{-\sigma} - \beta^b \hbar (c^b - \hbar c^b)^{-\sigma}$$

$$\Rightarrow \lambda^b = (1 - \beta^b \hbar)(c^b - \hbar c^b)^{-\sigma}$$

$$\Rightarrow \lambda^b(b^b) = (1 - \beta^b \hbar)(1 - \hbar)^{-\sigma} c^b(b^b)^{-\sigma}$$

$$\lambda_t^b - \mu_t^b = \beta^b \mathbb{E}_t \left[\lambda_{t+1}^b \frac{R_{b,t}^b}{\pi_{t+1}}\right] \tag{9}$$

$$\Rightarrow \lambda^b - \mu^b = \beta^b \lambda^b \frac{R_b^b}{\pi}$$
$$\Rightarrow \mu^b = \lambda^b - \beta^b \lambda^b \frac{R_b^b}{\pi}$$
$$\Rightarrow \mu^b(b^b) = \left(1 - \beta^b \frac{R_b^b}{\pi}\right) \lambda^b(b^b)$$

이제 다음 식을 만족하는 b^b를 구한다.

$$\lambda_t^b q_t = \frac{\zeta_t^h \xi}{h_t^b} + \beta^b \mathbb{E}_t \left[\lambda_{t+1}^b q_{t+1}\right] + \mu_t^b m_{H,t} \mathbb{E}_t \left[\frac{q_{t+1}\pi_{t+1}}{R_{b,t}^b}\right] \tag{10}$$
$$\Rightarrow \lambda^b q = \frac{\zeta^h \xi}{h^b} + \beta^b \left[\lambda^b q\right] + \mu^b m_H \left[\frac{q\pi}{R_b^b}\right]$$
$$\Rightarrow \lambda^b(b^b) q(b^b) - \frac{\xi}{h^b(b^b)} - \beta^b \left[\lambda^b(b^b) q(b^b)\right] - \mu^b(b^b) m_H \left[\frac{q(b^b)\pi}{R_b^b}\right] = 0$$

위 식을 만족하는 b^b를 구했다면 b, d, div, c^s, λ^s, q, h^s, h^b, c^b, λ^b, μ^b는 결정된 b^b의 함수이므로 쉽게 구할 수 있다.

$$c_t = \omega_s c_t^s + \omega_b c_t^b + c_t^e \tag{32}$$
$$\Rightarrow c = \omega_s c^s + \omega_b c^b + c^e$$

$$d_t = \omega_s d_t^s \tag{35}$$
$$\Rightarrow d = \omega_s d^s$$
$$\Rightarrow d^s = \frac{d}{\omega_s}$$

$$\lambda_t^e = (c_t^e - \chi_e c_{t-1}^e)^{-\sigma} - \beta^e \chi_e \mathbb{E}_t \left[(c_{t+1}^e - \chi_e c_t^e)^{-\sigma}\right] \tag{15}$$
$$\Rightarrow \lambda^e = (c^e - \chi_e c^e)^{-\sigma} - \beta^e \chi_e (c^e - \chi_e c^e)^{-\sigma}$$
$$\Rightarrow \lambda^e = (1 - \beta^e \chi_e)(c^e - \chi_e c^e)^{-\sigma}$$
$$\Rightarrow \lambda^e = (1 - \beta^e \chi_e)(1 - \chi_e)^{-\sigma}(c^e)^{-\sigma}$$

$$\Pi_{b,t} = (R_{b,t}^b - 1)(\omega_b b_t^b) + (R_{b,t}^e - 1) b_t^e - (R_{d,t}^D - 1) d_t - \frac{\kappa_b}{2} \left(\frac{k_{b,t}}{b_t} - \nu \zeta_t^\nu\right)^2 k_{b,t}$$
$$- \frac{\kappa_{Lb}}{2} \left(\frac{R_{b,t}^b}{R_{b,t-1}^b} - 1\right)^2 R_{b,t}^b (\omega_b b_t^b) - \frac{\kappa_{Le}}{2} \left(\frac{R_{b,t}^e}{R_{b,t-1}^e} - 1\right)^2 R_{b,t}^e b_t^e - \frac{\kappa_D}{2} \left(\frac{R_{d,t}^D}{R_{d,t-1}^D} - 1\right)^2 R_{d,t}^D d_t \tag{30}$$

$$\Rightarrow \Pi_b = (R_b^b - 1)(\omega_b b^b) + (R_b^e - 1) b^e - (R_d^D - 1) d - \frac{\kappa_b}{2} \left(\frac{k_b}{b} - \nu \zeta^\nu\right)^2 k_b$$

$$
\begin{aligned}
&- \frac{\kappa_{Lb}}{2}\left(\frac{R_b^b}{R_b^b}-1\right)^2 R_b^b(\omega_b b^b) - \frac{\kappa_{Le}}{2}\left(\frac{R_b^e}{R_b^e}-1\right)^2 R_b^e b^e - \frac{\kappa_D}{2}\left(\frac{R_d^D}{R_d^D}-1\right)^2 R_d^D d \\
\Rightarrow &\Pi_b = (R_b^b-1)(\omega_b b^b) + (R_b^e-1)b^e - (R_d^D-1)d
\end{aligned}
$$

한편 위 식에서 Π_b가 결정되었으므로 식 (31)은 결정된 Π_b가 되도록 하는 δ_b를 구하는데 사용한다.

$$
\begin{aligned}
k_{b,t} &= (1-\delta_b)\frac{k_{b,t-1}}{\pi_t} + \Pi_{b,t-1} + \zeta_t^{k_b} \qquad (31)\\
\Rightarrow k_b &= (1-\delta_b)\frac{k_b}{\pi} + \Pi_b + \underbrace{\zeta^{k_b}}_{=0} \\
\Rightarrow k_b &= \frac{k_b}{\pi} - \delta_b\frac{k_b}{\pi} + \Pi_b \\
\Rightarrow \delta_b\frac{k_b}{\pi} &= \frac{k_b}{\pi} - k_b + \Pi_b
\end{aligned}
$$

마지막으로 n^s와 n^b를 가능케하는 χ_s와 χ_b를 결정한다.

$$
\begin{aligned}
\chi_s(n_t^s)^\phi &= \lambda_t^s w_t^s \qquad (5)\\
\Rightarrow \chi_s(n^s)^\phi &= \lambda^s w^s \\
\Rightarrow \chi_s &= \frac{\lambda^s w^s}{(n^s)^\phi}
\end{aligned}
$$

$$
\begin{aligned}
\chi_b(n_t^b)^\phi &= \lambda_t^b w_t^b \qquad (11)\\
\Rightarrow \chi_b(n^b)^\phi &= \lambda^b w^b \\
\Rightarrow \chi_b &= \frac{\lambda^b w^b}{(n^b)^\phi}
\end{aligned}
$$

6 제15장 소비재와 투자재: 2부문 모형

소비재, 투자재, 임금의 인플레이션 π_c, π_i, π_w, 그리고 노동시간 n은 주어졌다고 가정한다. π_c, π_i, π_w는 중앙은행의 물가안정목표로 설정한다. 일반적으로 분기모형에서는 연율이 2%가 되도록 $(1+0.02)^{\frac{1}{4}}$로 설정한다. 노동 n은 하루의 1/3(24시간 중 8시간)을 노동에 사용한다는 의미에서 모형의 파라미터 χ 값을 $n=1/3$이 되도록 설정한다. 또한 소비재와 투자재의 기술수준 $z_{c,t}, z_{i,t}$, 그리고 투자특화기술수준 μ_c, μ_i의 정상상태 값은 1로 정규화한다.

$$
\begin{aligned}
\lambda_t &= \beta \mathbb{E}_t \left[\lambda_{t+1} \frac{R_t}{\pi_{c,t+1}} \right] \qquad (7)\\
\Rightarrow \lambda &= \beta \lambda \frac{R}{\pi_c} \\
\Rightarrow 1 &= \beta \frac{R}{\pi_c} \\
\Rightarrow R &= \frac{\pi_c}{\beta}
\end{aligned}
$$

$$
\begin{aligned}
Q_{c,t,t+1} &\equiv \beta \frac{\lambda_{t+1}}{\lambda_t} \qquad (33)\\
\Rightarrow Q_c &\equiv \beta \frac{\lambda}{\lambda} \\
\Rightarrow Q_c &\equiv \beta
\end{aligned}
$$

$$
\begin{aligned}
Q_{i,t,t+1} &\equiv Q_{c,t,t+1} \frac{q_{t+1}}{q_t} \qquad (34)\\
\Rightarrow Q_i &\equiv Q_c \frac{q}{q} \\
\Rightarrow Q_i &\equiv Q_c
\end{aligned}
$$

$$
\begin{aligned}
1 &= (1-\theta_c) \left(p_{c,t}^{opt} \right)^{1-\varepsilon_c} + \theta_c \left[\frac{(\pi_{c,t-1}^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}}{\pi_{c,t}} \right]^{1-\varepsilon_c} \qquad (18)\\
\Rightarrow 1 &= (1-\theta_c) \left(p_c^{opt} \right)^{1-\varepsilon_c} + \theta_c \left[\frac{(\pi_c^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}}{\pi_c} \right]^{1-\varepsilon_c} \\
\Rightarrow 1 &= (1-\theta_c) \left(p_c^{opt} \right)^{1-\varepsilon_c} + \theta_c \pi_c^{(\gamma_c - 1)(1-\varepsilon_c)} \\
\Rightarrow (1-\theta_c) \left(p_c^{opt} \right)^{1-\varepsilon_c} &= 1 - \theta_c \pi_c^{(\gamma_c - 1)(1-\varepsilon_c)} \\
\Rightarrow \left(p_c^{opt} \right)^{1-\varepsilon_c} &= \frac{1 - \theta_c \pi_c^{(\gamma_c - 1)(1-\varepsilon_c)}}{1-\theta_c} \\
\Rightarrow p_c^{opt} &= \left[\frac{1 - \theta_c \pi_c^{(\gamma_c - 1)(1-\varepsilon_c)}}{1-\theta_c} \right]^{\frac{1}{1-\varepsilon_c}}
\end{aligned}
$$

$$1 = (1-\theta_i)\left(p_{i,t}^{opt}\right)^{1-\varepsilon_i} + \theta_i\left[\frac{(\pi_{i,t-1}^{\iota_i}\pi_c^{1-\iota_i})^{\gamma_i}}{\pi_{i,t}}\right]^{1-\varepsilon_i} \tag{19}$$

$$\Rightarrow 1 = (1-\theta_i)\left(p_i^{opt}\right)^{1-\varepsilon_i} + \theta_i\left[\frac{(\pi_i^{\iota_i}\pi_c^{1-\iota_i})^{\gamma_i}}{\pi_i}\right]^{1-\varepsilon_i}$$

$$\Rightarrow 1 = (1-\theta_i)\left(p_i^{opt}\right)^{1-\varepsilon_i} + \theta_i\pi_i^{(\gamma_i-1)(1-\varepsilon_i)}$$

$$\Rightarrow (1-\theta_i)\left(p_i^{opt}\right)^{1-\varepsilon_i} = 1-\theta_i\pi_i^{(\gamma_i-1)(1-\varepsilon_i)}$$

$$\Rightarrow \left(p_i^{opt}\right)^{1-\varepsilon_i} = \frac{1-\theta_i\pi_i^{(\gamma_i-1)(1-\varepsilon_i)}}{1-\theta_i}$$

$$\Rightarrow p_i^{opt} = \left[\frac{1-\theta_i\pi_i^{(\gamma_i-1)(1-\varepsilon_i)}}{1-\theta_i}\right]^{\frac{1}{1-\varepsilon_i}}$$

$$1 = (1-\theta_w)\left(\tilde{w}_t^{opt}\right)^{1-\varepsilon_w} + \theta_w\left[\frac{(\pi_{w,t-1}^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w}}{\pi_{w,t}}\right]^{1-\varepsilon_w} \tag{20}$$

$$\Rightarrow 1 = (1-\theta_w)\left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} + \theta_w\left[\frac{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w}}{\pi_w}\right]^{1-\varepsilon_w}$$

$$\Rightarrow 1 = (1-\theta_w)\left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} + \theta_w\pi_w^{(\gamma_w-1)(1-\varepsilon_w)}$$

$$\Rightarrow (1-\theta_w)\left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} = 1-\theta_w\pi_w^{(\gamma_w-1)(1-\varepsilon_w)}$$

$$\Rightarrow \left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} = \frac{1-\theta_w\pi_w^{(\gamma_w-1)(1-\varepsilon_w)}}{1-\theta_w}$$

$$\Rightarrow \tilde{w}^{opt} = \left[\frac{1-\theta_w\pi_w^{(\gamma_w-1)(1-\varepsilon_w)}}{1-\theta_w}\right]^{\frac{1}{1-\varepsilon_w}}$$

$$\Delta_{c,t} = (1-\theta_c)(p_{c,t}^{opt})^{-\varepsilon_c} + \theta_c\left(\frac{(\pi_{c,t-1}^{\iota_c}\pi_c^{1-\iota_c})^{\gamma_c}}{\pi_{c,t}}\right)^{-\varepsilon_c}\Delta_{c,t-1} \tag{22}$$

$$\Rightarrow \Delta_c = (1-\theta_c)(p_c^{opt})^{-\varepsilon_c} + \theta_c\left(\frac{(\pi_c^{\iota_c}\pi_c^{1-\iota_c})^{\gamma_c}}{\pi_c}\right)^{-\varepsilon_c}\Delta_c$$

$$\Rightarrow \Delta_c = (1-\theta_c)(p_c^{opt})^{-\varepsilon_c} + \theta_c\pi_c^{-\varepsilon_c(\gamma_c-1)}\Delta_c$$

$$\Rightarrow \left[1-\theta_c\pi_c^{-\varepsilon_c(\gamma_c-1)}\right]\Delta_c = (1-\theta_c)(p_c^{opt})^{-\varepsilon_c}$$

$$\Rightarrow \Delta_c = \frac{(1-\theta_c)(p_c^{opt})^{-\varepsilon_c}}{1-\theta_c\pi_c^{-\varepsilon_c(\gamma_c-1)}}$$

$$\Delta_{i,t} = (1-\theta_i)(p_{i,t}^{opt})^{-\varepsilon_i} + \theta_i\left(\frac{(\pi_{i,t-1}^{\iota_i}\pi_c^{1-\iota_i})^{\gamma_i}}{\pi_{i,t}}\right)^{-\varepsilon_i}\Delta_{i,t-1} \tag{24}$$

$$\Rightarrow \Delta_i = (1-\theta_i)(p_i^{opt})^{-\varepsilon_i} + \theta_i \left(\frac{(\pi_i^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}}{\pi_i} \right)^{-\varepsilon_i} \Delta_i$$
$$\Rightarrow \Delta_i = (1-\theta_i)(p_i^{opt})^{-\varepsilon_i} + \theta_i \pi_i^{-\varepsilon_i(\gamma_i-1)} \Delta_i$$
$$\Rightarrow \left[1 - \theta_i \pi_i^{-\varepsilon_i(\gamma_i-1)} \right] \Delta_i = (1-\theta_i)(p_i^{opt})^{-\varepsilon_i}$$
$$\Rightarrow \Delta_i = \frac{(1-\theta_i)(p_i^{opt})^{-\varepsilon_i}}{1 - \theta_i \pi_i^{-\varepsilon_i(\gamma_i-1)}}$$

$$\Delta_{w,t} = (1-\theta_w)(\tilde{w}_t^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_{w,t-1}^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}}{\pi_{w,t}} \right)^{-\varepsilon_w} \Delta_{w,t-1} \tag{26}$$
$$\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_w^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}}{\pi_w} \right)^{-\varepsilon_w} \Delta_w$$
$$\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \pi_w^{-\varepsilon_w(\gamma_w-1)} \Delta_w$$
$$\Rightarrow \left[1 - \theta_w \pi_w^{-\varepsilon_w(\gamma_w-1)} \right] \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w}$$
$$\Rightarrow \Delta_w = \frac{(1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w}}{1 - \theta_w \pi_w^{-\varepsilon_w(\gamma_w-1)}}$$

$$x_{c,n,t} = y_{c,t} mc_{c,t} + \theta_c \mathbb{E}_t \left[Q_{c,t,t+1} \left(\frac{\pi_{c,t+1}}{(\pi_{c,t}^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c} x_{c,n,t+1} \right] \tag{16}$$
$$\Rightarrow x_{c,n} = y_c mc_c + \theta_c Q_c \left(\frac{\pi_c}{(\pi_c^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c} x_{c,n}$$
$$\Rightarrow x_{c,n} = y_c mc_c + \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)} x_{c,n}$$
$$\Rightarrow \left[1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)} \right] x_{c,n} = y_c mc_c$$
$$\Rightarrow x_{c,n} = \frac{y_c mc_c}{1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}}$$

$$x_{c,d,t} = y_{c,t} + \theta_c \mathbb{E}_t \left[Q_{c,t,t+1} \left(\frac{\pi_{c,t+1}}{(\pi_{c,t}^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c - 1} x_{c,d,t+1} \right] \tag{17}$$
$$\Rightarrow x_{c,d} = y_c + \theta_c Q_c \left(\frac{\pi_c}{(\pi_c^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c - 1} x_{c,d}$$
$$\Rightarrow x_{c,d} = y_c + \theta_c Q_c \pi_c^{(1-\gamma_c)(\varepsilon_c-1)} x_{c,d}$$
$$\Rightarrow \left[1 - \theta_c Q_c \pi_c^{(1-\gamma_c)(\varepsilon_c-1)} \right] x_{c,d} = y_c$$
$$\Rightarrow x_{c,d} = \frac{y_c}{1 - \theta_c Q_c \pi_c^{(1-\gamma_c)(\varepsilon_c-1)}}$$

$$p_{c,t}^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1} \frac{x_{c,n,t}}{x_{c,d,t}} \tag{15}$$

$$
\begin{aligned}
&\Rightarrow p_c^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1}\frac{x_{c,n}}{x_{c,d}} \\
&\Rightarrow p_c^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1}\frac{\frac{y_c mc_c}{1-\theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}}}{\frac{y_c}{1-\theta_c Q_c \pi_c^{(\varepsilon_c-1)(1-\gamma_c)}}} \\
&\Rightarrow p_c^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1}\frac{1-\theta_c Q_c \pi_c^{(\varepsilon_c-1)(1-\gamma_c)}}{1-\theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}} mc_c \\
&\Rightarrow mc_c = \frac{\varepsilon_c - 1}{\varepsilon_c}\frac{1-\theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}}{1-\theta_c Q_c \pi_c^{(\varepsilon_c-1)(1-\gamma_c)}} p_c^{opt}
\end{aligned}
$$

$$
x_{i,n,t} = y_{i,t} mc_{i,t} + \theta_i \mathbb{E}_t \left[Q_{i,t,t+1} \left(\frac{\pi_{i,t+1}}{(\pi_{i,t}^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i} x_{i,n,t+1} \right] \tag{2}
$$

$$
\begin{aligned}
&\Rightarrow x_{i,n} = y_i mc_i + \theta_i Q_i \left(\frac{\pi_i}{(\pi_i^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i} x_{i,n} \\
&\Rightarrow x_{i,n} = y_i mc_i + \theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)} x_{i,n} \\
&\Rightarrow \left[1 - \theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)} \right] x_{i,n} = y_i mc_i \\
&\Rightarrow x_{i,n} = \frac{y_i mc_i}{1-\theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)}}
\end{aligned}
$$

$$
x_{i,d,t} = y_{i,t} + \theta_i \mathbb{E}_t \left[Q_{i,t,t+1} \left(\frac{\pi_{i,t+1}}{(\pi_{i,t}^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i - 1} x_{i,d,t+1} \right] \tag{3}
$$

$$
\begin{aligned}
&\Rightarrow x_{i,d} = y_i + \theta_i Q_i \left(\frac{\pi_i}{(\pi_i^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i - 1} x_{i,d} \\
&\Rightarrow x_{i,d} = y_i + \theta_i Q_i \pi_i^{(\varepsilon_i-1)(1-\gamma_i)} x_{i,d} \\
&\Rightarrow \left[1 - \theta_i Q_i \pi_i^{(\varepsilon_i-1)(1-\gamma_i)} \right] x_{i,d} = y_i \\
&\Rightarrow x_{i,d} = \frac{y_i}{1-\theta_i Q_i \pi_i^{(\varepsilon_i-1)(1-\gamma_i)}}
\end{aligned}
$$

$$
\begin{aligned}
p_{i,t}^{opt} &= \frac{\varepsilon_i}{\varepsilon_i - 1}\frac{x_{i,n,t}}{x_{i,d,t}} \qquad (1) \\
&\Rightarrow p_i^{opt} = \frac{\varepsilon_i}{\varepsilon_i - 1}\frac{x_{i,n}}{x_{i,d}} \\
&\Rightarrow p_i^{opt} = \frac{\varepsilon_i}{\varepsilon_i - 1}\frac{\frac{y_i mc_i}{1-\theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)}}}{\frac{y_i}{1-\theta_i Q_i \pi_i^{(\varepsilon_i-1)(1-\gamma_i)}}} \\
&\Rightarrow p_i^{opt} = \frac{\varepsilon_i}{\varepsilon_i - 1}\frac{1-\theta_i Q_i \pi_i^{(\varepsilon_i-1)(1-\gamma_i)}}{1-\theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)}} mc_i
\end{aligned}
$$

$$\Rightarrow mc_i = \frac{\varepsilon_i - 1}{\varepsilon_i} \frac{1 - \theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)}}{1 - \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)}} p_i^{opt}$$

$$n_t^{eff} = \frac{n_t}{\Delta_{w,t}} \tag{25}$$
$$\Rightarrow n^{eff} = \frac{n}{\Delta_w}$$

$$r_{c,t}^k = mc_{c,t} \alpha_c \frac{y_{c,t} \Delta_{c,t}}{k_{c,t}} \tag{29}$$
$$\Rightarrow r_c^k = mc_c \alpha_c \frac{y_c \Delta_c}{k_c}$$
$$\Rightarrow k_c = mc_c \alpha_c \frac{y_c \Delta_c}{r_c^k}$$

$$w_t = mc_{c,t}(1 - \alpha_c) \frac{y_{c,t} \Delta_{c,t}}{n_{c,t}^{eff}} \tag{30}$$
$$\Rightarrow w = mc_c(1 - \alpha_c) \frac{y_c \Delta_c}{n_c^{eff}}$$
$$\Rightarrow n_c^{eff} = mc_c(1 - \alpha_c) \frac{y_c \Delta_c}{w}$$

$$y_{c,t} = \frac{z_{c,t}(k_{c,t})^{\alpha_c}(n_{c,t}^{eff})^{1-\alpha_c}}{\Delta_{c,t}} \tag{21}$$
$$\Rightarrow y_c = \frac{z_c(k_c)^{\alpha_c}(n_c^{eff})^{1-\alpha_c}}{\Delta_c}$$
$$\Rightarrow \Delta_c y_c = z_c \left(mc_c \alpha_c \frac{y_c \Delta_c}{r_c^k} \right)^{\alpha_c} \left(mc_c(1 - \alpha_c) \frac{y_c \Delta_c}{w} \right)^{1-\alpha_c}$$
$$\Rightarrow 1 = mc_c z_c \left(\frac{\alpha_c}{r_c^k} \right)^{\alpha_c} \left(\frac{1 - \alpha_c}{w} \right)^{1-\alpha_c}$$
$$\Rightarrow mc_c = \frac{1}{z_c} \left(\frac{\alpha_c}{r_c^k} \right)^{-\alpha_c} \left(\frac{1 - \alpha_c}{w} \right)^{-(1-\alpha_c)}$$
$$\Rightarrow mc_c = \frac{1}{z_c} \frac{(r_c^k)^{\alpha_c} w^{1-\alpha_c}}{\alpha_c^{\alpha_c}(1 - \alpha_c)^{1-\alpha_c}}$$
$$\Rightarrow w^{1-\alpha_c} = mc_c z_c \alpha_c^{\alpha_c}(1 - \alpha_c)^{1-\alpha_c}(r_c^k)^{-\alpha_c}$$
$$\Rightarrow w = \left[mc_c z_c \alpha_c^{\alpha_c}(1 - \alpha_c)^{1-\alpha_c} \right]^{\frac{1}{1-\alpha_c}} (r_c^k)^{-\frac{\alpha_c}{1-\alpha_c}} \tag{*}$$

$$r_{i,t}^k = \alpha_i mc_{i,t} q_t \frac{y_{i,t} \Delta_{i,t}}{k_{i,t}} \tag{31}$$

$$\Rightarrow r_i^k = \alpha_i mc_i q \frac{y_i \Delta_i}{k_i}$$
$$\Rightarrow k_i = \alpha_i mc_i q \frac{y_i \Delta_i}{r_i^k}$$

$$w_t = (1-\alpha_i) mc_{i,t} q_t \frac{y_{i,t} \Delta_{i,t}}{n_{i,t}^{eff}} \tag{32}$$
$$\Rightarrow w = (1-\alpha_i) mc_i q \frac{y_i \Delta_i}{n_i^{eff}}$$
$$\Rightarrow n_i^{eff} = (1-\alpha_i) mc_i q \frac{y_i \Delta_i}{w}$$

$$y_{i,t} = \frac{z_{i,t}(k_{i,t})^{\alpha_i}(n_{i,t}^{eff})^{1-\alpha_i}}{\Delta_{i,t}} \tag{23}$$
$$\Rightarrow y_i = \frac{z_i(k_i)^{\alpha_i}(n_i^{eff})^{1-\alpha_i}}{\Delta_i}$$
$$\Rightarrow \Delta_i y_i = z_i \left(\alpha_i mc_i q \frac{y_i \Delta_i}{r_i^k}\right)^{\alpha_i} \left((1-\alpha_i) mc_i q \frac{y_i \Delta_i}{w}\right)^{1-\alpha_i}$$
$$\Rightarrow 1 = mc_i q z_i \left(\frac{\alpha_i}{r_i^k}\right)^{\alpha_i} \left(\frac{1-\alpha_i}{w}\right)^{1-\alpha_i}$$
$$\Rightarrow mc_i = \frac{1}{z_i}\frac{1}{q}\left(\frac{\alpha_i}{r_i^k}\right)^{-\alpha_i} \left(\frac{1-\alpha_i}{w}\right)^{-(1-\alpha_i)}$$
$$\Rightarrow mc_i = \frac{1}{z_i}\frac{1}{q}\frac{(r_i^k)^{\alpha_i} w^{1-\alpha_i}}{\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}}$$
$$\Rightarrow w^{1-\alpha_i} = mc_i z_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}(r_i^k)^{-\alpha_i}$$
$$\Rightarrow w = \left[mc_i z_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} (r_i^k)^{-\frac{\alpha_i}{1-\alpha_i}} \tag{**}$$

$$\lambda_t q_t = \xi_{c,t}\left\{\left[1 - \frac{\psi_c}{2}\left(\frac{i_{c,t}}{i_{c,t-1}} - 1\right)^2\right]\mu_{c,t} - \psi_c\left(\frac{i_{c,t}}{i_{c,t-1}} - 1\right)\frac{1}{i_{c,t-1}}\mu_{c,t} i_{c,t}\right\}$$
$$+ \beta \mathbb{E}_t\left\{\xi_{c,t+1}\left[\psi_c\left(\frac{i_{c,t+1}}{i_{c,t}} - 1\right)\left(\frac{i_{c,t+1}}{i_{c,t}}\right)^2 \mu_{c,t+1}\right]\right\} \tag{8}$$
$$\Rightarrow \lambda q = \xi_c\left\{\left[1 - \frac{\psi_c}{2}\left(\frac{i_c}{i_c} - 1\right)^2\right]\mu_c - \psi_c\left(\frac{i_c}{i_c} - 1\right)\frac{1}{i_c}\mu_c i_c\right\}$$
$$+ \beta\xi_c\left[\psi_c\left(\frac{i_c}{i_c} - 1\right)\left(\frac{i_c}{i_c}\right)^2 \mu_c\right]$$
$$\Rightarrow \lambda q = \xi_c \mu_c$$
$$\Rightarrow \xi_c = \frac{\lambda}{\mu_c} q$$

$$\lambda_t q_t = \xi_{i,t}\left\{\left[1-\frac{\psi_i}{2}\left(\frac{i_{i,t}}{i_{i,t-1}}-1\right)^2\right]\mu_{i,t}-\psi_i\left(\frac{i_{i,t}}{i_{i,t-1}}-1\right)\frac{1}{i_{i,t-1}}\mu_{i,t}i_{i,t}\right\}$$
$$+\beta\mathbb{E}_t\left\{\xi_{i,t+1}\left[\psi_i\left(\frac{i_{i,t+1}}{i_{i,t}}-1\right)\left(\frac{i_{i,t+1}}{i_{i,t}}\right)^2\mu_{i,t+1}\right]\right\} \tag{9}$$
$$\Rightarrow\lambda q = \xi_i\left\{\left[1-\frac{\psi_i}{2}\left(\frac{i_i}{i_i}-1\right)^2\right]\mu_i-\psi_i\left(\frac{i_i}{i_i}-1\right)\frac{1}{i_i}\mu_i i_i\right\}$$
$$+\beta\xi_i\left[\psi_i\left(\frac{i_i}{i_i}-1\right)\left(\frac{i_i}{i_i}\right)^2\mu_i\right]$$
$$\Rightarrow\lambda q = \xi_i\mu_i$$
$$\Rightarrow\xi_i = \frac{\lambda}{\mu_i}q$$

정상상태에서 $\mu_c = \mu_i$ 이면, $\xi_c = \xi_i$ 가 된다.

$$\xi_{c,t} = \beta\mathbb{E}_t\left\{\lambda_{t+1}r^k_{c,t+1}+\xi_{c,t+1}(1-\delta_c)\right\} \tag{10}$$
$$\Rightarrow\xi_c = \beta\left(\lambda r^k_c+\xi_c(1-\delta_c)\right)$$
$$\Rightarrow\xi_c = \beta\lambda r^k_c+\beta\xi_c(1-\delta_c)$$
$$\Rightarrow\frac{\lambda}{\mu_c}q = \beta\lambda r^k_c+\beta\frac{\lambda}{\mu_c}q(1-\delta_c)$$
$$\Rightarrow\frac{1}{\mu_c}q = \beta r^k_c+\beta\frac{1}{\mu_c}q(1-\delta_c)$$
$$\Rightarrow\beta r^k_c = \frac{1}{\mu_c}q-\beta\frac{1}{\mu_c}q(1-\delta_c)$$
$$\Rightarrow r^k_c = \frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]q \tag{***}$$

$$\xi_{i,t} = \beta\mathbb{E}_t\left\{\lambda_{t+1}r^k_{i,t+1}+\xi_{i,t+1}(1-\delta_i)\right\} \tag{11}$$
$$\Rightarrow\xi_i = \beta\left(\lambda r^k_i+\xi_i(1-\delta_i)\right)$$
$$\Rightarrow\xi_i = \beta\lambda r^k_i+\beta\xi_i(1-\delta_i)$$
$$\Rightarrow\frac{\lambda}{\mu_i}q = \beta\lambda r^k_i+\beta\frac{\lambda}{\mu_i}q(1-\delta_i)$$
$$\Rightarrow\frac{1}{\mu_i}q = \beta r^k_i+\beta\frac{1}{\mu_i}q(1-\delta_i)$$
$$\Rightarrow\beta r^k_i = \frac{1}{\mu_i}q-\beta\frac{1}{\mu_i}q(1-\delta_i)$$
$$\Rightarrow r^k_i = \frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]q \tag{***}$$

정상상태에서 $\mu_c = \mu_i$ 이고 $\delta_c = \delta_i$ 이면, $r_c^k = r_i^k$ 가 된다.

식 (*), (**), (***)로부터 q의 정상상태 값을 도출할 수 있다.

$$\left[mc_i z_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} (r_i^k)^{-\frac{\alpha_i}{1-\alpha_i}} = \left[mc_c z_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} (r_c^k)^{-\frac{\alpha_c}{1-\alpha_c}}$$

$$\Rightarrow \left[mc_i z_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right] q\right)^{-\frac{\alpha_i}{1-\alpha_i}}$$

$$= \left[mc_c z_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right] q\right)^{-\frac{\alpha_c}{1-\alpha_c}}$$

$$\Rightarrow q^{\frac{1-\alpha_i}{1-\alpha_i}} q^{\frac{\alpha_c}{1-\alpha_c}} \left[mc_i z_i \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}$$

$$= \left[mc_c z_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}$$

$$\Rightarrow q^{\frac{1}{1-\alpha_c}} = \frac{\left[mc_c z_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}}{\left[mc_i z_i \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}}$$

$$\Rightarrow q = \left\{\frac{\left[mc_c z_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}}{\left[mc_i z_i \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}}\right\}^{1-\alpha_c}$$

만일 $\alpha_c = \alpha_i = \alpha$ 라면, q는 다음과 같다.

$$q = \left\{\frac{\left[mc_c z_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}}{\left[mc_i z_i \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}}\right\}^{1-\alpha_c}$$

$$\Rightarrow q = \left\{\frac{\left[mc_c z_c \alpha^{\alpha}(1-\alpha)^{1-\alpha}\right]^{\frac{1}{1-\alpha}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\frac{\alpha}{1-\alpha}}}{\left[mc_i z_i \alpha^{\alpha}(1-\alpha)^{1-\alpha}\right]^{\frac{1}{1-\alpha}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\frac{\alpha}{1-\alpha}}}\right\}^{1-\alpha}$$

$$\Rightarrow q = \frac{\left[mc_c z_c \alpha^{\alpha}(1-\alpha)^{1-\alpha}\right] \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\alpha}}{\left[mc_i z_i \alpha^{\alpha}(1-\alpha)^{1-\alpha}\right] \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\alpha}}$$

$$\Rightarrow q = \frac{\left[mc_c z_c\right] \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta}\right]\right)^{-\alpha}}{\left[mc_i z_i\right] \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta}\right]\right)^{-\alpha}}$$

$$\Rightarrow q = \frac{mc_c z_c}{mc_i z_i}\left(\frac{\mu_i}{\mu_c}\frac{1-\beta(1-\delta_c)}{1-\beta(1-\delta_i)}\right)^{-\alpha}$$

만일 소비재 부문과 투자재 부문이 완전 대칭이라면(즉 $\alpha_c = \alpha_i = \alpha$, $mc_c = mc_i$, $z_c = z_i$, $\mu_c = \mu_i$, $\delta_c = \delta_i$), $q = 1$ 이 된다.

식 (29), (30)로부터 다음이 성립한다.

$$\begin{aligned}
\frac{w_t}{r^k_{c,t}} &= \frac{mc_{c,t}(1-\alpha_c)\frac{y_{c,t}\Delta_{c,t}}{n^{eff}_{c,t}}}{mc_{c,t}\alpha_c\frac{y_{c,t}\Delta_{c,t}}{k_{c,t}}} \\
\Rightarrow \frac{w_t}{r^k_{c,t}} &= \frac{1-\alpha_c}{\alpha_c}\frac{k_{c,t}}{n^{eff}_{c,t}} \\
\Rightarrow \frac{k_{c,t}}{n^{eff}_{c,t}} &= \frac{\alpha_c}{1-\alpha_c}\frac{w_t}{r^k_{c,t}} \\
\Rightarrow \frac{k_c}{n^{eff}_c} &= \underbrace{\frac{\alpha_c}{1-\alpha_c}\frac{w}{r^k_c}}_{\equiv\kappa_c} \\
\Rightarrow \frac{k_c}{n^{eff}_c} &= \kappa_c \\
\Rightarrow k_c &= \kappa_c n^{eff}_c
\end{aligned}$$

식 (31), (32)로부터 다음이 성립한다.

$$\begin{aligned}
\frac{w_t}{r^k_{i,t}} &= \frac{mc_{i,t}(1-\alpha_i)q_t\frac{y_{i,t}\Delta_{i,t}}{n^{eff}_{i,t}}}{mc_{i,t}\alpha_i q_t\frac{y_{i,t}\Delta_{i,t}}{k_{i,t}}} \\
\Rightarrow \frac{w_t}{r^k_{i,t}} &= \frac{1-\alpha_i}{\alpha_i}\frac{k_{i,t}}{n^{eff}_{i,t}} \\
\Rightarrow \frac{k_{i,t}}{n^{eff}_{i,t}} &= \frac{\alpha_i}{1-\alpha_i}\frac{w_t}{r^k_{i,t}} \\
\Rightarrow \frac{k_i}{n^{eff}_i} &= \underbrace{\frac{\alpha_i}{1-\alpha_i}\frac{w}{r^k_i}}_{\equiv\kappa_i} \\
\Rightarrow \frac{k_i}{n^{eff}_i} &= \kappa_i \\
\Rightarrow k_i &= \kappa_i n^{eff}_i
\end{aligned}$$

식 (38), (40)으로부터 다음을 알 수 있다.

$$\begin{aligned}
n^{eff}_t &= n^{eff}_{c,t} + n^{eff}_{i,t} \\
\Rightarrow n^{eff} &= n^{eff}_c + n^{eff}_i \\
\Rightarrow n^{eff}_c &= n^{eff} - n^{eff}_i
\end{aligned} \tag{40}$$

$$\begin{aligned}
k_t &= k_{c,t} + k_{i,t} \\
\Rightarrow k &= k_c + k_i \\
\Rightarrow k &= \kappa_c n^{eff}_c + \kappa_i n^{eff}_i
\end{aligned} \tag{38}$$

$$
\begin{aligned}
&\Rightarrow k = \kappa_c \underbrace{\left(n^{eff} - n_i^{eff}\right)}_{=n_c^{eff}} + \kappa_i n_i^{eff} \\
&\Rightarrow k = \kappa_c n^{eff} - \kappa_c n_i^{eff} + \kappa_i n_i^{eff} \\
&\Rightarrow k = \kappa_c n^{eff} + (\kappa_i - \kappa_c) n_i^{eff}
\end{aligned}
$$

이제 식 (4), (5), (23), (41), (42)로 부터 n_i^{eff} 를 도출할 수 있다.

$$
\begin{aligned}
y_{c,t} &= c_t \\
\Rightarrow y_c &= c
\end{aligned} \tag{42}
$$

$$
\begin{aligned}
c_t + q_t i_t &= y_{c,t} + q_t y_{i,t} \\
\Rightarrow c + qi &= y_c + q y_i \\
\Rightarrow y_i &= i
\end{aligned} \tag{41}
$$

$$
\begin{aligned}
y_{i,t} &= \frac{z_{i,t}(k_{i,t})^{\alpha_i}(n_{i,t}^{eff})^{1-\alpha_i}}{\Delta_{i,t}} \\
\Rightarrow y_i &= \frac{z_i(k_i)^{\alpha_i}(n_i^{eff})^{1-\alpha_i}}{\Delta_i} \\
\Rightarrow y_i &= \frac{1}{\Delta_i} z_i \left(\frac{k_i}{n_i^{eff}}\right)^{\alpha_i} n_i^{eff} \\
\Rightarrow y_i &= \frac{1}{\Delta_i} z_i \kappa_i^{\alpha_i} n_i^{eff}
\end{aligned} \tag{23}
$$

$$
\begin{aligned}
k_{c,t+1} &= (1-\delta_c)k_{c,t} + \left[1 - \frac{\psi_c}{2}\left(\frac{i_{c,t}}{i_{c,t-1}} - 1\right)^2\right] \mu_{c,t} i_{c,t} \\
\Rightarrow k_c &= (1-\delta_c)k_c + \left[1 - \frac{\psi_c}{2}\left(\frac{i_c}{i_c} - 1\right)^2\right] \mu_c i_c \\
\Rightarrow k_c &= (1-\delta_c)k_c + \mu_c i_c \\
\Rightarrow \delta_c k_c &= \mu_c i_c \\
\Rightarrow i_c &= \frac{\delta_c}{\mu_c} k_c
\end{aligned} \tag{4}
$$

$$
k_{i,t+1} = (1-\delta_i)k_{i,t} + \left[1 - \frac{\psi_i}{2}\left(\frac{i_{i,t}}{i_{i,t-1}} - 1\right)^2\right] \mu_{i,t} i_{i,t} \tag{5}
$$

$$
\begin{aligned}
&\Rightarrow k_i = (1-\delta_i)k_i + \left[1 - \frac{\psi_i}{2}\left(\frac{i_i}{i_i} - 1\right)^2\right]\mu_i i_i \\
&\Rightarrow k_i = (1-\delta_i)k_i + \mu_i i_i \\
&\Rightarrow \delta_i k_i = \mu_i i_i \\
&\Rightarrow i_i = \frac{\delta_i}{\mu_i} k_i
\end{aligned}
$$

$$
\begin{aligned}
&\underbrace{i_c + i_i}_{=i} = \frac{\delta_c}{\mu_c}k_c + \frac{\delta_i}{\mu_i}k_i \\
&\Rightarrow \underbrace{i}_{=y_i} = \frac{\delta_c}{\mu_c}\kappa_c n_c^{eff} + \frac{\delta_i}{\mu_i}\kappa_i n_i^{eff} \\
&\Rightarrow y_i = \frac{\delta_c}{\mu_c}\kappa_c \underbrace{(n^{eff} - n_i^{eff})}_{=n_c^{eff}} + \frac{\delta_i}{\mu_i}\kappa_i n_i^{eff} \\
&\Rightarrow \underbrace{\frac{1}{\Delta_i} z_i \kappa_i^{\alpha_i} n_i^{eff}}_{=y_i} = \frac{\delta_c}{\mu_c}\kappa_c n^{eff} - \frac{\delta_c}{\mu_c}\kappa_c n_i^{eff} + \frac{\delta_i}{\mu_i}\kappa_i n_i^{eff} \\
&\Rightarrow \frac{1}{\Delta_i} z_i \kappa_i^{\alpha_i} n_i^{eff} + \frac{\delta_c}{\mu_c}\kappa_c n_i^{eff} - \frac{\delta_i}{\mu_i}\kappa_i n_i^{eff} = \frac{\delta_c}{\mu_c}\kappa_c n^{eff} \\
&\Rightarrow \left[\frac{1}{\Delta_i} z_i \kappa_i^{\alpha_i} + \frac{\delta_c}{\mu_c}\kappa_c - \frac{\delta_i}{\mu_i}\kappa_i\right] n_i^{eff} = \frac{\delta_c}{\mu_c}\kappa_c n^{eff} \\
&\Rightarrow n_i^{eff} = \frac{\frac{\delta_c}{\mu_c}\kappa_c}{\frac{1}{\Delta_i} z_i \kappa_i^{\alpha_i} + \frac{\delta_c}{\mu_c}\kappa_c - \frac{\delta_i}{\mu_i}\kappa_i} n^{eff}
\end{aligned}
$$

n_i^{eff} 가 주어지면 앞에서 설명했듯이 다음을 구할 수 있다.

$$
\begin{aligned}
n_c^{eff} &= n^{eff} - n_i^{eff} \\
k_i &= \kappa_i n_i^{eff} \\
k_c &= \kappa_c n_c^{eff} \\
k &= k_c + k_i \\
y_i &= \frac{1}{\Delta_i} z_i \kappa_i^{\alpha_i} n_i^{eff}
\end{aligned}
$$

$$
\begin{aligned}
y_{c,t} &= \frac{z_{c,t}(k_{c,t})^{\alpha_c}(n_{c,t}^{eff})^{1-\alpha_c}}{\Delta_{c,t}} \\
\Rightarrow y_c &= \frac{z_c(k_c)^{\alpha_c}(n_c^{eff})^{1-\alpha_c}}{\Delta_c}
\end{aligned} \tag{21}
$$

$$
y_{c,t} = c_t \tag{42}
$$

$$\Rightarrow y_c = c$$
$$\Rightarrow c = y_c$$

$$\begin{aligned} \lambda_t &= (c_t - \hbar c_{t-1})^{-\sigma} - \beta\hbar\mathbb{E}_t\left[(c_{t+1} - \hbar c_t)^{-\sigma}\right] \\ \Rightarrow\lambda &= (c - \hbar c)^{-\sigma} - \beta\hbar(c - \hbar c)^{-\sigma} \\ \Rightarrow\lambda &= c^{-\sigma}(1 - \hbar)^{-\sigma} - \beta\hbar c^{-\sigma}(1 - \hbar)^{-\sigma} \\ \Rightarrow\lambda &= c^{-\sigma}(1 - \hbar)^{-\sigma}(1 - \beta\hbar) \end{aligned} \tag{6}$$

$$\begin{aligned} y_t &\equiv y_{c,t} + q_t y_{i,t} \\ \Rightarrow y &= y_c + q y_i \end{aligned} \tag{37}$$

$$\begin{aligned} d_{c,t} &= y_{c,t}\left(1 - mc_{c,t}\Delta_{c,t}\right) \\ \Rightarrow d_c &= y_c\left(1 - mc_c\Delta_c\right) \end{aligned} \tag{27}$$

$$\begin{aligned} d_{i,t} &= y_{i,t}\left(1 - mc_{i,t}\Delta_{i,t}\right) \\ \Rightarrow d_i &= y_i\left(1 - mc_i\Delta_i\right) \end{aligned} \tag{28}$$

$$x_{w,d,t} = \lambda_t w_t n_t^{eff} + \beta\theta_w\mathbb{E}_t\left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w - 1} x_{w,d,t+1}\right] \tag{14}$$

$$\begin{aligned} \Rightarrow x_{w,d} &= \lambda w n^{eff} + \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w}}\right)^{\varepsilon_w - 1} x_{w,d} \\ \Rightarrow x_{w,d} &= \lambda w n^{eff} + \beta\theta_w\pi_w^{(1-\gamma_w)(\varepsilon_w - 1)} x_{w,d} \\ \Rightarrow &\left[1 - \beta\theta_w\pi_w^{(1-\gamma_w)(\varepsilon_w - 1)}\right] x_{w,d} = \lambda w n^{eff} \\ \Rightarrow x_{w,d} &= \frac{\lambda w n^{eff}}{1 - \beta\theta_w\pi_w^{(1-\gamma_w)(\varepsilon_w - 1)}} \end{aligned}$$

$$\begin{aligned} (\tilde{w}_t^{opt})^{1+\varepsilon_w\phi} &= \frac{\varepsilon_w}{\varepsilon_w - 1}\frac{x_{w,n,t}}{x_{w,d,t}} \\ \Rightarrow(\tilde{w}^{opt})^{1+\varepsilon_w\phi} &= \frac{\varepsilon_w}{\varepsilon_w - 1}\frac{x_{w,n}}{x_{w,d}} \\ \Rightarrow x_{w,n} &= \frac{\varepsilon_w - 1}{\varepsilon_w} x_{w,d}(\tilde{w}^{opt})^{1+\varepsilon_w\phi} \end{aligned} \tag{12}$$

마지막으로 $n = 1/3$이 되도록 하는 χ는 다음과 같다.

$$x_{w,n,t} = \chi \left(n_t^{eff} \right)^{1+\phi} + \beta \theta_w \mathbb{E}_t \left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}} \right)^{\varepsilon_w(1+\phi)} x_{w,n,t+1} \right] \tag{13}$$

$$\Rightarrow x_{w,n} = \chi \left(n^{eff} \right)^{1+\phi} + \beta \theta_w \left(\frac{\pi_w}{(\pi_w^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}} \right)^{\varepsilon_w(1+\phi)} x_{w,n}$$

$$\Rightarrow \chi \left(n^{eff} \right)^{1+\phi} = \left[1 - \beta \theta_w \pi_w^{(1-\gamma_w)\varepsilon_w(1+\phi)} \right] x_{w,n}$$

$$\Rightarrow \chi = \left[1 - \beta \theta_w \pi_w^{(1-\gamma_w)\varepsilon_w(1+\phi)} \right] \frac{x_{w,n}}{\left(n^{eff} \right)^{1+\phi}}$$

식 (35), (36), (43), (44), (45), (46), (47)는 정상상태 값 도출에 사용하지 않았다. 이는 다음 제약을 사용한데 기인한다.

- $\pi_c = \pi_w = \pi_i = (1 + 0.02)^{\frac{1}{4}}$
- $z_c = z_i = \mu_c = \mu_i = 1$
- $n = 1/3$

7 제16장 추세와 균형성장경로

제16장의 세 모형(추세 RBC, 2부문 2추세, 1부문 GHK)의 정상상태를 단계별로 유도한다. 모든 정상상태는 닫힌 형태이며, 정상화 변수의 물결표는 본문 표기를 따르되 아래에서는 생략한다. 캘리브레이션은 본문과 같다: $\beta = 0.99$, $\sigma = 2$, $\phi = 1$, $\alpha = 0.33$, $\delta = 0.025$, 노동 목표 $n = 1/3$(χ 역산).

7.1 추세 RBC 모형

균형조건은 본문 §16.3의 9개이며, 정상상태에서 $z = 1$, $g = \bar{g} = 1.005$이다.

1단계: 임대료율. 오일러 방정식(본문 식 (6))의 정상상태에서

$$1 = \beta\,\bar{g}^{-\sigma}(r^k + 1 - \delta) \;\Rightarrow\; r^k = \frac{\bar{g}^{\sigma}}{\beta} - (1-\delta) = \frac{1.005^2}{0.99} - 0.975 = 0.045227$$

2단계: 자본-노동 비율. 자본수요 $r^k_t = \alpha \tilde{y}_t g_t / \tilde{k}_{t-1}$의 정상상태에 생산함수(식 (4))를 대입한다. 유효자본 $\hat{k} \equiv k/\bar{g}$(생산에 투입되는 추세 대비 자본)와 그 노동 비율 $\kappa \equiv \hat{k}/n$으로 쓰면 $y = \kappa^{\alpha} n$이고

$$r^k = \alpha\,\frac{y}{\hat{k}} = \alpha\,\kappa^{\alpha-1} \;\Rightarrow\; \kappa = \left(\frac{r^k}{\alpha}\right)^{\frac{1}{\alpha-1}} = \left(\frac{0.045227}{0.33}\right)^{-1/0.67} = 19.4191$$

3단계: 수량 캐스케이드. $n = 1/3$ 목표에서

$$k = \bar{g}\,\kappa n = 1.005 \times 19.4191 \times 0.33333 = 6.50540\,y \quad = \kappa^{\alpha} n = 19.4191^{0.33} \times 0.33333 = 0.88714\,i = k\left[1 - \frac{1-\delta}{\bar{g}}\right] =$$

유효 감가율 $1 - (1-\delta)/\bar{g} = 0.029851$은 물리적 감가($\delta = 0.025$)에 추세 성장 몫($\approx \bar{g} - 1$)이 더해진 값이다.

4단계: χ 역산. 노동공급(식 (8))에서

$$\chi = \frac{c^{-\sigma} w}{n^{\phi}} = \frac{0.69295^{-2} \times 1.78316}{0.33333} = 11.1405$$

$\bar{g} \to 1$ 극한에서 위 모든 값이 제3장 무추세 RBC의 정상상태로 되돌아감은 본문 §16.3에서 기계정밀도로 검증하였다.

7.2 2부문 2추세 모형

균형조건은 본문 §16.4의 22개이며, 정상상태에서 $z_c = z_i = 1$, $g_{a_c} = \bar{g}_{a_c} = 1.003$, $g_{a_i} = \bar{g}_{a_i} = 1.008$, $g_x = \bar{g}_{a_c}^{1-\alpha} \bar{g}_{a_i}^{\alpha} = 1.004647$이다.

1단계: 상대가격. 경쟁 균형에서 각 부문의 가격은 단위비용과 같다. 콥-더글라스 기술의 단위비용은 $\mathrm{uc}_j = (1/z_j)(r^k/\alpha)^{\alpha}\{w/(1-\alpha)\}^{1-\alpha}$ (정상화 변수 기준; 요소가격은 부문 간 균등화되어 공통)이므로

$$1 = \frac{1}{\bar{z}_c}\left(\frac{r^k}{\alpha}\right)^{\alpha}\left(\frac{w}{1-\alpha}\right)^{1-\alpha}, \qquad q = \frac{1}{\bar{z}_i}\left(\frac{r^k}{\alpha}\right)^{\alpha}\left(\frac{w}{1-\alpha}\right)^{1-\alpha}$$

둘째 식을 첫째 식으로 나누면 요소가격 항이 전부 소거되고

$$q = \frac{\bar{z}_c}{\bar{z}_i} = 1$$

— 본문 식 (36)이다. α가 부문 간에 같다는 가정이 이 소거의 전부다.

2단계: 임대료율. 자본 오일러 방정식(본문 식 (30))의 정상상태에서 $B \equiv \beta\,\bar{g}_x^{1-\sigma}/\bar{g}_{a_i}$로 놓으면

$$q = B\left(r^k + (1-\delta)q\right) \;\Rightarrow\; r^k = q\left(\frac{1}{B} - (1-\delta)\right) = q\left(\frac{\bar{g}_{a_i}\,\bar{g}_x^{\sigma-1}}{\beta} - (1-\delta)\right) = 0.047914$$

3단계: 공통 자본-노동 비율. 소비재 부문 자본수요와 생산함수에서 (배분된 유효자본의 노동 비율 κ)

$$r^k = \alpha\,\bar{z}_c\,\kappa^{\alpha-1} \;\Rightarrow\; \kappa = \left(\frac{r^k}{\alpha\,\bar{z}_c}\right)^{\frac{1}{\alpha-1}} = 17.8168$$

투자재 부문 조건 $r^k = q\,\alpha\,\bar{z}_i\,\kappa_i^{\alpha-1}$은 $q = \bar{z}_c/\bar{z}_i$ 덕분에 같은 κ를 주므로 1단계와 정합적이다.

4단계: 노동 배분. 자본 축적식의 정상상태 $k\,(1-(1-\delta)/\bar{g}_{a_i}) = y_i$에 유효자본 $k/\bar{g}_{a_i} = \kappa n$과 투자재 생산함수 $y_i = \bar{z}_i\kappa^\alpha n_i$를 대입하면

$$\bar{g}_{a_i}\,\kappa n\left(1 - \frac{1-\delta}{\bar{g}_{a_i}}\right) = \bar{z}_i\,\kappa^\alpha n_i \;\Rightarrow\; n_i = \frac{\kappa^{1-\alpha}\,(\bar{g}_{a_i} - 1 + \delta)}{\bar{z}_i}\,n$$

수치로는 $n_i = 17.8168^{0.67} \times 0.033 \times 0.33333 = 0.075761$, $n_c = n - n_i = 0.257572$ — 노동의 22.7%가 투자재 부문에 고용된다. 괄호 항 $\bar{g}_{a_i} - 1 + \delta = 0.033$이 유효 투자 소요(성장 몫 + 감가상각)이다.

5단계: 수량 캐스케이드와 χ.

$$k_c = \kappa\,n_c = 4.58910, \qquad k_i = \kappa\,n_i = 1.34982, \qquad k = \bar{g}_{a_i}(k_c + k_i) = 5.98644\,y_c \quad = \bar{z}_c\,k_c^\alpha n_c^{1-\alpha} = 0.66630 = c,$$

균형성장경로의 함의를 수치로 확인하면: 총산출은 분기 $\ln \bar{g}_x = 0.464\%$(연 1.87%), 실질 투자는 $\ln \bar{g}_{a_i} = 0.797\%$(연 3.24%)로 자라고, 상대가격은 $(1-\alpha)(\ln \bar{g}_{a_c} - \ln \bar{g}_{a_i}) = -0.333\%$(연 -1.33%)로 하락한다.

7.3 1부문 GHK 표현과 동일성 확인

GHK 표현의 정상상태는 임대료율 앵커만 다르다. 오일러 방정식의 정상상태에서

$$1 = \beta\,\frac{\bar{g}_x^{-\sigma}}{\bar{g}_v}\,(r_v^k + 1 - \delta) \;\Rightarrow\; r_v^k = \frac{\bar{g}_v\,\bar{g}_x^\sigma}{\beta} - (1-\delta)$$

이것이 2부문 표현의 r^k(2단계, $q=1$)와 같음을 지수 대수로 확인한다. $\bar{g}_v = (\bar{g}_{a_i}/\bar{g}_{a_c})^{1-\alpha}$와 $\bar{g}_x = \bar{g}_{a_c}^{1-\alpha}\bar{g}_{a_i}^{\alpha}$를 대입하면

$$\bar{g}_v\,\bar{g}_x^\sigma = \bar{g}_{a_c}^{-(1-\alpha)}\bar{g}_{a_i}^{1-\alpha} \cdot \bar{g}_{a_c}^{(1-\alpha)\sigma}\bar{g}_{a_i}^{\alpha\sigma} = \bar{g}_{a_c}^{(1-\alpha)(\sigma-1)}\,\bar{g}_{a_i}^{1+\alpha(\sigma-1)}$$

한편 2부문 쪽 계수는

$$\bar{g}_{a_i}\,\bar{g}_x^{\sigma-1} = \bar{g}_{a_i} \cdot \bar{g}_{a_c}^{(1-\alpha)(\sigma-1)}\bar{g}_{a_i}^{\alpha(\sigma-1)} = \bar{g}_{a_c}^{(1-\alpha)(\sigma-1)}\,\bar{g}_{a_i}^{1+\alpha(\sigma-1)}$$

로 정확히 같다. 따라서 $r_v^k = r^k$이고, 이하의 캐스케이드(κ와 모든 수량)는 2부문 표현과 동일한 값을 준다 — 본문 §16.5의 기계 정밀도 IRF 일치가 정상상태 수준에서도 성립하는 이유다.

8 제17장 재정정책과 제약가계

정상상태에서 소비재, 투자재, 임금의 인플레이션율 π_c, π_i, π_w와 노동시간 n은 외생적으로 주어진다고 가정한다. 세 가지 인플레이션율은 중앙은행이 설정하는 물가안정목표에 의해 결정되며, 분기모형에서는 연율 2%가 되도록 $(1+0.02)^{\frac{1}{4}}$로 설정하는 것이 일반적이다. 노동시간 n은 하루의 1/3(24시간 중 8시간)을 노동에 사용한다는 해석에 따라, 모형의 파라미터 χ를 조정하여 $n=1/3$이 되도록 한다.

소비재와 투자재의 기술수준 $z_{c,t}$, $z_{i,t}$, 그리고 투자특화기술 $\mu_{c,t}$, $\mu_{i,t}$, μ_g는 모두 정상상태에서 1로 정규화한다.

세율의 정상상태 값 τ_c, τ_n, τ_k, τ_r, τ_d 역시 외생적으로 주어진다.

또한 정부의 지출 항목은 GDP 대비 비중을 기준으로 규정하며, 정부소비 s_{g_c}, 정부투자 $s_{\tilde{g}_i}$, 이전지출 s_{tr}은 모두 정상상태에서 주어진 상수로 가정한다. 또한 정액세 τ^L과 이전지출과 정액세의 제약가계몫 α^T, α^L도 주어져있다.

우선 계산의 편의를 위해 $\tilde{z}_c$와 $\tilde{z}_i$는 주어진 것으로 가정하자.

$$\begin{aligned} \tilde{z}_{c,t} &\equiv z_{c,t}(k_{g,t})^{\alpha_g} \\ \Rightarrow \tilde{z}_c &= z_c(k_g)^{\alpha_g} \end{aligned} \tag{78}$$

$$\begin{aligned} \tilde{z}_{i,t} &\equiv z_{i,t}(k_{g,t})^{\alpha_g} \\ \Rightarrow \tilde{z}_i &= z_i(k_g)^{\alpha_g} \end{aligned} \tag{79}$$

이는 k_g를 주어진 것으로 가정하는 것과 같다.

$$\begin{aligned} \lambda_t &= \beta \mathbb{E}_t \left\{ \lambda_{t+1} \left[\frac{R_t}{\pi_{c,t+1}} - \tau_{r,t+1} \left(\frac{R_t - 1}{\pi_{c,t+1}} \right) \right] \right\} \\ \Rightarrow \lambda &= \beta \lambda \left(\frac{R}{\pi_c} - \tau_r \frac{R-1}{\pi_c} \right) \\ \Rightarrow \frac{1}{\beta} &= \left(\frac{R}{\pi_c} - \tau_r \frac{R-1}{\pi_c} \right) \\ \Rightarrow \frac{\pi_c}{\beta} &= R - \tau_r (R-1) \\ \Rightarrow \frac{\pi_c}{\beta} &= (1-\tau_r)R + \tau_r \\ \Rightarrow (1-\tau_r)R &= \frac{\pi_c}{\beta} - \tau_r \\ \Rightarrow R &= \frac{1}{1-\tau_r} \left(\frac{\pi_c}{\beta} - \tau_r \right) \end{aligned} \tag{24}$$

$$Q_{c,t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t} \tag{67}$$

$$\Rightarrow Q_c = \beta \frac{\lambda}{\lambda}$$
$$\Rightarrow Q_c = \beta$$

$$Q_{i,t,t+1} \equiv Q_{c,t,t+1} \frac{q_{t+1}}{q_t} \tag{68}$$
$$\Rightarrow Q_i \equiv Q_c \frac{q}{q}$$
$$\Rightarrow Q_i = \beta$$

$$1 = (1-\theta_c)\left(p_{c,t}^{opt}\right)^{1-\varepsilon_c} + \theta_c \left[\frac{\left(\pi_{c,t-1}^{\iota_c}\pi_c^{1-\iota_c}\right)^{\gamma_c}}{\pi_{c,t}}\right]^{1-\varepsilon_c} \tag{38}$$
$$\Rightarrow 1 = (1-\theta_c)\left(p_c^{opt}\right)^{1-\varepsilon_c} + \theta_c \left[\frac{\left(\pi_c^{\iota_c}\pi_c^{1-\iota_c}\right)^{\gamma_c}}{\pi_c}\right]^{1-\varepsilon_c}$$
$$\Rightarrow 1 = (1-\theta_c)\left(p_c^{opt}\right)^{1-\varepsilon_c} + \theta_c \pi_c^{(\gamma_c-1)(1-\varepsilon_c)}$$
$$\Rightarrow (1-\theta_c)\left(p_c^{opt}\right)^{1-\varepsilon_c} = 1 - \theta_c \pi_c^{(\gamma_c-1)(1-\varepsilon_c)}$$
$$\Rightarrow \left(p_c^{opt}\right)^{1-\varepsilon_c} = \frac{1-\theta_c \pi_c^{(\gamma_c-1)(1-\varepsilon_c)}}{1-\theta_c}$$
$$\Rightarrow p_c^{opt} = \left[\frac{1-\theta_c \pi_c^{(\gamma_c-1)(1-\varepsilon_c)}}{1-\theta_c}\right]^{\frac{1}{1-\varepsilon_c}}$$

$$1 = (1-\theta_i)\left(p_{i,t}^{opt}\right)^{1-\varepsilon_i} + \theta_i \left[\frac{\left(\pi_{i,t-1}^{\iota_i}\pi_c^{1-\iota_i}\right)^{\gamma_i}}{\pi_{i,t}}\right]^{1-\varepsilon_i} \tag{39}$$
$$\Rightarrow 1 = (1-\theta_i)\left(p_i^{opt}\right)^{1-\varepsilon_i} + \theta_i \left[\frac{\left(\pi_i^{\iota_i}\pi_c^{1-\iota_i}\right)^{\gamma_i}}{\pi_i}\right]^{1-\varepsilon_i}$$
$$\Rightarrow 1 = (1-\theta_i)\left(p_i^{opt}\right)^{1-\varepsilon_i} + \theta_i \pi_i^{(\gamma_i-1)(1-\varepsilon_i)}$$
$$\Rightarrow (1-\theta_i)\left(p_i^{opt}\right)^{1-\varepsilon_i} = 1 - \theta_i \pi_i^{(\gamma_i-1)(1-\varepsilon_i)}$$
$$\Rightarrow \left(p_i^{opt}\right)^{1-\varepsilon_i} = \frac{1-\theta_i \pi_i^{(\gamma_i-1)(1-\varepsilon_i)}}{1-\theta_i}$$
$$\Rightarrow p_i^{opt} = \left[\frac{1-\theta_i \pi_i^{(\gamma_i-1)(1-\varepsilon_i)}}{1-\theta_i}\right]^{\frac{1}{1-\varepsilon_i}}$$

$$1 = (1-\theta_w)\left(\tilde{w}_t^{opt}\right)^{1-\varepsilon_w} + \theta_w \left[\frac{\left(\pi_{w,t-1}^{\iota_w}\pi_c^{1-\iota_w}\right)^{\gamma_w}}{\pi_{w,t}}\right]^{1-\varepsilon_w} \tag{40}$$
$$\Rightarrow 1 = (1-\theta_w)\left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} + \theta_w \left[\frac{\left(\pi_w^{\iota_w}\pi_c^{1-\iota_w}\right)^{\gamma_w}}{\pi_w}\right]^{1-\varepsilon_w}$$

$$
\begin{aligned}
&\Rightarrow 1 = (1-\theta_w)\left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} + \theta_w \pi_w^{(\gamma_w-1)(1-\varepsilon_w)} \\
&\Rightarrow (1-\theta_w)\left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} = 1 - \theta_w \pi_w^{(\gamma_w-1)(1-\varepsilon_w)} \\
&\Rightarrow \left(\tilde{w}^{opt}\right)^{1-\varepsilon_w} = \frac{1-\theta_w \pi_w^{(\gamma_w-1)(1-\varepsilon_w)}}{1-\theta_w} \\
&\Rightarrow \tilde{w}^{opt} = \left[\frac{1-\theta_w \pi_w^{(\gamma_w-1)(1-\varepsilon_w)}}{1-\theta_w}\right]^{\frac{1}{1-\varepsilon_w}}
\end{aligned}
$$

$$
\Delta_{c,t} = (1-\theta_c)(p_{c,t}^{opt})^{-\varepsilon_c} + \theta_c \left(\frac{(\pi_{c,t-1}^{\iota_c}\pi_c^{1-\iota_c})^{\gamma_c}}{\pi_{c,t}}\right)^{-\varepsilon_c} \Delta_{c,t-1} \tag{42}
$$

$$
\begin{aligned}
&\Rightarrow \Delta_c = (1-\theta_c)(p_c^{opt})^{-\varepsilon_c} + \theta_c \left(\frac{(\pi_c^{\iota_c}\pi_c^{1-\iota_c})^{\gamma_c}}{\pi_c}\right)^{-\varepsilon_c} \Delta_c \\
&\Rightarrow \Delta_c = (1-\theta_c)(p_c^{opt})^{-\varepsilon_c} + \theta_c \pi_c^{-\varepsilon_c(\gamma_c-1)} \Delta_c \\
&\Rightarrow \left[1 - \theta_c \pi_c^{-\varepsilon_c(\gamma_c-1)}\right] \Delta_c = (1-\theta_c)(p_c^{opt})^{-\varepsilon_c} \\
&\Rightarrow \Delta_c = \frac{(1-\theta_c)(p_c^{opt})^{-\varepsilon_c}}{1-\theta_c \pi_c^{-\varepsilon_c(\gamma_c-1)}}
\end{aligned}
$$

$$
\Delta_{i,t} = (1-\theta_i)(p_{i,t}^{opt})^{-\varepsilon_i} + \theta_i \left(\frac{(\pi_{i,t-1}^{\iota_i}\pi_c^{1-\iota_i})^{\gamma_i}}{\pi_{i,t}}\right)^{-\varepsilon_i} \Delta_{i,t-1} \tag{44}
$$

$$
\begin{aligned}
&\Rightarrow \Delta_i = (1-\theta_i)(p_i^{opt})^{-\varepsilon_i} + \theta_i \left(\frac{(\pi_i^{\iota_i}\pi_c^{1-\iota_i})^{\gamma_i}}{\pi_i}\right)^{-\varepsilon_i} \Delta_i \\
&\Rightarrow \Delta_i = (1-\theta_i)(p_i^{opt})^{-\varepsilon_i} + \theta_i \pi_i^{-\varepsilon_i(\gamma_i-1)} \Delta_i \\
&\Rightarrow \left[1 - \theta_i \pi_i^{-\varepsilon_i(\gamma_i-1)}\right] \Delta_i = (1-\theta_i)(p_i^{opt})^{-\varepsilon_i} \\
&\Rightarrow \Delta_i = \frac{(1-\theta_i)(p_i^{opt})^{-\varepsilon_i}}{1-\theta_i \pi_i^{-\varepsilon_i(\gamma_i-1)}}
\end{aligned}
$$

$$
\Delta_{w,t} = (1-\theta_w)(\tilde{w}_t^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_{w,t-1}^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w}}{\pi_{w,t}}\right)^{-\varepsilon_w} \Delta_{w,t-1} \tag{46}
$$

$$
\begin{aligned}
&\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w}}{\pi_w}\right)^{-\varepsilon_w} \Delta_w \\
&\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \pi_w^{-\varepsilon_w(\gamma_w-1)} \Delta_w \\
&\Rightarrow \left[1 - \theta_w \pi_w^{-\varepsilon_w(\gamma_w-1)}\right] \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} \\
&\Rightarrow \Delta_w = \frac{(1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w}}{1-\theta_w \pi_w^{-\varepsilon_w(\gamma_w-1)}}
\end{aligned}
$$

$$
x_{c,n,t} = y_{c,t} mc_{c,t} + \theta_c \mathbb{E}_t \left[Q_{c,t,t+1} \left(\frac{\pi_{c,t+1}}{(\pi_{c,t}^{\iota_c}\pi_c^{1-\iota_c})^{\gamma_c}}\right)^{\varepsilon_c} x_{c,n,t+1}\right] \tag{33}
$$

$$\Rightarrow x_{c,n} = y_c mc_c + \theta_c Q_c \left(\frac{\pi_c}{(\pi_c^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c} x_{c,n}$$
$$\Rightarrow x_{c,n} = y_c mc_c + \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)} x_{c,n}$$
$$\Rightarrow \left[1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)} \right] x_{c,n} = y_c mc_c$$
$$\Rightarrow x_{c,n} = \frac{y_c mc_c}{1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}}$$

$$x_{c,d,t} = y_{c,t} + \theta_c \mathbb{E}_t \left[Q_{c,t,t+1} \left(\frac{\pi_{c,t+1}}{(\pi_{c,t}^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c - 1} x_{c,d,t+1} \right] \tag{34}$$
$$\Rightarrow x_{c,d} = y_c + \theta_c Q_c \left(\frac{\pi_c}{(\pi_c^{\iota_c} \pi_c^{1-\iota_c})^{\gamma_c}} \right)^{\varepsilon_c - 1} x_{c,d}$$
$$\Rightarrow x_{c,d} = y_c + \theta_c Q_c \pi_c^{(1-\gamma_c)(\varepsilon_c - 1)} x_{c,d}$$
$$\Rightarrow \left[1 - \theta_c Q_c \pi_c^{(1-\gamma_c)(\varepsilon_c - 1)} \right] x_{c,d} = y_c$$
$$\Rightarrow x_{c,d} = \frac{y_c}{1 - \theta_c Q_c \pi_c^{(1-\gamma_c)(\varepsilon_c - 1)}}$$

$$p_{c,t}^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1} \frac{x_{c,n,t}}{x_{c,d,t}} \tag{32}$$
$$\Rightarrow p_c^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1} \frac{x_{c,n}}{x_{c,d}}$$
$$\Rightarrow p_c^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1} \frac{\frac{y_c mc_c}{1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}}}{\frac{y_c}{1 - \theta_c Q_c \pi_c^{(\varepsilon_c - 1)(1-\gamma_c)}}}$$
$$\Rightarrow p_c^{opt} = \frac{\varepsilon_c}{\varepsilon_c - 1} \frac{1 - \theta_c Q_c \pi_c^{(\varepsilon_c - 1)(1-\gamma_c)}}{1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}} mc_c$$
$$\Rightarrow mc_c = \frac{\varepsilon_c - 1}{\varepsilon_c} \frac{1 - \theta_c Q_c \pi_c^{\varepsilon_c(1-\gamma_c)}}{1 - \theta_c Q_c \pi_c^{(\varepsilon_c - 1)(1-\gamma_c)}} p_c^{opt}$$

$$x_{i,n,t} = y_{i,t} mc_{i,t} + \theta_i \mathbb{E}_t \left[Q_{i,t,t+1} \left(\frac{\pi_{i,t+1}}{(\pi_{i,t}^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i} x_{i,n,t+1} \right] \tag{36}$$
$$\Rightarrow x_{i,n} = y_i mc_i + \theta_i Q_i \left(\frac{\pi_i}{(\pi_i^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i} x_{i,n}$$
$$\Rightarrow x_{i,n} = y_i mc_i + \theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)} x_{i,n}$$
$$\Rightarrow \left[1 - \theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)} \right] x_{i,n} = y_i mc_i$$
$$\Rightarrow x_{i,n} = \frac{y_i mc_i}{1 - \theta_i Q_i \pi_i^{\varepsilon_i(1-\gamma_i)}}$$

$$x_{i,d,t} = y_{i,t} + \theta_i \mathbb{E}_t \left[Q_{i,t,t+1} \left(\frac{\pi_{i,t+1}}{(\pi_{i,t}^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i - 1} x_{i,d,t+1} \right] \tag{36}$$

$$
\begin{aligned}
&\Rightarrow x_{i,d} = y_i + \theta_i Q_i \left(\frac{\pi_i}{(\pi_i^{\iota_i} \pi_c^{1-\iota_i})^{\gamma_i}} \right)^{\varepsilon_i - 1} x_{i,d} \\
&\Rightarrow x_{i,d} = y_i + \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)} x_{i,d} \\
&\Rightarrow \left[1 - \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)} \right] x_{i,d} = y_i \\
&\Rightarrow x_{i,d} = \frac{y_i}{1 - \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)}}
\end{aligned}
$$

$$
\begin{aligned}
p_{i,t}^{opt} &= \frac{\varepsilon_i}{\varepsilon_i - 1} \frac{x_{i,n,t}}{x_{i,d,t}} \\
\Rightarrow p_i^{opt} &= \frac{\varepsilon_i}{\varepsilon_i - 1} \frac{x_{i,n}}{x_{i,d}} \\
\Rightarrow p_i^{opt} &= \frac{\varepsilon_i}{\varepsilon_i - 1} \frac{\frac{y_i mc_i}{1 - \theta_i Q_i \pi_i^{\varepsilon_i (1-\gamma_i)}}}{\frac{y_i}{1 - \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)}}} \\
\Rightarrow p_i^{opt} &= \frac{\varepsilon_i}{\varepsilon_i - 1} \frac{1 - \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)}}{1 - \theta_i Q_i \pi_i^{\varepsilon_i (1-\gamma_i)}} mc_i \\
\Rightarrow mc_i &= \frac{\varepsilon_i - 1}{\varepsilon_i} \frac{1 - \theta_i Q_i \pi_i^{\varepsilon_i (1-\gamma_i)}}{1 - \theta_i Q_i \pi_i^{(\varepsilon_i - 1)(1-\gamma_i)}} p_i^{opt}
\end{aligned} \tag{35}
$$

$$
\begin{aligned}
n_t^{eff} &= \frac{n_t}{\Delta_{w,t}} \\
\Rightarrow n^{eff} &= \frac{n}{\Delta_w}
\end{aligned} \tag{45}
$$

$$
\begin{aligned}
n_t^{c,eff} &= n_t^{o,eff} \\
\Rightarrow n^{c,eff} &= n^{o,eff}
\end{aligned} \tag{66}
$$

$$
\begin{aligned}
n_t^{eff} &= \omega n_t^{c,eff} + (1-\omega) n_t^{o,eff} \\
\Rightarrow n^{eff} &= \omega n^{c,eff} + (1-\omega) n^{o,eff} \\
\Rightarrow n^{eff} &= \omega n^{c,eff} + (1-\omega) n^{c,eff} \\
\Rightarrow n^{eff} &= n^{c,eff}
\end{aligned} \tag{54}
$$

결국 $n^{eff} = n^{c,eff} = n^{o,eff}$ 이다.

$$
r_{c,t}^k = mc_{c,t} \alpha_c \frac{y_{c,t} \Delta_{c,t}}{k_{c,t}} \tag{49}
$$

$$\Rightarrow r_c^k = mc_c\alpha_c\frac{y_c\Delta_c}{k_c}$$
$$\Rightarrow k_c = mc_c\alpha_c\frac{y_c\Delta_c}{r_c^k}$$

$$w_t = mc_{c,t}(1-\alpha_c)\frac{y_{c,t}\Delta_{c,t}}{n_{c,t}^{eff}} \tag{50}$$
$$\Rightarrow w = mc_c(1-\alpha_c)\frac{y_c\Delta_c}{n_c^{eff}}$$
$$\Rightarrow n_c^{eff} = mc_c(1-\alpha_c)\frac{y_c\Delta_c}{w}$$

$$y_{c,t} = \frac{\tilde{z}_{c,t}(k_{c,t})^{\alpha_c}(n_{c,t}^{eff})^{1-\alpha_c}}{\Delta_{c,t}} \tag{41}$$
$$\Rightarrow y_c = \frac{\tilde{z}_c(k_c)^{\alpha_c}(n_c^{eff})^{1-\alpha_c}}{\Delta_c}$$
$$\Rightarrow \Delta_c y_c = \tilde{z}_c\left(mc_c\alpha_c\frac{y_c\Delta_c}{r_c^k}\right)^{\alpha_c}\left(mc_c(1-\alpha_c)\frac{y_c\Delta_c}{w}\right)^{1-\alpha_c}$$
$$\Rightarrow 1 = mc_c\tilde{z}_c\left(\frac{\alpha_c}{r_c^k}\right)^{\alpha_c}\left(\frac{1-\alpha_c}{w}\right)^{1-\alpha_c}$$
$$\Rightarrow mc_c = \frac{1}{\tilde{z}_c}\left(\frac{\alpha_c}{r_c^k}\right)^{-\alpha_c}\left(\frac{1-\alpha_c}{w}\right)^{-(1-\alpha_c)}$$
$$\Rightarrow mc_c = \frac{1}{\tilde{z}_c}\frac{(r_c^k)^{\alpha_c}w^{1-\alpha_c}}{\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}}$$
$$\Rightarrow w^{1-\alpha_c} = mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}(r_c^k)^{-\alpha_c}$$
$$\Rightarrow w = \left[mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}}(r_c^k)^{-\frac{\alpha_c}{1-\alpha_c}} \tag{41*}$$

$$r_{i,t}^k = \alpha_i mc_{i,t}q_t\frac{y_{i,t}\Delta_{i,t}}{k_{i,t}} \tag{51}$$
$$\Rightarrow r_i^k = \alpha_i mc_i q\frac{y_i\Delta_i}{k_i}$$
$$\Rightarrow k_i = \alpha_i mc_i q\frac{y_i\Delta_i}{r_i^k}$$

$$w_t = (1-\alpha_i)mc_{i,t}q_t\frac{y_{i,t}\Delta_{i,t}}{n_{i,t}^{eff}} \tag{52}$$
$$\Rightarrow w = (1-\alpha_i)mc_i q\frac{y_i\Delta_i}{n_i^{eff}}$$
$$\Rightarrow n_i^{eff} = (1-\alpha_i)mc_i q\frac{y_i\Delta_i}{w}$$

$$y_{i,t} = \frac{\tilde{z}_{i,t}(k_{i,t})^{\alpha_i}(n_{i,t}^{eff})^{1-\alpha_i}}{\Delta_{i,t}} \tag{43}$$

$$\Rightarrow y_i = \frac{\tilde{z}_i(k_i)^{\alpha_i}(n_i^{eff})^{1-\alpha_i}}{\Delta_i}$$

$$\Rightarrow \Delta_i y_i = \tilde{z}_i \left(\alpha_i mc_i q \frac{y_i \Delta_i}{r_i^k}\right)^{\alpha_i} \left((1-\alpha_i) mc_i q \frac{y_i \Delta_i}{w}\right)^{1-\alpha_i}$$

$$\Rightarrow 1 = mc_i q \tilde{z}_i \left(\frac{\alpha_i}{r_i^k}\right)^{\alpha_i} \left(\frac{1-\alpha_i}{w}\right)^{1-\alpha_i}$$

$$\Rightarrow mc_i = \frac{1}{\tilde{z}_i}\frac{1}{q}\left(\frac{\alpha_i}{r_i^k}\right)^{-\alpha_i} \left(\frac{1-\alpha_i}{w}\right)^{-(1-\alpha_i)}$$

$$\Rightarrow mc_i = \frac{1}{\tilde{z}_i}\frac{1}{q}\frac{(r_i^k)^{\alpha_i} w^{1-\alpha_i}}{\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}}$$

$$\Rightarrow w^{1-\alpha_i} = mc_i \tilde{z}_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}(r_i^k)^{-\alpha_i}$$

$$\Rightarrow w = \left[mc_i \tilde{z}_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} (r_i^k)^{-\frac{\alpha_i}{1-\alpha_i}} \tag{43*}$$

$$\lambda_t q_t = \xi_{c,t}\left\{\left[1 - \frac{\psi_c}{2}\left(\frac{i_{c,t}^o}{i_{c,t-1}^o} - 1\right)^2\right]\mu_{c,t} - \psi_c\left(\frac{i_{c,t}^o}{i_{c,t-1}^o} - 1\right)\frac{1}{i_{c,t-1}^o}\mu_{c,t} i_{c,t}^o\right\}$$
$$+ \beta\mathbb{E}_t\left\{\xi_{c,t+1}\left[\psi_c\left(\frac{i_{c,t+1}^o}{i_{c,t}^o} - 1\right)\left(\frac{i_{c,t+1}^o}{i_{c,t}^o}\right)^2 \mu_{c,t+1}\right]\right\} \tag{25}$$

$$\Rightarrow \lambda q = \xi_c\left\{\left[1 - \frac{\psi_c}{2}\left(\frac{i_c^o}{i_c^o} - 1\right)^2\right]\mu_c - \psi_c\left(\frac{i_c^o}{i_c^o} - 1\right)\frac{1}{i_c^o}\mu_c i_c^o\right\}$$
$$+ \beta\xi_c\left[\psi_c\left(\frac{i_c^o}{i_c^o} - 1\right)\left(\frac{i_c^o}{i_c^o}\right)^2 \mu_c\right]$$

$$\Rightarrow \lambda q = \xi_c \mu_c$$

$$\Rightarrow \xi_c = \frac{\lambda}{\mu_c} q$$

$$\lambda_t q_t = \xi_{i,t}\left\{\left[1 - \frac{\psi_i}{2}\left(\frac{i_{i,t}^o}{i_{i,t-1}^o} - 1\right)^2\right]\mu_{i,t} - \psi_i\left(\frac{i_{i,t}^o}{i_{i,t-1}^o} - 1\right)\frac{1}{i_{i,t-1}^o}\mu_{i,t} i_{i,t}^o\right\}$$
$$+ \beta\mathbb{E}_t\left\{\xi_{i,t+1}\left[\psi_i\left(\frac{i_{i,t+1}^o}{i_{i,t}^o} - 1\right)\left(\frac{i_{i,t+1}^o}{i_{i,t}^o}\right)^2 \mu_{i,t+1}\right]\right\} \tag{26}$$

$$\Rightarrow \lambda q = \xi_i\left\{\left[1 - \frac{\psi_i}{2}\left(\frac{i_i^o}{i_i^o} - 1\right)^2\right]\mu_i - \psi_i\left(\frac{i_i^o}{i_i^o} - 1\right)\frac{1}{i_i^o}\mu_i i_i^o\right\}$$

$$+\beta\xi_i\left[\psi_i\left(\frac{i_i^o}{i_i^o}-1\right)\left(\frac{i_i^o}{i_i^o}\right)^2\mu_i\right]$$

$$\Rightarrow \lambda q = \xi_i \mu_i$$

$$\Rightarrow \xi_i = \frac{\lambda}{\mu_i} q$$

정상상태에서 $\mu_c = \mu_i$ 이면, $\xi_c = \xi_i$ 가 된다.

$$\xi_{c,t} = \beta \mathbb{E}_t \left\{ \lambda_{t+1}(1-\tau_{k,t+1}) r^k_{c,t+1} + \xi_{c,t+1}(1-\delta_c) \right\} \tag{27}$$

$$\Rightarrow \xi_c = \beta \left(\lambda(1-\tau_k) r^k_c + \xi_c(1-\delta_c) \right)$$

$$\Rightarrow \xi_c = \beta\lambda(1-\tau_k) r^k_c + \beta\xi_c(1-\delta_c)$$

$$\Rightarrow \frac{\lambda}{\mu_c} q = \beta\lambda(1-\tau_k) r^k_c + \beta\frac{\lambda}{\mu_c} q(1-\delta_c)$$

$$\Rightarrow \frac{1}{\mu_c} q = \beta(1-\tau_k) r^k_c + \beta\frac{1}{\mu_c} q(1-\delta_c)$$

$$\Rightarrow \beta(1-\tau_k) r^k_c = \frac{1}{\mu_c} q - \beta\frac{1}{\mu_c} q(1-\delta_c)$$

$$\Rightarrow r^k_c = \frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right] q \tag{27*}$$

$$\xi_{i,t} = \beta \mathbb{E}_t \left\{ \lambda_{t+1}(1-\tau_{k,t+1}) r^k_{i,t+1} + \xi_{i,t+1}(1-\delta_i) \right\} \tag{28}$$

$$\Rightarrow \xi_i = \beta \left(\lambda(1-\tau_k) r^k_i + \xi_i(1-\delta_i) \right)$$

$$\Rightarrow \xi_i = \beta\lambda(1-\tau_k) r^k_i + \beta\xi_i(1-\delta_i)$$

$$\Rightarrow \frac{\lambda}{\mu_i} q = \beta\lambda(1-\tau_k) r^k_i + \beta\frac{\lambda}{\mu_i} q(1-\delta_i)$$

$$\Rightarrow \frac{1}{\mu_i} q = \beta(1-\tau_k) r^k_i + \beta\frac{1}{\mu_i} q(1-\delta_i)$$

$$\Rightarrow \beta(1-\tau_k) r^k_i = \frac{1}{\mu_i} q - \beta\frac{1}{\mu_i} q(1-\delta_i)$$

$$\Rightarrow r^k_i = \frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right] q \tag{28*}$$

정상상태에서 $\mu_c = \mu_i$ 이고 $\delta_c = \delta_i$ 이면, $r^k_c = r^k_i$ 가 된다.

식 (27*), (28*), (41*), (43*)로부터 q의 정상상태 값을 도출할 수 있다.

$$\underbrace{\left[mc_i \tilde{z}_i q \alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} (r^k_i)^{-\frac{\alpha_i}{1-\alpha_i}}}_{\text{식 (36*)}} = \underbrace{\left[mc_c \tilde{z}_c \alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} (r^k_c)^{-\frac{\alpha_c}{1-\alpha_c}}}_{\text{식 (34*)}}$$

$$\Rightarrow \left[mc_i\tilde{z}_i q\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\underbrace{\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]q}_{=r_i^k:\text{식 (21*)}}\right)^{-\frac{\alpha_i}{1-\alpha_i}}$$

$$= \left[mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\underbrace{\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]q}_{=r_c^k:\text{식 (20*)}}\right)^{-\frac{\alpha_c}{1-\alpha_c}}$$

$$\Rightarrow q^{\frac{1-\alpha_i}{1-\alpha_i}} q^{\frac{\alpha_c}{1-\alpha_c}} \left[mc_i\tilde{z}_i\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}$$

$$= \left[mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}$$

$$\Rightarrow q^{\frac{1}{1-\alpha_c}} = \frac{\left[mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}}{\left[mc_i\tilde{z}_i\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}}$$

$$\Rightarrow q = \left\{\frac{\left[mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}}{\left[mc_i\tilde{z}_i\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}}\right\}^{1-\alpha_c}$$

만일 $\alpha_c = \alpha_i = \alpha$ 라면, q는 다음과 같다.

$$q = \left\{\frac{\left[mc_c\tilde{z}_c\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}\right]^{\frac{1}{1-\alpha_c}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_c}{1-\alpha_c}}}{\left[mc_i\tilde{z}_i\alpha_i^{\alpha_i}(1-\alpha_i)^{1-\alpha_i}\right]^{\frac{1}{1-\alpha_i}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha_i}{1-\alpha_i}}}\right\}^{1-\alpha_c}$$

$$\Rightarrow q = \left\{\frac{\left[mc_c\tilde{z}_c\alpha^{\alpha}(1-\alpha)^{1-\alpha}\right]^{\frac{1}{1-\alpha}} \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha}{1-\alpha}}}{\left[mc_i\tilde{z}_i\alpha^{\alpha}(1-\alpha)^{1-\alpha}\right]^{\frac{1}{1-\alpha}} \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\frac{\alpha}{1-\alpha}}}\right\}^{1-\alpha}$$

$$\Rightarrow q = \frac{\left[mc_c\tilde{z}_c\alpha^{\alpha}(1-\alpha)^{1-\alpha}\right] \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\alpha}}{\left[mc_i\tilde{z}_i\alpha^{\alpha}(1-\alpha)^{1-\alpha}\right] \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\alpha}}$$

$$\Rightarrow q = \frac{\left[mc_c\tilde{z}_c\right] \left(\frac{1}{\mu_c}\left[\frac{1-\beta(1-\delta_c)}{\beta(1-\tau_k)}\right]\right)^{-\alpha}}{\left[mc_i\tilde{z}_i\right] \left(\frac{1}{\mu_i}\left[\frac{1-\beta(1-\delta_i)}{\beta(1-\tau_k)}\right]\right)^{-\alpha}}$$

$$\Rightarrow q = \frac{mc_c\tilde{z}_c}{mc_i\tilde{z}_i}\left(\frac{\mu_i}{\mu_c}\frac{1-\beta(1-\delta_c)}{1-\beta(1-\delta_i)}\right)^{-\alpha}$$

만일 소비재 부문과 투자재 부문이 완전 대칭이라면(즉 $\alpha_c = \alpha_i = \alpha$, $mc_c = mc_i$, $\tilde{z}_c = \tilde{z}_i$, $\mu_c = \mu_i$, $\delta_c = \delta_i$), $q = 1$ 이 된다.

q를 알면 식 (27*), (28*)로부터 r_c^k, r_i^k를 구할 수 있다. 한편 q와 r_c^k, r_i^k를 알면 식 (41*), (43*)로 부터 w

를 알 수 있다. 그러나, $\tilde{z}_c$와 $\tilde{z}_i$는 여전히 미지수이므로 q역시 아직 미지수임에 유의해야 한다.

식 (49), (50)으로부터 다음이 성립한다.

$$\begin{aligned}
&\frac{w_t}{r^k_{c,t}} = \frac{mc_{c,t}(1-\alpha_c)\frac{y_{c,t}\Delta_{c,t}}{n^{eff}_{c,t}}}{mc_{c,t}\alpha_c\frac{y_{c,t}\Delta_{c,t}}{k_{c,t}}} \\
\Rightarrow &\frac{w_t}{r^k_{c,t}} = \frac{1-\alpha_c}{\alpha_c}\frac{k_{c,t}}{n^{eff}_{c,t}} \\
\Rightarrow &\frac{k_{c,t}}{n^{eff}_{c,t}} = \frac{\alpha_c}{1-\alpha_c}\frac{w_t}{r^k_{c,t}} \\
\Rightarrow &\frac{k_c}{n^{eff}_c} = \underbrace{\frac{\alpha_c}{1-\alpha_c}\frac{w}{r^k_c}}_{\equiv\kappa_c} \\
\Rightarrow &\frac{k_c}{n^{eff}_c} = \kappa_c \\
\Rightarrow &k_c = \kappa_c n^{eff}_c
\end{aligned}$$

식 (51), (52)로부터 다음이 성립한다.

$$\begin{aligned}
&\frac{w_t}{r^k_{i,t}} = \frac{mc_{i,t}(1-\alpha_i)q_t\frac{y_{i,t}\Delta_{i,t}}{n^{eff}_{i,t}}}{mc_{i,t}\alpha_i q_t\frac{y_{i,t}\Delta_{i,t}}{k_{i,t}}} \\
\Rightarrow &\frac{w_t}{r^k_{i,t}} = \frac{1-\alpha_i}{\alpha_i}\frac{k_{i,t}}{n^{eff}_{i,t}} \\
\Rightarrow &\frac{k_{i,t}}{n^{eff}_{i,t}} = \frac{\alpha_i}{1-\alpha_i}\frac{w_t}{r^k_{i,t}} \\
\Rightarrow &\frac{k_i}{n^{eff}_i} = \underbrace{\frac{\alpha_i}{1-\alpha_i}\frac{w}{r^k_i}}_{\equiv\kappa_i} \\
\Rightarrow &\frac{k_i}{n^{eff}_i} = \kappa_i \\
\Rightarrow &k_i = \kappa_i n^{eff}_i
\end{aligned}$$

식 (71), (73)으로 부터 다음을 알 수 있다.

$$\begin{aligned}
n^{eff}_t &= n^{eff}_{c,t} + n^{eff}_{i,t} \\
\Rightarrow n^{eff} &= n^{eff}_c + n^{eff}_i \\
\Rightarrow n^{eff}_c &= n^{eff} - n^{eff}_i
\end{aligned} \tag{73}$$

$$k_t = k_{c,t} + k_{i,t} \tag{71}$$

$$
\begin{aligned}
&\Rightarrow k = k_c + k_i \\
&\Rightarrow k = \kappa_c n_c^{eff} + \kappa_i n_i^{eff} \\
&\Rightarrow k = \kappa_c \underbrace{\left(n^{eff} - n_i^{eff}\right)}_{=n_c^{eff}} + \kappa_i n_i^{eff} \\
&\Rightarrow k = \kappa_c n^{eff} - \kappa_c n_i^{eff} + \kappa_i n_i^{eff} \\
&\Rightarrow k = \kappa_c n^{eff} + (\kappa_i - \kappa_c) n_i^{eff}
\end{aligned}
$$

이제 y_c, y_i, y, g_i를 모두 n_i^{eff}의 함수로 나타내고 투자재 부문의 청산 조건에 대입하면 n_i^{eff}를 도출할 수 있다.

$$
\begin{aligned}
y_{c,t} &= \frac{\tilde{z}_{c,t}(k_{c,t})^{\alpha_c}(n_{c,t}^{eff})^{1-\alpha_c}}{\Delta_{c,t}} \qquad (41) \\
\Rightarrow y_c &= \frac{\tilde{z}_c(k_c)^{\alpha_c}(n_c^{eff})^{1-\alpha_c}}{\Delta_c} \\
\Rightarrow y_c &= \frac{\tilde{z}_c}{\Delta_c}\left(\underbrace{\frac{k_c}{n_c^{eff}}}_{\equiv \kappa_c}\right)^{\alpha_c} n_c^{eff} \\
\Rightarrow y_c &= \frac{\tilde{z}_c}{\Delta_c}\left(\kappa_c\right)^{\alpha_c} n_c^{eff} \\
\Rightarrow y_c &= \frac{\tilde{z}_c}{\Delta_c}\left(\kappa_c\right)^{\alpha_c}\left(n^{eff} - n_i^{eff}\right)
\end{aligned}
$$

$$
\begin{aligned}
y_{i,t} &= \frac{\tilde{z}_{i,t}(k_{i,t})^{\alpha_i}(n_{i,t}^{eff})^{1-\alpha_i}}{\Delta_{i,t}} \qquad (43) \\
\Rightarrow y_i &= \frac{\tilde{z}_i(k_i)^{\alpha_i}(n_i^{eff})^{1-\alpha_i}}{\Delta_i} \\
\Rightarrow y_i &= \frac{\tilde{z}_i}{\Delta_i}\left(\underbrace{\frac{k_i}{n_i^{eff}}}_{\equiv \kappa_i}\right)^{\alpha_i} n_i^{eff} \\
\Rightarrow y_i &= \frac{\tilde{z}_i}{\Delta_i}\left(\kappa_i\right)^{\alpha_i} n_i^{eff}
\end{aligned}
$$

$$
\begin{aligned}
y_t &\equiv y_{c,t} + q_t y_{i,t} \qquad (75) \\
\Rightarrow y &= y_c + q y_i \\
\Rightarrow y &= \underbrace{\frac{\tilde{z}_c}{\Delta_c}\left(\kappa_c\right)^{\alpha_c}\left(n^{eff} - n_i^{eff}\right)}_{=y_c} + q\frac{\tilde{z}_i}{\Delta_i}\left(\kappa_i\right)^{\alpha_i} n_i^{eff}
\end{aligned}
$$

$$\tilde{g}_{i,t} \equiv q_t g_{i,t} \tag{7}$$
$$\Rightarrow \tilde{g}_i = q g_i$$

$$s_{\tilde{g}_i,t} \equiv \frac{\tilde{g}_{i,t}}{y_t} \tag{6}$$
$$\Rightarrow s_{\tilde{g}_i} \equiv \frac{\tilde{g}_i}{y}$$
$$\Rightarrow \tilde{g}_i = s_{\tilde{g}_i} y$$
$$\Rightarrow q g_i = s_{\tilde{g}_i} y$$
$$\Rightarrow g_i = \frac{s_{\tilde{g}_i}}{q} y$$
$$\Rightarrow g_i = \frac{s_{\tilde{g}_i}}{q} \underbrace{\left[\frac{\tilde{z}_c}{\Delta_c} (\kappa_c)^{\alpha_c} \left(n^{eff} - n_i^{eff} \right) + q \frac{\tilde{z}_i}{\Delta_i} (\kappa_i)^{\alpha_i} n_i^{eff} \right]}_{=y}$$

$$k^o_{c,t+1} = (1-\delta_c) k^o_{c,t} + \left[1 - \frac{\psi_c}{2} \left(\frac{i^o_{c,t}}{i^o_{c,t-1}} - 1 \right)^2 \right] \mu_{c,t} i^o_{c,t} \tag{21}$$
$$\Rightarrow k^o_c = (1-\delta_c) k^o_c + \left[1 - \frac{\psi_c}{2} \left(\frac{i^o_c}{i^o_c} - 1 \right)^2 \right] \mu_c i^o_c$$
$$\Rightarrow k^o_c = (1-\delta_c) k^o_c + \mu_c i^o_c$$
$$\Rightarrow \delta_c k^o_c = \mu_c i^o_c$$
$$\Rightarrow i^o_c = \frac{\delta_c}{\mu_c} k^o_c$$
$$\Rightarrow \omega i^o_c = \frac{\delta_c}{\mu_c} \omega k^o_c$$
$$\Rightarrow i_c = \frac{\delta_c}{\mu_c} k_c \tag{21*}$$

$$k^o_{i,t+1} = (1-\delta_i) k^o_{i,t} + \left[1 - \frac{\psi_i}{2} \left(\frac{i^o_{i,t}}{i^o_{i,t-1}} - 1 \right)^2 \right] \mu_{i,t} i^o_{i,t} \tag{22}$$
$$\Rightarrow k^o_i = (1-\delta_i) k^o_i + \left[1 - \frac{\psi_i}{2} \left(\frac{i^o_i}{i^o_i} - 1 \right)^2 \right] \mu_i i^o_i$$
$$\Rightarrow k^o_i = (1-\delta_i) k^o_i + \mu_i i^o_i$$
$$\Rightarrow \delta_i k^o_i = \mu_i i^o_i$$
$$\Rightarrow i^o_i = \frac{\delta_i}{\mu_i} k^o_i$$

$$\Rightarrow \omega i_i^o = \frac{\delta_i}{\mu_i}\omega k_i^o$$
$$\Rightarrow i_i = \frac{\delta_i}{\mu_i}k_i \tag{22*}$$

$$\begin{aligned} i_t &= i_{c,t} + i_{i,t} \\ \Rightarrow i &= i_c + i_i \end{aligned} \tag{72}$$

식 (21*)와 (22*)를 합하면 다음과 같다.

$$\begin{aligned} i_c + i_i &= \frac{\delta_c}{\mu_c}k_c + \frac{\delta_i}{\mu_i}k_i \\ \Rightarrow \underbrace{i_c + i_i}_{=i} &= \frac{\delta_c}{\mu_c}\underbrace{\kappa_c n_c^{eff}}_{=k_c} + \frac{\delta_i}{\mu_i}\underbrace{\kappa_i n_i^{eff}}_{=k_i} \\ \Rightarrow i &= \frac{\delta_c}{\mu_c}\kappa_c \underbrace{\left(n^{eff} - n_i^{eff}\right)}_{=n_c^{eff}} + \frac{\delta_i}{\mu_i}\kappa_i n_i^{eff} \\ \Rightarrow i &= \frac{\delta_c}{\mu_c}\kappa_c n^{eff} + \left(\frac{\delta_i}{\mu_i}\kappa_i - \frac{\delta_c}{\mu_c}\kappa_c\right) n_i^{eff} \end{aligned}$$

$$y_{i,t} = i_t + g_{i,t} \tag{77}$$
$$\Rightarrow y_i = i + g_i$$
$$\begin{aligned} \Rightarrow \underbrace{\frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i} n_i^{eff}}_{=y_i} &= \underbrace{\frac{\delta_c}{\mu_c}\kappa_c n^{eff} + \left(\frac{\delta_i}{\mu_i}\kappa_i - \frac{\delta_c}{\mu_c}\kappa_c\right) n_i^{eff}}_{=i} \\ &\quad + \underbrace{\frac{s_{\tilde{g}_i}}{q}\underbrace{\left[\frac{\tilde{z}_c}{\Delta_c}(\kappa_c)^{\alpha_c}\left(n^{eff} - n_i^{eff}\right) + q\frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i} n_i^{eff}\right]}_{=y}}_{=g_i} \end{aligned}$$
$$\Rightarrow \frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i} n_i^{eff} = \frac{\delta_c}{\mu_c}\kappa_c n^{eff} + \left(\frac{\delta_i}{\mu_i}\kappa_i - \frac{\delta_c}{\mu_c}\kappa_c\right) n_i^{eff} + \frac{s_{\tilde{g}_i}}{q}\frac{\tilde{z}_c}{\Delta_c}(\kappa_c)^{\alpha_c}\left(n^{eff} - n_i^{eff}\right) + \frac{s_{\tilde{g}_i}}{q} q \frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i} n_i^{eff}$$
$$\Rightarrow \left[\frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i} - \left(\frac{\delta_i}{\mu_i}\kappa_i - \frac{\delta_c}{\mu_c}\kappa_c\right) + \frac{s_{\tilde{g}_i}}{q}\frac{\tilde{z}_c}{\Delta_c}(\kappa_c)^{\alpha_c} - s_{\tilde{g}_i}\frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i}\right] n_i^{eff} = \frac{\delta_c}{\mu_c}\kappa_c n^{eff} + \frac{s_{\tilde{g}_i}}{q}\frac{\tilde{z}_c}{\Delta_c}(\kappa_c)^{\alpha_c} n^{eff}$$
$$\Rightarrow n_i^{eff} = \frac{\frac{\delta_c}{\mu_c}\kappa_c n^{eff} + \frac{s_{\tilde{g}_i}}{q}\frac{\tilde{z}_c}{\Delta_c}(\kappa_c)^{\alpha_c} n^{eff}}{\frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i} - \left(\frac{\delta_i}{\mu_i}\kappa_i - \frac{\delta_c}{\mu_c}\kappa_c\right) + \frac{s_{\tilde{g}_i}}{q}\frac{\tilde{z}_c}{\Delta_c}(\kappa_c)^{\alpha_c} - s_{\tilde{g}_i}\frac{\tilde{z}_i}{\Delta_i}(\kappa_i)^{\alpha_i}}$$

지금까지 $\tilde{z}_c, \tilde{z}_i$는 계산 편의를 위해 주어진 것으로 가정하였다. 그러나, 아직 $\tilde{z}_c, \tilde{z}_i$는 확정되지 않은 미지수이다. q와 n_i^{eff} 모두 정상상태 값이 두 변수 $\tilde{z}_c, \tilde{z}_i$의 함수이므로 두 변수의 정상상태 값을 알아야 다른 변수들의 정상상태 값들을 구할 수 있다. 특히 $\tilde{z}_c, \tilde{z}_i$는 정부자본 k_g의 함수이므로 k_g를 알아야 한다.

우선 k_g는 정부의 자본축적식으로부터 다음과 같이 y의 함수로 나타낼 수 있다.

$$
\begin{aligned}
& k_{g,t+1} = (1-\delta_g)k_{g,t} + \mu_{g,t}g_{i,t} - \frac{\psi_g}{2}\left(\frac{g_{i,t}}{g_{i,t-1}} - 1\right)^2 \mu_{g,t}g_{i,t} \qquad (2) \\
\Rightarrow & k_g = (1-\delta_g)k_g + \mu_g g_i - \frac{\psi_g}{2}\left(\frac{g_i}{g_i} - 1\right)^2 \mu_g g_i \\
\Rightarrow & \delta_g k_g = \mu_g g_i \\
\Rightarrow & k_g = \frac{\mu_g}{\delta_g} g_i \\
\Rightarrow & k_g = \frac{\mu_g}{\delta_g} \underbrace{\frac{s_{\tilde{g}_i}}{q} y}_{=g_i:\text{식 (7)}}
\end{aligned}
$$

이 식에서 보는 것 처럼 k_g는 q와 y의 함수이지만 앞에서 살펴보았듯이 q와 y 안에는 $\tilde{z}_c$, $\tilde{z}_i$가 포함되어 있고 $\tilde{z}_c$, $\tilde{z}_i$는 다시 k_g의 함수이므로, k_g에 대한 해석적 폐쇄해(closed solution)는 존재하지 않는 것이 일반적이다[1]. 따라서 수치적 방법을 이용하여 k_g 값을 구한다.

즉, $q = q(\tilde{z}_c, \tilde{z}_i)$이고, $y = y(\tilde{z}_c, \tilde{z}_i)$이므로, k_g는 다음 고정점 방정식의 해로 규정된다.

$$
F(k_g) \equiv k_g - \frac{\mu_g}{\delta_g} \frac{s_{\tilde{g}_i}}{q\left(\tilde{z}_c(k_g), \tilde{z}_i(k_g)\right)} y\left(\tilde{z}_c(k_g), \tilde{z}_i(k_g)\right) = 0
$$

여기서

$$
\tilde{z}_c(k_g) = z_c k_g^{\alpha_g}, \quad \tilde{z}_i(k_g) = z_i k_g^{\alpha_g}
$$

위 방식으로 k_g를 구했다고 가정하면, $\tilde{z}_c$, $\tilde{z}_i$를 구할 수 있고, 이를 통해 q, y, n_i^{eff}가 결정된다. 순차적으로 r_c^k, r_i^k, w, i, g_i, $\tilde{g}_i$, y_c, y_i, k, n_c^{eff}, k_c, k_i, i_c, i_i, $x_{c,n}$, $x_{c,d}$, $x_{i,n}$, $x_{i,d}$ 등 나머지 변수들의 정상상태 값도 구할 수 있다.

$$
\begin{aligned}
& k_{c,t} = (1-\omega)k_{c,t}^o \qquad (57) \\
\Rightarrow & k_c = (1-\omega)k_c^o \\
\Rightarrow & k_c^o = \frac{k_c}{1-\omega}
\end{aligned}
$$

$$
\begin{aligned}
& k_{i,t} = (1-\omega)k_{i,t}^o \qquad (58) \\
\Rightarrow & k_i = (1-\omega)k_i^o \\
\Rightarrow & k_i^o = \frac{k_i}{1-\omega}
\end{aligned}
$$

$$
i_{c,t} = (1-\omega)i_{c,t}^o \qquad (59)
$$

[1] α_g가 특정한 값을 갖는 경우 폐쇄해가 존재할 수 있다.

$$\Rightarrow i_c = (1-\omega) i_c^o$$
$$\Rightarrow i_c^o = \frac{i_c}{1-\omega}$$

$$i_{i,t} = (1-\omega) i_{i,t}^o \tag{60}$$
$$\Rightarrow i_i = (1-\omega) i_i^o$$
$$\Rightarrow i_i^o = \frac{i_i}{1-\omega}$$

$$d_{c,t} = y_{c,t}\left(1 - mc_{c,t}\Delta_{c,t}\right) \tag{47}$$
$$\Rightarrow d_c = y_c\left(1 - mc_c\Delta_c\right)$$

$$d_{i,t} = y_{i,t}\left(1 - mc_{i,t}\Delta_{i,t}\right) \tag{48}$$
$$\Rightarrow d_i = y_i\left(1 - mc_i\Delta_i\right)$$

$$d_t = d_{c,t} + q_t d_{i,t} \tag{74}$$
$$\Rightarrow d = d_c + q d_i$$

$$d_{c,t} = (1-\omega) d_{c,t}^o \tag{62}$$
$$\Rightarrow d_c = (1-\omega) d_c^o$$
$$\Rightarrow d_c^o = \frac{d_c}{1-\omega}$$

$$d_{i,t} = (1-\omega) d_{i,t}^o \tag{63}$$
$$\Rightarrow d_i = (1-\omega) d_i^o$$
$$\Rightarrow d_i^o = \frac{d_i}{1-\omega}$$

$$s_{g_c,t} \equiv \frac{g_{c,t}}{y_t} \tag{4}$$
$$\Rightarrow s_{g_c} = \frac{g_c}{y}$$
$$\Rightarrow g_c = s_{g_c} y$$

$$s_{tr,t} \equiv \frac{tr_t}{y_t} \tag{9}$$

$$\Rightarrow s_{tr} = \frac{tr}{y}$$
$$\Rightarrow tr = s_{tr} y$$

$$\begin{aligned} y_{c,t} &= c_t + g_{c,t} \\ \Rightarrow y_c &= c + g_c \\ \Rightarrow c &= y_c - g_c \end{aligned} \tag{76}$$

$$\begin{aligned} g_{c,t} + q_t g_{i,t} + tr_t + \frac{R_{t-1}}{\pi_{c,t}} b_{t-1} &= \tau_t + b_t \\ \Rightarrow g_c + q g_i + tr + \frac{R}{\pi_c} b &= \tau + b \\ \Rightarrow \tau &= g_c + q g_i + tr + \frac{R}{\pi_c} b - b \end{aligned} \tag{1}$$

$$\begin{aligned} \tau_t &= \tau_{c,t} c_t + \tau_{n,t} w_t n_t^{eff} + \tau_{k,t}(r_{c,t}^k k_{c,t} + r_{i,t}^k k_{i,t}) \\ &\quad + \tau_{d,t} d_{c,t} + \tau_{d,t} q_t d_{i,t} + \tau_{r,t} \left(\frac{R_{t-1} - 1}{\pi_{c,t}} \right) b_{t-1} + \tau_t^L \end{aligned} \tag{3}$$

$$\Rightarrow \tau = \tau_c c + \tau_n w n^{eff} + \tau_k(r_c^k k_c + r_i^k k_i) + \tau_d d_c + \tau_d q d_i + \tau_r \left(\frac{R-1}{\pi_c} \right) b + \tau^L$$

$$\Rightarrow \underbrace{g_c + q g_i + tr + \frac{R}{\pi_c} b - b}_{=\tau}$$
$$= \tau_c c + \tau_n w n^{eff} + \tau_k(r_c^k k_c + r_i^k k_i) + \tau_d d_c + \tau_d q d_i + \tau_r \left(\frac{R-1}{\pi_c} \right) b + \tau^L$$

$$\Rightarrow \frac{R}{\pi_c} b - b - \tau_r \left(\frac{R-1}{\pi_c} \right) b$$
$$= \tau_c c + \tau_n w n^{eff} + \tau_k(r_c^k k_c + r_i^k k_i) + \tau_d d_c + \tau_d q d_i + \tau^L - g_c - q g_i - tr$$

$$\Rightarrow b = \frac{\tau_c c + \tau_n w n^{eff} + \tau_k(r_c^k k_c + r_i^k k_i) + \tau_d d_c + \tau_d q d_i + \tau^L - g_c - q g_i - tr}{\frac{R - \tau_r(R-1)}{\pi_c} - 1}$$

여기서 보듯이 성장이 없는 정태분석에서 정부부채(b)는 정부 기초수지(세입-지출)가 흑자(>0)일때만 양(+)의 값을 갖는다. 여기서, 기초수지란 정부부채로 인한 세입 또는 세출의 영향을 제외한 수지를 의미한다. 따라서, 성장이 없는 정태분석에서는 미국 등 대부분의 선진국이 경험하고 있는 기초수지 적자(<0), 정부부채 >0를 가정할 수 없다. 본 장에서는 정부부채를 양(+)의 숫자로 만들기 위해 정액세의 정상상태 값(τ^L)을 실제보다 크게 높여 기초수지가 흑자가 되도록 하였다.

$$\text{정부 기초수지} = \underbrace{\tau_c c + \tau_n w n^{eff} + \tau_k(r_c^k k_c + r_i^k k_i) + \tau_d d_c + \tau_d q d_i + \tau^L}_{\text{기초 세입}} - \underbrace{(g_c + q g_i + tr)}_{\text{기초 지출}}$$

성장이 없는 정태에서는 세후 실질 순이자율($\frac{R - \tau_r(R-1)}{\pi_c} - 1$)이 양수일 때 정부부채가 양수이면 반드시 기초수지가 흑자다. 그러나 균형성장 정태를 고려하면 이야기가 달라진다. 산출이 매기 성장하면

부채도 함께 커질 수 있으므로 부채 수준 대신 부채/산출 비율을 일정하게 두고 본다. 이때 필요한 기초수지의 부호는 이자–성장 격차로 결정되어, 성장률이 세후 실질 이자율보다 높으면 기초수지가 적자이면서도 양의 부채비율이 안정적으로 유지된다. 실무적으로는 목표 부채비율을 정해 두고 정액세의 평균이나 일부 세목·지출의 평균이 아주 완만하게 조정되도록 설계하면, 성장 환경에서 관찰되는 "기초적자 + 양의 부채"를 정상상태 가정과 양립시킬 수 있다.

$$
\begin{aligned}
s_{b,t} &\equiv \frac{b_t}{y_t} \\
\Rightarrow s_b &= \frac{b}{y}
\end{aligned}
\tag{17}
$$

$$
\begin{aligned}
b_t &= (1-\omega) b_t^o \\
\Rightarrow b &= (1-\omega) b^o \\
\Rightarrow b^o &= \frac{b}{1-\omega}
\end{aligned}
\tag{61}
$$

$$
\begin{aligned}
\omega tr_t^c &= \alpha_t^T tr_t \\
\Rightarrow \omega tr^c &= \alpha^T tr \\
\Rightarrow tr^c &= \frac{\alpha^T tr}{\omega}
\end{aligned}
\tag{64}
$$

$$
\begin{aligned}
tr_t &= \omega tr_t^c + (1-\omega) tr_t^o \\
\Rightarrow tr &= \omega tr^c + (1-\omega) tr^o \\
\Rightarrow (1-\omega) tr^o &= tr - \omega tr^c \\
\Rightarrow tr^o &= \frac{tr - \omega tr^c}{1-\omega}
\end{aligned}
\tag{56}
$$

$$
\begin{aligned}
\omega \tau_t^{c,L} &= \alpha_t^L \tau_t^L \\
\Rightarrow \omega \tau^{c,L} &= \alpha^L \tau^L \\
\Rightarrow \tau^{c,L} &= \frac{\alpha^L \tau^L}{\omega}
\end{aligned}
\tag{65}
$$

$$
\begin{aligned}
\tau_t^L &= \omega \tau_t^{c,L} + (1-\omega) \tau_t^{o,L} \\
\Rightarrow \tau^L &= \omega \tau^{c,L} + (1-\omega) \tau^{o,L} \\
\Rightarrow (1-\omega) \tau^{o,L} &= \tau^L - \omega \tau^{c,L}
\end{aligned}
\tag{55}
$$

$$\Rightarrow \tau^{o,L} = \frac{\tau^L - \omega \tau^{c,L}}{1-\omega}$$

$$\begin{aligned} c_t^c &= \frac{1}{1+\tau_{c,t}} \left[(1-\tau_{n,t}) w_t n_t^{c,eff} + tr_t^c - \tau_t^{c,L} \right] \\ \Rightarrow c^c &= \frac{1}{1+\tau_c} \left[(1-\tau_n) w n^{c,eff} + tr^c - \tau^{c,L} \right] \end{aligned} \tag{20}$$

$$\begin{aligned} c_t &= \omega c_t^c + (1-\omega) c_t^o \\ \Rightarrow c &= \omega c^c + (1-\omega) c^o \\ \Rightarrow (1-\omega) c^o &= c - \omega c^c \\ \Rightarrow c^o &= \frac{c - \omega c^c}{1-\omega} \end{aligned} \tag{53}$$

$$\begin{aligned} \lambda_t (1+\tau_{c,t}) &= (c_t^o - \hbar c_{t-1}^o)^{-\sigma} - \hbar \beta \mathbb{E}_t \left[(c_{t+1}^o - \hbar c_t^o)^{-\sigma} \right] \\ \Rightarrow \lambda (1+\tau_c) &= (c^o - \hbar c^o)^{-\sigma} - \hbar \beta (c^o - \hbar c^o)^{-\sigma} \\ \Rightarrow \lambda (1+\tau_c) &= (1-\hbar\beta)(c^o - \hbar c^o)^{-\sigma} \\ \Rightarrow \lambda (1+\tau_c) &= (c^o)^{-\sigma} (1-\hbar\beta)(1-\hbar)^{-\sigma} \\ \Rightarrow \lambda &= \frac{(c^o)^{-\sigma}(1-\hbar\beta)(1-\hbar)^{-\sigma}}{1+\tau_c} \end{aligned} \tag{23}$$

$$\begin{aligned} x_{w,d,t} &= \lambda_t (1-\tau_{n,t}) w_t n_t^{eff} + \beta \theta_w \mathbb{E}_t \left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}} \right)^{\varepsilon_w - 1} x_{w,d,t+1} \right] \\ \Rightarrow x_{w,d} &= \lambda (1-\tau_n) w n^{eff} + \beta \theta_w \left(\frac{\pi_w}{(\pi_w^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}} \right)^{\varepsilon_w - 1} x_{w,d} \\ \Rightarrow x_{w,d} &= \lambda (1-\tau_n) w n^{eff} + \beta \theta_w \pi_w^{(1-\gamma_w)(\varepsilon_w - 1)} x_{w,d} \\ \Rightarrow & \left[1 - \beta \theta_w \pi_w^{(1-\gamma_w)(\varepsilon_w - 1)} \right] x_{w,d} = \lambda (1-\tau_n) w n^{eff} \\ \Rightarrow x_{w,d} &= \frac{\lambda (1-\tau_n) w n^{eff}}{1 - \beta \theta_w \pi_w^{(1-\gamma_w)(\varepsilon_w - 1)}} \end{aligned} \tag{31}$$

$$\begin{aligned} (\tilde{w}_t^{opt})^{1+\varepsilon_w \phi} &= \frac{\varepsilon_w}{\varepsilon_w - 1} \frac{x_{w,n,t}}{x_{w,d,t}} \\ \Rightarrow (\tilde{w}^{opt})^{1+\varepsilon_w \phi} &= \frac{\varepsilon_w}{\varepsilon_w - 1} \frac{x_{w,n}}{x_{w,d}} \\ \Rightarrow x_{w,n} &= \frac{\varepsilon_w - 1}{\varepsilon_w} x_{w,d} (\tilde{w}_t^{opt})^{1+\varepsilon_w \phi} \end{aligned} \tag{29}$$

마지막으로 $n = 1/3$ 이 되도록 하는 χ 는 다음과 같다.

$$x_{w,n,t} = \chi \left(n_t^{eff} \right)^{1+\phi} + \beta \theta_w \mathbb{E}_t \left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}} \right)^{\varepsilon_w (1+\phi)} x_{w,n,t+1} \right] \tag{30}$$

$$\Rightarrow x_{w,n} = \chi \left(n^{eff} \right)^{1+\phi} + \beta \theta_w \left(\frac{\pi_w}{(\pi_w^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w}} \right)^{\varepsilon_w (1+\phi)} x_{w,n}$$

$$\Rightarrow \chi \left(n^{eff} \right)^{1+\phi} = \left[1 - \beta \theta_w \pi_w^{(1-\gamma_w)\varepsilon_w (1+\phi)} \right] x_{w,n}$$

$$\Rightarrow \chi = \left[1 - \beta \theta_w \pi_w^{(1-\gamma_w)\varepsilon_w (1+\phi)} \right] \frac{x_{w,n}}{\left(n^{eff} \right)^{1+\phi}}$$

9 제18장 중규모 개방경제 DSGE 모형

본 모형은 생산성($\tilde{z}$)이 정부 자본(k_g)에 의존하고, 자본은 다시 생산(y)에 의존하는 순환 구조를 가지므로, 폐쇄해(closed-form solution)를 한 번에 구할 수 없다. 따라서 생산량 y를 미지수로 두고 시장청산 조건을 만족하는 해를 찾는 수치적 접근이 필요하다.

외생변수 및 파라미터 설정

다음 변수들은 정상상태에서 고정된 값으로 주어진다.

- 인플레이션 및 성장: $\pi_c = \pi$ (목표 인플레이션)
- 노동시간: $n = 1/3$ (목표치, χ로 보정)
- 실질환율: $rer = 1$ (목표치, y^*로 보정)
- 자본 가동률: $u = 1$ (정규화, $\Psi(1) = 0$ 가정)
- 해외변수: π^*, R^*, q_o^*
- 재정 및 통화: $\tau_c, \tau_n, \tau_k, \tau_d, \tau_r, s_{g_c}, s_{\tilde{g}_i}, s_{tr}, s_b, s_{bf}$ 등 비율 변수
- 충격 평균: $z_d = 1, \mu_i = 1, \mu_g = 1, v = 1$ 등

9.1 인플레이션, 이자율, 할인인자

$$
\begin{aligned}
\pi_{d,t} &\equiv \frac{q_{d,t}}{q_{d,t-1}}\pi_{c,t} \\
\Rightarrow \pi_d &= \frac{q_d}{q_d}\pi_c \\
\Rightarrow \pi_d &= \pi_c
\end{aligned}
\tag{98}
$$

$$
\begin{aligned}
\pi_{f,t} &\equiv \frac{q_{f,t}}{q_{f,t-1}}\pi_{c,t} \\
\Rightarrow \pi_f &= \frac{q_f}{q_f}\pi_c \\
\Rightarrow \pi_f &= \pi_c
\end{aligned}
\tag{99}
$$

$$
\begin{aligned}
\pi_{i,t} &\equiv \frac{q_{i,t}}{q_{i,t-1}}\pi_{c,t} \\
\Rightarrow \pi_i &= \frac{q_i}{q_i}\pi_c \\
\Rightarrow \pi_i &= \pi_c
\end{aligned}
\tag{100}
$$

$$\pi_{x,t} \equiv \frac{q_{x,t}}{q_{x,t-1}}\pi_{c,t} \tag{101}$$
$$\Rightarrow \pi_x = \frac{q_x}{q_x}\pi_c$$
$$\Rightarrow \pi_x = \pi_c$$

$$\pi_{w,t} \equiv \frac{w_t}{w_{t-1}}\pi_{c,t} \tag{103}$$
$$\Rightarrow \pi_w = \frac{w}{w}\pi_c$$
$$\Rightarrow \pi_w = \pi_c$$

$$\frac{rer_t}{rer_{t-1}} = \tilde{e}_t\frac{\pi_t^*}{\pi_{c,t}} \tag{104}$$
$$\Rightarrow \frac{rer}{rer} = \tilde{e}\frac{\pi^*}{\pi_c}$$
$$\Rightarrow \tilde{e} = \frac{\pi_c}{\pi^*}$$

$$\pi_{x,t}^* \equiv \frac{q_{x,t}^*}{q_{x,t-1}^*}\frac{\pi_{c,t}}{\tilde{e}_t} \tag{102}$$
$$\Rightarrow \pi_x^* = \frac{q_x^*}{q_x^*}\frac{\pi_c}{\tilde{e}}$$
$$\Rightarrow \pi_x^* = \frac{\pi_c}{\tilde{e}}$$

$$\lambda_t = \beta\mathbb{E}_t\left\{\lambda_{t+1}\tilde{e}_{t+1}\left[\frac{\Phi_t R_t^* - \tau_{r,t+1}\left(\Phi_t R_t^* - 1\right)}{\pi_{c,t+1}}\right]\right\} \tag{7}$$
$$\Rightarrow \lambda = \beta\lambda\tilde{e}\left[\frac{\Phi R^* - \tau_r\left(\Phi R^* - 1\right)}{\pi_c}\right]$$
$$\Rightarrow \Phi R^* - \tau_r\left(\Phi R^* - 1\right) = \frac{\pi_c}{\beta\tilde{e}}$$
$$\Rightarrow (1-\tau_r)\Phi R^* = \frac{\pi_c}{\beta\tilde{e}} - \tau_r$$
$$\Rightarrow \Phi = \frac{1}{R^*}\left[\frac{1}{1-\tau_r}\left(\frac{\pi_c}{\beta\tilde{e}} - \tau_r\right)\right]$$

$$\lambda_t = \beta\mathbb{E}_t\left\{\lambda_{t+1}\left[\frac{R_t - \tau_{r,t+1}\left(R_t - 1\right)}{\pi_{c,t+1}}\right]\right\} \tag{8}$$

$$
\begin{aligned}
&\Rightarrow \lambda = \beta\lambda\left[\frac{R-\tau_r\left(R-1\right)}{\pi_c}\right] \\
&\Rightarrow R-\tau_r\left(R-1\right) = \frac{\pi_c}{\beta} \\
&\Rightarrow (1-\tau_r)R = \frac{\pi_c}{\beta}-\tau_r \\
&\Rightarrow R = \frac{1}{1-\tau_r}\left(\frac{\pi_c}{\beta}-\tau_r\right)
\end{aligned}
$$

$$
\begin{aligned}
Q_{d,t,t+1} &\equiv \beta\frac{\lambda_{t+1}}{\lambda_t}\frac{q_{d,t+1}}{q_{d,t}} \\
\Rightarrow Q_d &= \beta\frac{\lambda}{\lambda}\frac{q_d}{q_d} \\
\Rightarrow Q_d &= \beta
\end{aligned} \tag{95}
$$

$$
\begin{aligned}
Q_{f,t,t+1} &\equiv \beta\frac{\lambda_{t+1}}{\lambda_t}\frac{q_{f,t+1}}{q_{f,t}} \\
\Rightarrow Q_f &= \beta\frac{\lambda}{\lambda}\frac{q_f}{q_f} \\
\Rightarrow Q_f &= \beta
\end{aligned} \tag{96}
$$

$$
\begin{aligned}
Q_{x,t,t+1} &\equiv \beta\frac{\lambda_{t+1}}{\lambda_t}\frac{q^*_{x,t+1}}{q^*_{x,t}} \\
\Rightarrow Q_x &= \beta\frac{\lambda}{\lambda}\frac{q^*_x}{q^*_x} \\
\Rightarrow Q_x &= \beta
\end{aligned} \tag{97}
$$

9.2 최적 가격/임금 및 분산/유효 노동/상대가격

$$
\begin{aligned}
1 &= (1-\theta_d)\left(\tilde{p}^{opt}_{d,t}\right)^{1-\varepsilon_d} + \theta_d\left[\frac{(\pi^{\iota_d}_{d,t-1}\pi_d^{1-\iota_d})^{\gamma^{idx}_d}}{\pi_{d,t}}\right]^{1-\varepsilon_d} \\
\Rightarrow 1 &= (1-\theta_d)\left(\tilde{p}^{opt}_{d}\right)^{1-\varepsilon_d} + \theta_d\left[\frac{(\pi^{\iota_d}_{d}\pi_d^{1-\iota_d})^{\gamma^{idx}_d}}{\pi_{d}}\right]^{1-\varepsilon_d} \\
\Rightarrow (1-\theta_d)\left(\tilde{p}^{opt}_{d}\right)^{1-\varepsilon_d} &= 1-\theta_d\pi_d^{(\gamma^{idx}_d-1)(1-\varepsilon_d)} \\
\Rightarrow \left(\tilde{p}^{opt}_{d}\right)^{1-\varepsilon_d} &= \frac{1-\theta_d\pi_d^{(\gamma^{idx}_d-1)(1-\varepsilon_d)}}{1-\theta_d} \\
\Rightarrow \tilde{p}^{opt}_{d} &= \left[\frac{1-\theta_d\pi_d^{(\gamma^{idx}_d-1)(1-\varepsilon_d)}}{1-\theta_d}\right]^{\frac{1}{1-\varepsilon_d}}
\end{aligned} \tag{64}
$$

$$x_{d,n,t} = m_{d,t} mc_t^{(d)} + \theta_d \mathbb{E}_t \left[Q_{d,t,t+1} \left(\frac{\pi_{d,t+1}}{(\pi_{d,t}^{\iota_d} \pi_d^{1-\iota_d})^{\gamma_d^{idx}}} \right)^{\varepsilon_d} x_{d,n,t+1} \right] \tag{16}$$

$$\Rightarrow x_{d,n} = m_d mc^{(d)} + \theta_d Q_d \left(\frac{\pi_d}{(\pi_d^{\iota_d} \pi_d^{1-\iota_d})^{\gamma_d^{idx}}} \right)^{\varepsilon_d} x_{d,n}$$

$$\Rightarrow x_{d,n} = m_d mc^{(d)} + \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})} x_{d,n}$$

$$\Rightarrow \left(1 - \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})} \right) x_{d,n} = m_d mc^{(d)}$$

$$\Rightarrow x_{d,n} = \frac{m_d mc^{(d)}}{1 - \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})}}$$

$$x_{d,d,t} = m_{d,t} + \theta_d \mathbb{E}_t \left[Q_{d,t,t+1} \left(\frac{\pi_{d,t+1}}{(\pi_{d,t}^{\iota_d} \pi_d^{1-\iota_d})^{\gamma_d^{idx}}} \right)^{\varepsilon_d - 1} x_{d,d,t+1} \right] \tag{17}$$

$$\Rightarrow x_{d,d} = m_d + \theta_d Q_d \left(\frac{\pi_d}{(\pi_d^{\iota_d} \pi_d^{1-\iota_d})^{\gamma_d^{idx}}} \right)^{\varepsilon_d - 1} x_{d,d}$$

$$\Rightarrow x_{d,d} = m_d + \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})} x_{d,d}$$

$$\Rightarrow \left(1 - \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})} \right) x_{d,d} = m_d$$

$$\Rightarrow x_{d,d} = \frac{m_d}{1 - \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})}}$$

$$\tilde{p}_{d,t}^{opt} = \left(\frac{\varepsilon_d}{\varepsilon_d - 1} \mu_{d,t} \right) \frac{x_{d,n,t}}{x_{d,d,t}} \tag{15}$$

$$\Rightarrow \tilde{p}_d^{opt} = \left(\frac{\varepsilon_d}{\varepsilon_d - 1} \underbrace{\mu_d}_{=1} \right) \frac{x_{d,n}}{x_{d,d}}$$

$$\Rightarrow \tilde{p}_d^{opt} = \frac{\varepsilon_d}{\varepsilon_d - 1} \frac{\frac{m_d mc^{(d)}}{1 - \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})}}}{\frac{m_d}{1 - \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})}}}$$

$$\Rightarrow \tilde{p}_d^{opt} = \frac{\varepsilon_d}{\varepsilon_d - 1} \frac{1 - \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})}}{1 - \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})}} mc^{(d)}$$

$$\Rightarrow mc^{(d)} = \tilde{p}_d^{opt} \left(\frac{\varepsilon_d - 1}{\varepsilon_d} \right) \left(\frac{1 - \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})}}{1 - \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})}} \right)$$

$$\Delta_{d,t} = (1 - \theta_d)(\tilde{p}_{d,t}^{opt})^{-\varepsilon_d} + \theta_d \left(\frac{(\pi_{d,t-1}^{\iota_d} \pi_d^{1-\iota_d})^{\gamma_d^{idx}}}{\pi_{d,t}} \right)^{-\varepsilon_d} \Delta_{d,t-1} \tag{70}$$

$$
\begin{aligned}
&\Rightarrow \Delta_d = (1-\theta_d)(\tilde{p}_d^{opt})^{-\varepsilon_d} + \theta_d \left(\frac{(\pi_d^{\iota_d} \pi_d^{1-\iota_d})^{\gamma_d^{idx}}}{\pi_d} \right)^{-\varepsilon_d} \Delta_d \\
&\Rightarrow \Delta_d = (1-\theta_d)(\tilde{p}_d^{opt})^{-\varepsilon_d} + \theta_d \left(\pi_d\right)^{\varepsilon_d(1-\gamma_d^{idx})} \Delta_d \\
&\Rightarrow \left[1 - \theta_d \left(\pi_d\right)^{\varepsilon_d(1-\gamma_d^{idx})} \right] \Delta_d = (1-\theta_d)(\tilde{p}_d^{opt})^{-\varepsilon_d} \\
&\Rightarrow \Delta_d = \frac{(1-\theta_d)}{1 - \theta_d \left(\pi_d\right)^{\varepsilon_d(1-\gamma_d^{idx})}} (\tilde{p}_d^{opt})^{-\varepsilon_d}
\end{aligned}
$$

정상상태에서 $\pi_f = \pi_c$ 이므로, 다음이 성립한다.

$$
\begin{aligned}
&1 = (1-\theta_f) \left(\tilde{p}_{f,t}^{opt} \right)^{1-\varepsilon_f} + \theta_f \left[\frac{(\pi_{f,t-1}^{\iota_f} \pi_c^{1-\iota_f})^{\gamma_f^{idx}}}{\pi_{f,t}} \right]^{1-\varepsilon_f} \qquad (65) \\
&\Rightarrow 1 = (1-\theta_f) \left(\tilde{p}_f^{opt} \right)^{1-\varepsilon_f} + \theta_f \left[\frac{(\pi_f^{\iota_f} \pi_c^{1-\iota_f})^{\gamma_f^{idx}}}{\pi_f} \right]^{1-\varepsilon_f} \\
&\Rightarrow \left(\tilde{p}_f^{opt} \right)^{1-\varepsilon_f} = \frac{1 - \theta_f \pi_f^{(\gamma_f^{idx}-1)(1-\varepsilon_f)}}{1-\theta_f} \\
&\Rightarrow \tilde{p}_f^{opt} = \left[\frac{1 - \theta_f \pi_f^{(\gamma_f^{idx}-1)(1-\varepsilon_f)}}{1-\theta_f} \right]^{\frac{1}{1-\varepsilon_f}}
\end{aligned}
$$

$$
\begin{aligned}
&x_{f,n,t} = m_{f,t} \left(\frac{rer_t}{q_{f,t}} \right) + \theta_f \mathbb{E}_t \left[Q_{f,t,t+1} \left(\frac{\pi_{f,t+1}}{(\pi_{f,t}^{\iota_f} \pi_c^{1-\iota_f})^{\gamma_f^{idx}}} \right)^{\varepsilon_f} x_{f,n,t+1} \right] \qquad (19) \\
&\Rightarrow x_{f,n} = m_f \left(\frac{rer}{q_f} \right) + \theta_f Q_f \left(\frac{\pi_f}{(\pi_f^{\iota_f} \pi_c^{1-\iota_f})^{\gamma_f^{idx}}} \right)^{\varepsilon_f} x_{f,n} \\
&\Rightarrow x_{f,n} = m_f \left(\frac{rer}{q_f} \right) + \theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})} x_{f,n} \\
&\Rightarrow \left[1 - \theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})} \right] x_{f,n} = m_f \left(\frac{rer}{q_f} \right) \\
&\Rightarrow x_{f,n} = \frac{m_f \left(\frac{rer}{q_f} \right)}{1 - \theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})}}
\end{aligned}
$$

$$
\begin{aligned}
&x_{f,d,t} = m_{f,t} + \theta_f \mathbb{E}_t \left[Q_{f,t,t+1} \left(\frac{\pi_{f,t+1}}{(\pi_{f,t}^{\iota_f} \pi_c^{1-\iota_f})^{\gamma_f^{idx}}} \right)^{\varepsilon_f - 1} x_{f,d,t+1} \right] \qquad (20) \\
&\Rightarrow x_{f,d} = m_f + \theta_f Q_f \left(\frac{\pi_f}{(\pi_f^{\iota_f} \pi_c^{1-\iota_f})^{\gamma_f^{idx}}} \right)^{\varepsilon_f - 1} x_{f,d}
\end{aligned}
$$

$$\Rightarrow x_{f,d} = m_f + \theta_f Q_f \pi_f^{(\varepsilon_f - 1)(1-\gamma_f^{idx})} x_{f,d}$$
$$\Rightarrow \left[1 - \theta_f Q_f \pi_f^{(\varepsilon_f - 1)(1-\gamma_f^{idx})}\right] x_{f,d} = m_f$$
$$\Rightarrow x_{f,d} = \frac{m_f}{1 - \theta_f Q_f \pi_f^{(\varepsilon_f - 1)(1-\gamma_f^{idx})}}$$

$$\tilde{p}_{f,t}^{opt} = \left(\frac{\varepsilon_f}{\varepsilon_f - 1}\mu_{f,t}\right)\frac{x_{f,n,t}}{x_{f,d,t}} \tag{18}$$
$$\Rightarrow \tilde{p}_f^{opt} = \left(\frac{\varepsilon_f}{\varepsilon_f - 1}\underbrace{\mu_f}_{=1}\right)\frac{x_{f,n}}{x_{f,d}}$$
$$\Rightarrow \tilde{p}_f^{opt} = \frac{\varepsilon_f}{\varepsilon_f - 1}\frac{\frac{m_f\left(\frac{rer}{q_f}\right)}{1-\theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})}}}{\frac{m_f}{1-\theta_f Q_f \pi_f^{(\varepsilon_f-1)(1-\gamma_f^{idx})}}}$$
$$\Rightarrow \tilde{p}_f^{opt} = \frac{\varepsilon_f}{\varepsilon_f - 1}\frac{rer}{q_f}\frac{1 - \theta_f Q_f \pi_f^{(\varepsilon_f-1)(1-\gamma_f^{idx})}}{1 - \theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})}}$$
$$\Rightarrow q_f = \frac{\varepsilon_f}{\varepsilon_f - 1}\frac{rer}{\tilde{p}_f^{opt}}\frac{1 - \theta_f Q_f \pi_f^{(\varepsilon_f-1)(1-\gamma_f^{idx})}}{1 - \theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})}}$$

$$\Delta_{f,t} = (1-\theta_f)(\tilde{p}_{f,t}^{opt})^{-\varepsilon_f} + \theta_f\left(\frac{(\pi_{f,t-1}^{\iota_f}\pi_c^{1-\iota_f})^{\gamma_f^{idx}}}{\pi_{f,t}}\right)^{-\varepsilon_f}\Delta_{f,t-1} \tag{72}$$
$$\Rightarrow \Delta_f = (1-\theta_f)(\tilde{p}_f^{opt})^{-\varepsilon_f} + \theta_f\left(\frac{(\pi_f^{\iota_f}\pi_c^{1-\iota_f})^{\gamma_f^{idx}}}{\pi_f}\right)^{-\varepsilon_f}\Delta_f$$
$$\Rightarrow \Delta_f = (1-\theta_f)(\tilde{p}_f^{opt})^{-\varepsilon_f} + \theta_f \pi_f^{-\varepsilon_f(\gamma_f^{idx}-1)}\Delta_f$$
$$\Rightarrow \left[1 - \theta_f \pi_f^{-\varepsilon_f(\gamma_f^{idx}-1)}\right]\Delta_f = (1-\theta_f)(\tilde{p}_f^{opt})^{-\varepsilon_f}$$
$$\Rightarrow \Delta_f = \frac{(1-\theta_f)}{1 - \theta_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})}}(\tilde{p}_f^{opt})^{-\varepsilon_f}$$

$$1 = (1-\theta_x)\left(\tilde{p}_{x,t}^{*,opt}\right)^{1-\varepsilon_x} + \theta_x\left[\frac{(\pi_{x,t-1}^{*\iota_x}\pi_x^{*(1-\iota_x)})^{\gamma_x^{idx}}}{\pi_{x,t}^*}\right]^{1-\varepsilon_x} \tag{66}$$
$$\Rightarrow 1 = (1-\theta_x)\left(\tilde{p}_x^{*,opt}\right)^{1-\varepsilon_x} + \theta_x\left[\frac{(\pi_x^{*\iota_x}\pi_x^{*(1-\iota_x)})^{\gamma_x^{idx}}}{\pi_x^*}\right]^{1-\varepsilon_x}$$

$$\Rightarrow 1 = (1-\theta_x)\left(\tilde{p}_x^{*,opt}\right)^{1-\varepsilon_x} + \theta_x \left(\pi_x^*\right)^{(\gamma_x^{idx}-1)(1-\varepsilon_x)}$$
$$\Rightarrow (1-\theta_x)\left(\tilde{p}_x^{*,opt}\right)^{1-\varepsilon_x} = 1 - \theta_x \left(\pi_x^*\right)^{(\gamma_x^{idx}-1)(1-\varepsilon_x)}$$
$$\Rightarrow \tilde{p}_x^{*,opt} = \left[\frac{1-\theta_x \left(\pi_x^*\right)^{(\gamma_x^{idx}-1)(1-\varepsilon_x)}}{1-\theta_x}\right]^{\frac{1}{1-\varepsilon_x}}$$

$$x_{x,n,t} = x_t \frac{q_{x,t}}{q_{x,t}^*} + \theta_x \mathbb{E}_t \left[Q_{x,t,t+1} \left(\frac{\pi_{x,t+1}^*}{(\pi_{x,t}^{*\iota_x} \pi_x^{*1-\iota_x})^{\gamma_x^{idx}}} \right)^{\varepsilon_x} x_{x,n,t+1} \right] \tag{41}$$
$$\Rightarrow x_{x,n} = x \frac{q_x}{q_x^*} + \theta_x Q_x \left(\frac{\pi_x^*}{(\pi_x^{*\iota_x} \pi_x^{*1-\iota_x})^{\gamma_x^{idx}}} \right)^{\varepsilon_x} x_{x,n}$$
$$\Rightarrow x_{x,n} = x \frac{q_x}{q_x^*} + \theta_x Q_x \left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})} x_{x,n}$$
$$\Rightarrow \left(1 - \theta_x Q_x \left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}\right) x_{x,n} = x \frac{q_x}{q_x^*}$$
$$\Rightarrow x_{x,n} = \frac{x \frac{q_x}{q_x^*}}{1 - \theta_x Q_x \left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}}$$

$$x_{x,d,t} = x_t + \theta_x \mathbb{E}_t \left[Q_{x,t,t+1} \left(\frac{\pi_{x,t+1}^*}{(\pi_{x,t}^{*\iota_x} \pi_x^{*1-\iota_x})^{\gamma_x^{idx}}} \right)^{\varepsilon_x - 1} x_{x,d,t+1} \right] \tag{42}$$
$$\Rightarrow x_{x,d} = x + \theta_x Q_x \left(\frac{\pi_x^*}{(\pi_x^{*\iota_x} \pi_x^{*1-\iota_x})^{\gamma_x^{idx}}} \right)^{\varepsilon_x - 1} x_{x,d}$$
$$\Rightarrow x_{x,d} = x + \theta_x Q_x \left(\pi_x^*\right)^{(\varepsilon_x - 1)(1-\gamma_x^{idx})} x_{x,d}$$
$$\Rightarrow \left(1 - \theta_x Q_x \left(\pi_x^*\right)^{(\varepsilon_x - 1)(1-\gamma_x^{idx})}\right) x_{x,d} = x$$
$$\Rightarrow x_{x,d} = \frac{x}{1 - \theta_x Q_x \left(\pi_x^*\right)^{(\varepsilon_x - 1)(1-\gamma_x^{idx})}}$$

$$\Delta_{x,t} = (1-\theta_x)(\tilde{p}_{x,t}^{*,opt})^{-\varepsilon_x} + \theta_x \left(\frac{(\pi_{x,t-1}^{*\iota_x} \pi_x^{*(1-\iota_x)})^{\gamma_x^{idx}}}{\pi_{x,t}^*} \right)^{-\varepsilon_x} \Delta_{x,t-1} \tag{74}$$
$$\Rightarrow s_x = (1-\theta_x)(\tilde{p}_x^{*,opt})^{-\varepsilon_x} + \theta_x \left(\frac{(\pi_x^{*\iota_x} \pi_x^{*(1-\iota_x)})^{\gamma_x^{idx}}}{\pi_x^*} \right)^{-\varepsilon_x} s_x$$
$$\Rightarrow s_x = (1-\theta_x)(\tilde{p}_x^{*,opt})^{-\varepsilon_x} + \theta_x \left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})} s_x$$
$$\Rightarrow \left[1 - \theta_x \left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}\right] s_x = (1-\theta_x)(\tilde{p}_x^{*,opt})^{-\varepsilon_x}$$
$$\Rightarrow s_x = \frac{(1-\theta_x)}{1 - \theta_x \left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}} (\tilde{p}_x^{*,opt})^{-\varepsilon_x}$$

정상상태에서 $\pi_w = \pi_c$ 이므로 다음이 성립한다.

$$1 = (1-\theta_w)(\tilde{w}_t^{opt})^{1-\varepsilon_w} + \theta_w \left[\frac{(\pi_{w,t-1}^{\iota_w} \pi_c^{1-\iota_w})^{\gamma_w^{idx}}}{\pi_{w,t}} \right]^{1-\varepsilon_w} \tag{63}$$

$$
\begin{aligned}
&\Rightarrow 1 = (1-\theta_w)(\tilde{w}^{opt})^{1-\varepsilon_w} + \theta_w \left[\frac{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w^{idx}}}{\pi_w}\right]^{1-\varepsilon_w} \\
&\Rightarrow (1-\theta_w)(\tilde{w}^{opt})^{1-\varepsilon_w} = 1 - \theta_w \pi_w^{(1-\varepsilon_w)(\gamma_w^{idx}-1)} \\
&\Rightarrow (\tilde{w}^{opt})^{1-\varepsilon_w} = \frac{1-\theta_w \pi_w^{(1-\varepsilon_w)(\gamma_w^{idx}-1)}}{1-\theta_w} \\
&\Rightarrow \tilde{w}^{opt} = \left[\frac{1-\theta_w \pi_w^{(1-\varepsilon_w)(\gamma_w^{idx}-1)}}{1-\theta_w}\right]^{\frac{1}{1-\varepsilon_w}}
\end{aligned}
$$

$$
\Delta_{w,t} = (1-\theta_w)(\tilde{w}_t^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_{w,t-1}^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w^{idx}}}{\pi_{w,t}}\right)^{-\varepsilon_w} \Delta_{w,t-1} \tag{68}
$$

$$
\begin{aligned}
&\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \left(\frac{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w^{idx}}}{\pi_w}\right)^{-\varepsilon_w} \Delta_w \\
&\Rightarrow \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} + \theta_w \pi_w^{-\varepsilon_w(\gamma_w^{idx}-1)} \Delta_w \\
&\Rightarrow \left[1 - \theta_w \pi_w^{-\varepsilon_w(\gamma_w^{idx}-1)}\right] \Delta_w = (1-\theta_w)(\tilde{w}^{opt})^{-\varepsilon_w} \\
&\Rightarrow \Delta_w = \frac{(1-\theta_w)}{1-\theta_w \pi_w^{-\varepsilon_w(\gamma_w^{idx}-1)}} (\tilde{w}^{opt})^{-\varepsilon_w}
\end{aligned}
$$

$$
\begin{aligned}
n_t^{eff} &= \frac{n_t}{\Delta_{w,t}} \\
\Rightarrow n^{eff} &= \frac{n}{\Delta_w}
\end{aligned} \tag{67}
$$

$$
\begin{aligned}
n_t^{c,eff} &= n_t^{o,eff} \\
\Rightarrow n^{c,eff} &= n^{o,eff}
\end{aligned} \tag{94}
$$

$$
\begin{aligned}
&n_t^{eff} = \omega n_t^{c,eff} + (1-\omega) n_t^{o,eff} \\
&\Rightarrow n^{eff} = \omega n^{c,eff} + (1-\omega) n^{o,eff} \\
&\Rightarrow n^{eff} = \omega n^{o,eff} + (1-\omega) n^{o,eff} = \omega n^{c,eff} + (1-\omega) n^{c,eff} \\
&\Rightarrow n^{o,eff} = n^{c,eff} = n^{eff}
\end{aligned} \tag{81}
$$

$$
\begin{aligned}
q_{o,t} &= rer_t q_{o,t}^* \\
\Rightarrow q_o &= rer q_o^*
\end{aligned} \tag{105}
$$

9.3 정부자본 k_g 도출

이 알고리즘은 단일 변수 k_g에 대한 비선형 방정식 $F(k_g) = 0$을 푸는 문제이다.

1단계: 공공자본 가정 (Guess k_g)

$$k_g^{guess} = \text{Initial Value (e.g., 0.5)}$$

2단계: 유효 총요소생산성($\tilde{z}$) 확정

k_g가 주어졌으므로, 모든 부문의 생산성이 고정된다.

$$\begin{aligned}
&\tilde{z}_d = z_d(k_g^{guess})^{\alpha_{g,d}},\\
&\tilde{z}_c^m = z_c^m(k_g^{guess})^{\alpha_{g,c}^m}, \quad \tilde{z}_i^m = z_i^m(k_g^{guess})^{\alpha_{g,i}^m}, \quad \tilde{z}_x^m = z_x^m(k_g^{guess})^{\alpha_{g,x}^m}\\
&\tilde{z}_c = z_c(k_g^{guess})^{\alpha_{g,c}}, \quad \tilde{z}_i = z_i(k_g^{guess})^{\alpha_{g,i}}, \quad \tilde{z}_x = z_x(k_g^{guess})^{\alpha_{g,x}}
\end{aligned}$$

3단계: 상대가격 시스템(q_d, q_i, q_x) 도출

생산성($\tilde{z}$)과 수입재/유가(q_f, q_o)가 모두 확정되었으므로, 최종재 가격 식을 역산하여 국내 중간재 가격 q_d를 찾아낸다.

$$\begin{aligned}
1 &= \frac{1}{\tilde{z}_{c,t}}\left[\gamma_c^o\left(q_{c,t}^m\right)^{1-\xi_c^o} + (1-\gamma_c^o)\left(q_{o,t}\right)^{1-\xi_c^o}\right]^{\frac{1}{1-\xi_c^o}} \qquad (26)\\
\Rightarrow 1 &= \frac{1}{\tilde{z}_c}\left[\gamma_c^o\left(q_c^m\right)^{1-\xi_c^o} + (1-\gamma_c^o)\left(q_o\right)^{1-\xi_c^o}\right]^{\frac{1}{1-\xi_c^o}}\\
\Rightarrow \tilde{z}_c &= \left[\gamma_c^o\left(q_c^m\right)^{1-\xi_c^o} + (1-\gamma_c^o)\left(q_o\right)^{1-\xi_c^o}\right]^{\frac{1}{1-\xi_c^o}}\\
\Rightarrow \tilde{z}_c^{1-\xi_c^o} &= \gamma_c^o\left(q_c^m\right)^{1-\xi_c^o} + (1-\gamma_c^o)\left(q_o\right)^{1-\xi_c^o}\\
\Rightarrow \gamma_c^o\left(q_c^m\right)^{1-\xi_c^o} &= \tilde{z}_c^{1-\xi_c^o} - (1-\gamma_c^o)\left(q_o\right)^{1-\xi_c^o}\\
\Rightarrow \left(q_c^m\right)^{1-\xi_c^o} &= \frac{\tilde{z}_c^{1-\xi_c^o} - (1-\gamma_c^o)\left(q_o\right)^{1-\xi_c^o}}{\gamma_c^o}\\
\Rightarrow q_c^m &= \left[\frac{\tilde{z}_c^{1-\xi_c^o} - (1-\gamma_c^o)\left(q_o\right)^{1-\xi_c^o}}{\gamma_c^o}\right]^{\frac{1}{1-\xi_c^o}}
\end{aligned}$$

$$\begin{aligned}
q_{c,t}^m &= \frac{1}{\tilde{z}_{c,t}^m}\left[\gamma_c(q_{d,t})^{1-\xi_c} + (1-\gamma_c)\left(q_{f,t}\right)^{1-\xi_c}\right]^{\frac{1}{1-\xi_c}} \qquad (23)\\
\Rightarrow q_c^m &= \frac{1}{\tilde{z}_c^m}\left[\gamma_c(q_d)^{1-\xi_c} + (1-\gamma_c)\left(q_f\right)^{1-\xi_c}\right]^{\frac{1}{1-\xi_c}}\\
\Rightarrow \tilde{z}_c^m q_c^m &= \left[\gamma_c(q_d)^{1-\xi_c} + (1-\gamma_c)\left(q_f\right)^{1-\xi_c}\right]^{\frac{1}{1-\xi_c}}\\
\Rightarrow \left(\tilde{z}_c^m q_c^m\right)^{1-\xi_c} &= \gamma_c(q_d)^{1-\xi_c} + (1-\gamma_c)\left(q_f\right)^{1-\xi_c}\\
\Rightarrow \gamma_c(q_d)^{1-\xi_c} &= \left(\tilde{z}_c^m q_c^m\right)^{1-\xi_c} - (1-\gamma_c)\left(q_f\right)^{1-\xi_c}\\
\Rightarrow (q_d)^{1-\xi_c} &= \frac{\left(\tilde{z}_c^m q_c^m\right)^{1-\xi_c} - (1-\gamma_c)\left(q_f\right)^{1-\xi_c}}{\gamma_c}
\end{aligned}$$

$$\Rightarrow q_d = \left[\frac{\left(\tilde{z}_c^m q_c^m\right)^{1-\xi_c} - (1-\gamma_c)\left(q_f\right)^{1-\xi_c}}{\gamma_c} \right]^{\frac{1}{1-\xi_c}}$$

$$q_{i,t}^m = \frac{1}{\tilde{z}_{i,t}^m} \left[\gamma_i (q_{d,t})^{1-\xi_i} + (1-\gamma_i)\left(q_{f,t}\right)^{1-\xi_i} \right]^{\frac{1}{1-\xi_i}} \tag{29}$$

$$\Rightarrow q_i^m = \frac{1}{\tilde{z}_i^m} \left[\gamma_i (q_d)^{1-\xi_i} + (1-\gamma_i)\left(q_f\right)^{1-\xi_i} \right]^{\frac{1}{1-\xi_i}}$$

$$q_{i,t} = \frac{1}{\tilde{z}_{i,t}} \left[\gamma_i^o \left(q_{i,t}^m\right)^{1-\xi_i^o} + (1-\gamma_i^o)\left(q_{o,t}\right)^{1-\xi_i^o} \right]^{\frac{1}{1-\xi_i^o}} \tag{32}$$

$$\Rightarrow q_i = \frac{1}{\tilde{z}_i} \left[\gamma_i^o \left(q_i^m\right)^{1-\xi_i^o} + (1-\gamma_i^o)\left(q_o\right)^{1-\xi_i^o} \right]^{\frac{1}{1-\xi_i^o}}$$

$$q_{x,t}^m = \frac{1}{\tilde{z}_{x,t}^m} \left[\gamma_x (q_{d,t})^{1-\xi_x} + (1-\gamma_x)\left(q_{f,t}\right)^{1-\xi_x} \right]^{\frac{1}{1-\xi_x}} \tag{35}$$

$$\Rightarrow q_x^m = \frac{1}{\tilde{z}_x^m} \left[\gamma_x (q_d)^{1-\xi_x} + (1-\gamma_x)\left(q_f\right)^{1-\xi_x} \right]^{\frac{1}{1-\xi_x}}$$

$$q_{x,t} = \frac{1}{\tilde{z}_{x,t}} \left[\gamma_x^o \left(q_{x,t}^m\right)^{1-\xi_x^o} + (1-\gamma_x^o)\left(q_{o,t}\right)^{1-\xi_x^o} \right]^{\frac{1}{1-\xi_x^o}} \tag{38}$$

$$\Rightarrow q_x = \frac{1}{\tilde{z}_x} \left[\gamma_x^o \left(q_x^m\right)^{1-\xi_x^o} + (1-\gamma_x^o)\left(q_o\right)^{1-\xi_x^o} \right]^{\frac{1}{1-\xi_x^o}}$$

4단계: 생산요소가격 및 민간 자본(k, $\bar{k}$)

$r^k \Rightarrow w \Rightarrow k$ 순으로 구한다.

$$\begin{aligned} \lambda_t q_{i,t} = {} & \xi_t \left\{ \left[1 - \frac{\psi}{2} \left(\frac{i_t^o}{i_{t-1}^o} - 1 \right)^2 \right] \mu_{i,t} - \psi \left(\frac{i_t^o}{i_{t-1}^o} - 1 \right) \frac{i_t^o}{i_{t-1}^o} \mu_{i,t} \right\} \\ & + \beta \mathbb{E}_t \left\{ \xi_{t+1} \left[\psi \left(\frac{i_{t+1}^o}{i_t^o} - 1 \right) \left(\frac{i_{t+1}^o}{i_t^o} \right)^2 \mu_{i,t+1} \right] \right\} \end{aligned} \tag{9}$$

$$\Rightarrow \lambda q_i = \xi \left\{ \left[1 - \frac{\psi}{2} \left(\frac{i^o}{i^o} - 1 \right)^2 \right] \mu_i - \psi \left(\frac{i^o}{i^o} - 1 \right) \frac{i^o}{i^o} \mu_i \right\} + \beta \xi \left[\psi \left(\frac{i^o}{i^o} - 1 \right) \left(\frac{i^o}{i^o} \right)^2 \mu_i \right]$$

$$\Rightarrow \lambda q_i = \xi \mu_i$$

$$\Rightarrow \xi = \frac{\lambda q_i}{\mu_i}$$

$$\xi_t = \beta \mathbb{E}_t \left[\lambda_{t+1} \left((1 - \tau_{k,t+1}) r_{t+1}^k u_{t+1} - \Psi(u_{t+1}) \right) + \xi_{t+1}(1-\delta) \right] \tag{11}$$

$$
\begin{aligned}
&\Rightarrow \xi = \beta\lambda \left((1-\tau_k) r^k \underbrace{u}_{=1} - \underbrace{\Psi(u)}_{=\Psi(1)=0} \right) + \beta\xi(1-\delta) \\
&\Rightarrow \beta\lambda(1-\tau_k) r^k = \xi \left[1 - \beta(1-\delta)\right] \\
&\Rightarrow \beta\lambda(1-\tau_k) r^k = \frac{\lambda q_i}{\mu_i} \left[1 - \beta(1-\delta)\right] \\
&\Rightarrow r^k = \frac{q_i}{\beta(1-\tau_k)} \frac{1}{\mu_i} \left[1 - \beta(1-\delta)\right]
\end{aligned}
$$

한편 균형조건 식에는 포함되지 않지만 본문의 한계비용 정의식으로 부터 다음을 알 수 있다[2].

$$
\begin{aligned}
mc_t^{(c)} &= \frac{1}{\tilde{z}_{d,t}} \left(\frac{r_t^k}{\alpha}\right)^{\alpha} \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha} \\
\Rightarrow q_{d,t} mc_t^{(d)} &= \frac{1}{\tilde{z}_{d,t}} \left(\frac{r_t^k}{\alpha}\right)^{\alpha} \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha} \\
\Rightarrow q_d mc^{(d)} &= \frac{1}{\tilde{z}_d} \left(\frac{r^k}{\alpha}\right)^{\alpha} \left(\frac{w}{1-\alpha}\right)^{1-\alpha} \\
\Rightarrow \left(\frac{w}{1-\alpha}\right)^{1-\alpha} &= q_d mc^{(d)} \tilde{z}_d \left(\frac{\alpha}{r^k}\right)^{\alpha} \\
\Rightarrow \frac{w}{1-\alpha} &= \left[q_d mc^{(d)} \tilde{z}_d \left(\frac{\alpha}{r^k}\right)^{\alpha}\right]^{\frac{1}{1-\alpha}} \\
\Rightarrow w &= (1-\alpha) \left[q_d mc^{(d)} \tilde{z}_d \left(\frac{\alpha}{r^k}\right)^{\alpha}\right]^{\frac{1}{1-\alpha}}
\end{aligned}
$$

$$
\begin{aligned}
r_t^k &= \alpha mc_t^{(d)} q_{d,t} \frac{m_{d,t}\Delta_{d,t}}{k_t} \\
\Rightarrow r^k &= \alpha mc^{(d)} q_d \frac{m_d \Delta_d}{k}
\end{aligned} \tag{75}
$$

$$
\begin{aligned}
w_t &= (1-\alpha) mc_t^{(d)} q_{d,t} \frac{m_{d,t}\Delta_{d,t}}{n_t^{eff}} \\
\Rightarrow w &= (1-\alpha) mc^{(d)} q_d \frac{m_d \Delta_d}{n^{eff}} \\
\Rightarrow w &= (1-\alpha) mc^{(d)} q_d \frac{m_d \Delta_d}{n^{eff}}
\end{aligned} \tag{76}
$$

위 두 식으로부터 $k, \bar{k}$를 구할 수 있다.

$$
\frac{w}{r^k} = \frac{(1-\alpha) mc^{(d)} q_d \frac{m_d \Delta_d}{n^{eff}}}{\alpha mc^{(d)} q_d \frac{m_d \Delta_d}{k}}
$$

[2] 한계비용 정의식은 다른 균형조건 식들로 나타낼 수 있으므로 중복을 피하기 위해 균형조건 식에 포함되지 않지만 정상상태 값을 구하는 데에는 사용할 수 있다.

$$
\begin{aligned}
&\Rightarrow \frac{w}{r^k} = \frac{(1-\alpha)}{\alpha}\frac{k}{n^{eff}} \\
&\Rightarrow \frac{k}{n^{eff}} = \frac{\alpha}{1-\alpha}\frac{w}{r^k} \\
&\Rightarrow k = \left(\frac{\alpha}{1-\alpha}\frac{w}{r^k}\right) n^{eff}
\end{aligned}
$$

$$
\begin{aligned}
k_t &= u_t \bar{k}_{t-1} \\
\Rightarrow k &= u\bar{k} \\
\Rightarrow \bar{k} &= \frac{k}{u}
\end{aligned} \tag{83}
$$

정상상태에서는 가동률 $u = 1$ 이고 가동비용 $\Psi(1) = 0$ 이므로, 유효 자본 k와 물리적 자본 $\bar{k}$는 동일하다. 또한 오일러 방정식에서 가동비용 항이 사라지므로 자본수익률 결정식은 단순 모형과 동일하다.

5단계: 총생산(y) 및 새로운 공공자본(k_g^{new}) 도출

$$
\begin{aligned}
y_t &\equiv m_{d,t} \\
\Rightarrow y &= m_d
\end{aligned} \tag{118}
$$

$$
\begin{aligned}
m_{d,t} &= \frac{\tilde{z}_{d,t} k_t^{\alpha} (n_t^{eff})^{1-\alpha}}{\Delta_{d,t}} \\
\Rightarrow \underbrace{m_d}_{=y} &= \frac{\tilde{z}_d k^{\alpha} (n^{eff})^{1-\alpha}}{\Delta_d} \\
\Rightarrow y &= \frac{\tilde{z}_d k^{\alpha} (n^{eff})^{1-\alpha}}{\Delta_d}
\end{aligned} \tag{69}
$$

$$
\begin{aligned}
\tilde{g}_{i,t} &\equiv q_{i,t} g_{i,t} \\
\Rightarrow \tilde{g}_i &= q_i g_i
\end{aligned} \tag{49}
$$

$$
\begin{aligned}
s_{\tilde{g}_i,t} &\equiv \frac{\tilde{g}_{i,t}}{q_{d,t} y_t} \\
\Rightarrow s_{\tilde{g}_i} &= \frac{\tilde{g}_i}{q_d y} \\
\Rightarrow \tilde{g}_i &= s_{\tilde{g}_i} q_d y \\
\Rightarrow q_i g_i &= s_{\tilde{g}_i} q_d y \\
\Rightarrow g_i &= \frac{s_{\tilde{g}_i} q_d}{q_i} y
\end{aligned} \tag{48}
$$

이제 정부투자 g_i를 통해 축적되는 새로운 공공자본 k_g^{new}을 구할 수 있다.

$$
\begin{aligned}
& k_{g,t+1} = (1-\delta_g)k_{g,t} + \mu_{g,t}g_{i,t} - \frac{\psi_g}{2}\left(\frac{g_{i,t}}{g_{i,t-1}} - 1\right)^2 \mu_{g,t}g_{i,t} \qquad (44) \\
\Rightarrow & k_g = (1-\delta_g)k_g + \mu_g g_i - \frac{\psi_g}{2}\left(\frac{g_i}{g_i} - 1\right)^2 \mu_g g_i \\
\Rightarrow & k_g = (1-\delta_g)k_g + \mu_g g_i \\
\Rightarrow & \delta_g k_g = \mu_g g_i \\
\Rightarrow & k_g^{new} = \frac{\mu_g}{\delta_g} g_i
\end{aligned}
$$

6단계: 수렴 확인 (Loop Closing)

잔차가 0에 가까운지 확인한다.

$$\text{잔차} = k_g^{guess} - k_g^{new}$$

만일 0에 가까우면 종료, 아니면 $k_g^{guess} = k_g^{new}$로 두고 2단계로 돌아간다.

7단계: 수렴 후 정상상태 값 계산

k_g가 수렴하면, 다음 정상상태 값들을 모두 알 수 있다.

- 유효 총요소생산성: $\tilde{z}_d$, $\tilde{z}_c^m$, $\tilde{z}_i^m$, $\tilde{z}_x^m$, $\tilde{z}_c$, $\tilde{z}_i$, $\tilde{z}_x$
- 상대가격: q_c^m, q_i^m, q_x^m, q_d, q_i, q_x
- 생산요소가격: r^k, w
- 민간자본/산출량/정부투자: k, $\bar{k}$, y, m_d, g_i, $\tilde{g}_i$

q_x가 주어지면 q_x^*를 알 수 있다.

$$
\begin{aligned}
& \tilde{p}_{x,t}^{*,opt} = \left(\frac{\varepsilon_x}{\varepsilon_x - 1}\mu_{x,t}\right)\frac{x_{x,n,t}}{x_{x,d,t}} \qquad (40) \\
\Rightarrow & \tilde{p}_x^{*,opt} = \left(\frac{\varepsilon_x}{\varepsilon_x - 1}\underbrace{\mu_x}_{=1}\right)\frac{x_{x,n}}{x_{x,d}} \\
\Rightarrow & \tilde{p}_x^{*,opt} = \frac{\varepsilon_x}{\varepsilon_x - 1}\frac{\frac{x\frac{q_x}{q_x^*}}{1-\theta_x Q_x(\pi_x^*)^{\varepsilon_x(1-\gamma_x^{idx})}}}{\frac{x}{1-\theta_x Q_x(\pi_x^*)^{(\varepsilon_x-1)(1-\gamma_x^{idx})}}} \\
\Rightarrow & \tilde{p}_x^{*,opt} = \frac{\varepsilon_x}{\varepsilon_x - 1}\frac{1-\theta_x Q_x\left(\pi_x^*\right)^{(\varepsilon_x-1)(1-\gamma_x^{idx})}}{1-\theta_x Q_x\left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}}\frac{q_x}{q_x^*} \\
\Rightarrow & \frac{q_x}{q_x^*} = \tilde{p}_x^{*,opt}\left(\frac{\varepsilon_x - 1}{\varepsilon_x}\right)\left(\frac{1-\theta_x Q_x\left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}}{1-\theta_x Q_x\left(\pi_x^*\right)^{(\varepsilon_x-1)(1-\gamma_x^{idx})}}\right)
\end{aligned}
$$

$$\Rightarrow q_x^* = \frac{q_x}{\tilde{p}_x^{*,opt}} \left(\frac{\varepsilon_x}{\varepsilon_x - 1} \right) \left(\frac{1 - \theta_x Q_x \left(\pi_x^* \right)^{(\varepsilon_x - 1)(1 - \gamma_x^{idx})}}{1 - \theta_x Q_x \left(\pi_x^* \right)^{\varepsilon_x (1 - \gamma_x^{idx})}} \right)$$

또한 q_d, k, y, m_d가 주어지면 다음을 알 수 있다.

앞에서 $s_{b^f} \equiv b^f / q_d y$는 데이터의 장기 평균 등으로 주어진 것으로 간주한다고 하였음.

$$\begin{aligned} s_{b^f,t} &\equiv \frac{b_t^f}{q_{d,t} y_t} \\ \Rightarrow s_{b^f} &= \frac{b^f}{q_d y} \\ \Rightarrow b^f &= s_{b^f} q_d y \end{aligned} \tag{120}$$

$$\begin{aligned} b_t^f &= (1 - \omega) b_t^{o,f} \\ \Rightarrow b^f &= (1 - \omega) b^{o,f} \\ \Rightarrow b^{o,f} &= \frac{b^f}{1 - \omega} \end{aligned} \tag{86}$$

$$\begin{aligned} \bar{k}_t &= (1 - \omega) \bar{k}_t^o \\ \Rightarrow \bar{k} &= (1 - \omega) \bar{k}^o \\ \Rightarrow \bar{k}^o &= \frac{\bar{k}}{1 - \omega} \end{aligned} \tag{82}$$

$$\begin{aligned} \bar{k}_{t+1}^o &= (1 - \delta) \bar{k}_t^o + \left[1 - \frac{\psi}{2} \left(\frac{i_t^o}{i_{t-1}^o} - 1 \right)^2 \right] \mu_{i,t} i_t^o \\ \Rightarrow \bar{k}^o &= (1 - \delta) \bar{k}^o + \left[1 - \frac{\psi}{2} \left(\frac{i^o}{i^o} - 1 \right)^2 \right] \mu_i i^o \\ \Rightarrow \bar{k}^o &= (1 - \delta) \bar{k}^o + \mu_i i^o \\ \Rightarrow \mu_i i^o &= \delta \bar{k}^o \\ \Rightarrow i^o &= \frac{\delta}{\mu_i} \bar{k}^o \end{aligned} \tag{4}$$

$$\begin{aligned} i_t &= (1 - \omega) i_t^o \\ \Rightarrow i &= (1 - \omega) i^o \end{aligned} \tag{84}$$

$$y_{i,t} = i_t + g_{i,t} \tag{114}$$

$$\Rightarrow y_i = i + g_i$$

$$d_{d,t} = m_{d,t}\left(1 - mc_t^{(d)}\Delta_{d,t}\right) \tag{75}$$
$$\Rightarrow d_d = m_d\left(1 - mc^{(d)}\Delta_d\right)$$

$$d_{d,t} = (1-\omega)d_{d,t}^o \tag{87}$$
$$\Rightarrow d_d = (1-\omega)d_d^o$$
$$\Rightarrow d_d^o = \frac{d_d}{1-\omega}$$

$$s_{g_c,t} \equiv \frac{g_{c,t}}{q_{d,t}y_t} \tag{46}$$
$$\Rightarrow s_{g_c} = \frac{g_c}{q_d y}$$
$$\Rightarrow g_c = s_{g_c} q_d y$$

$$s_{tr,t} \equiv \frac{tr_t}{q_{d,t}y_t} \tag{51}$$
$$\Rightarrow s_{tr} = \frac{tr}{q_d y}$$
$$\Rightarrow tr = s_{tr} q_d y$$

$$\omega tr_t^c = \alpha_t^T tr_t \tag{92}$$
$$\Rightarrow \omega tr^c = \alpha^T tr$$
$$\Rightarrow tr^c = \frac{\alpha^T}{\omega} tr$$

$$tr_t = \omega tr_t^c + (1-\omega)tr_t^o \tag{91}$$
$$\Rightarrow tr = \omega tr^c + (1-\omega)tr^o$$
$$\Rightarrow (1-\omega)tr^o = tr - \omega tr^c$$
$$\Rightarrow tr^o = \frac{tr - \omega tr^c}{1-\omega}$$

$$s_{b,t} \equiv \frac{b_t}{q_{d,t}y_t} \tag{59}$$

$$
\begin{aligned}
&\Rightarrow s_b = \frac{b}{q_d y} \\
&\Rightarrow b = s_b q_d y
\end{aligned}
$$

$$
\begin{aligned}
b_t &= (1-\omega) b_t^o \\
\Rightarrow b &= (1-\omega) b^o \\
\Rightarrow b^o &= \frac{b}{1-\omega}
\end{aligned} \tag{85}
$$

$$
\begin{aligned}
g_{c,t} + q_{i,t} g_{i,t} + tr_t + \frac{R_{t-1}}{\pi_{c,t}} b_{t-1} &= \tau_t + b_t \\
\Rightarrow g_c + q_i g_i + tr + \frac{R}{\pi_c} b &= \tau + b \\
\Rightarrow \tau &= g_c + q_i g_i + tr + \frac{R}{\pi_c} b - b
\end{aligned} \tag{43}
$$

$$
\begin{aligned}
x_{d,n} &= \frac{m_d m c^{(d)}}{1 - \theta_d Q_d \pi_d^{\varepsilon_d (1-\gamma_d^{idx})}} \\
x_{d,d} &= \frac{m_d}{1 - \theta_d Q_d \pi_d^{(\varepsilon_d - 1)(1-\gamma_d^{idx})}}
\end{aligned}
$$

r^k를 알면 $\Psi'(u)$을 알 수 있다.

$$
\begin{aligned}
(1-\tau_{k,t}) r_t^k &= \Psi'(u_t) \\
\Rightarrow (1-\tau_k) r^k &= \Psi'(u) \\
\Rightarrow \Psi'(u) &= (1-\tau_k) r^k
\end{aligned} \tag{10}
$$

9.4 소비와 수출

두 개의 식(국내시장 청산, 국제수지)과 두 개의 미지수(c, x)를 가지고 있다. 이를 $Ac + Bx = D$ 형태의 선형 시스템으로 정리하여 풀어야 한다.

먼저 각 최종재 생산(y_c, y_i, y_x) 1단위를 만드는 데 필요한 국내 중간재(m_d), 수입 중간재(m_f), 원유(o)의 투입 비율을 미리 정의한다.

소비재

$$
m_{d,t}^c = \gamma_c \left(\frac{q_{d,t}}{q_{c,t}^m} \right)^{-\xi_c} y_{c,t}^m \tag{21}
$$

$$\Rightarrow m_d^c = \gamma_c \left(\frac{q_d}{q_c^m} \right)^{-\xi_c} y_c^m$$
$$\Rightarrow y_c^m = \frac{1}{\gamma_c} \left(\frac{q_d}{q_c^m} \right)^{\xi_c} m_d^c$$

$$m_{f,t}^c = (1-\gamma_c) \left(\frac{q_{f,t}}{q_{c,t}^m} \right)^{-\xi_c} y_{c,t}^m \tag{22}$$
$$\Rightarrow m_f^c = (1-\gamma_c) \left(\frac{q_f}{q_c^m} \right)^{-\xi_c} y_c^m$$
$$\Rightarrow y_c^m = \frac{1}{(1-\gamma_c)} \left(\frac{q_f}{q_c^m} \right)^{\xi_c} m_f^c$$

$$y_{c,t}^m = \gamma_c^o \left(q_{c,t}^m \right)^{-\xi_c^o} y_{c,t} \tag{24}$$
$$\Rightarrow y_c^m = \gamma_c^o \left(q_c^m \right)^{-\xi_c^o} y_c$$
$$\Rightarrow \frac{1}{\gamma_c} \left(\frac{q_d}{q_c^m} \right)^{\xi_c} m_d^c = \gamma_c^o \left(q_c^m \right)^{-\xi_c^o} y_c$$
$$\Rightarrow \frac{m_d^c}{y_c} = \gamma_c^o \left(q_c^m \right)^{-\xi_c^o} \gamma_c \left(\frac{q_d}{q_c^m} \right)^{-\xi_c} \equiv \omega_c^d$$

$$y_{c,t}^m = \gamma_c^o \left(q_{c,t}^m \right)^{-\xi_c^o} y_{c,t} \tag{22, 24}$$
$$\Rightarrow y_c^m = \gamma_c^o \left(q_c^m \right)^{-\xi_c^o} y_c$$
$$\Rightarrow \frac{1}{(1-\gamma_c)} \left(\frac{q_f}{q_c^m} \right)^{\xi_c} m_f^c = \gamma_c^o \left(q_c^m \right)^{-\xi_c^o} y_c$$
$$\Rightarrow \frac{m_f^c}{y_c} = \gamma_c^o \left(q_c^m \right)^{-\xi_c^o} (1-\gamma_c) \left(\frac{q_f}{q_c^m} \right)^{-\xi_c} \equiv \omega_c^f$$

$$o_t^c = (1-\gamma_c^o) \left(q_{o,t} \right)^{-\xi_c^o} y_{c,t} \tag{25}$$
$$\Rightarrow o^c = (1-\gamma_c^o) \left(q_o \right)^{-\xi_c^o} y_c$$
$$\Rightarrow \frac{o^c}{y_c} = (1-\gamma_c^o) \left(q_o \right)^{-\xi_c^o} \equiv \omega_c^o$$

투자재

$$m_{d,t}^i = \gamma_i \left(\frac{q_{d,t}}{q_{i,t}^m} \right)^{-\xi_i} y_{i,t}^m \tag{27}$$
$$\Rightarrow m_d^i = \gamma_i \left(\frac{q_d}{q_i^m} \right)^{-\xi_i} y_i^m$$

$$\Rightarrow y_i^m = \frac{1}{\gamma_i}\left(\frac{q_d}{q_i^m}\right)^{\xi_i} m_d^i$$

$$m_{f,t}^i = (1-\gamma_i)\left(\frac{q_{f,t}}{q_{i,t}^m}\right)^{-\xi_i} y_{i,t}^m \tag{28}$$
$$\Rightarrow m_f^i = (1-\gamma_i)\left(\frac{q_f}{q_i^m}\right)^{-\xi_i} y_i^m$$
$$\Rightarrow y_i^m = \frac{1}{(1-\gamma_i)}\left(\frac{q_f}{q_i^m}\right)^{\xi_i} m_f^i$$

$$y_{i,t}^m = \gamma_i^o\left(\frac{q_{i,t}^m}{q_{i,t}}\right)^{-\xi_i^o} y_{i,t} \tag{30}$$
$$\Rightarrow y_i^m = \gamma_i^o\left(\frac{q_i^m}{q_i}\right)^{-\xi_i^o} y_i$$
$$\Rightarrow \frac{1}{\gamma_i}\left(\frac{q_d}{q_i^m}\right)^{\xi_i} m_d^i = \gamma_i^o\left(\frac{q_i^m}{q_i}\right)^{-\xi_i^o} y_i$$
$$\Rightarrow \frac{m_d^i}{y_i} = \gamma_i^o\left(\frac{q_i^m}{q_i}\right)^{-\xi_i^o}\gamma_i\left(\frac{q_d}{q_i^m}\right)^{-\xi_i} \equiv \omega_i^d$$

$$y_{i,t}^m = \gamma_i^o\left(\frac{q_{i,t}^m}{q_{i,t}}\right)^{-\xi_i^o} y_{i,t} \tag{28, 30}$$
$$\Rightarrow y_i^m = \gamma_i^o\left(\frac{q_i^m}{q_i}\right)^{-\xi_i^o} y_i$$
$$\Rightarrow \frac{1}{(1-\gamma_i)}\left(\frac{q_f}{q_i^m}\right)^{\xi_i} m_f^i = \gamma_i^o\left(\frac{q_i^m}{q_i}\right)^{-\xi_i^o} y_i$$
$$\Rightarrow \frac{m_f^i}{y_i} = \gamma_i^o\left(\frac{q_i^m}{q_i}\right)^{-\xi_i^o}(1-\gamma_i)\left(\frac{q_f}{q_i^m}\right)^{-\xi_i} \equiv \omega_i^f$$

$$o_t^i = (1-\gamma_i^o)\left(\frac{q_{o,t}}{q_{i,t}}\right)^{-\xi_i^o} y_{i,t} \tag{31}$$
$$\Rightarrow o^i = (1-\gamma_i^o)\left(\frac{q_o}{q_i}\right)^{-\xi_i^o} y_i$$
$$\Rightarrow \frac{o^i}{y_i} = (1-\gamma_i^o)\left(\frac{q_o}{q_i}\right)^{-\xi_i^o} \equiv \omega_i^o$$

수출재

$$m_{d,t}^x = \gamma_x\left(\frac{q_{d,t}}{q_{x,t}^m}\right)^{-\xi_x} y_{x,t}^m \tag{33}$$

$$\Rightarrow m_d^x = \gamma_x \left(\frac{q_d}{q_x^m}\right)^{-\xi_x} y_x^m$$
$$\Rightarrow y_x^m = \frac{1}{\gamma_x} \left(\frac{q_d}{q_x^m}\right)^{\xi_x} m_d^x$$

$$m_{f,t}^x = (1-\gamma_x) \left(\frac{q_{f,t}}{q_{x,t}^m}\right)^{-\xi_x} y_{x,t}^m \tag{34}$$
$$\Rightarrow m_f^x = (1-\gamma_x) \left(\frac{q_f}{q_x^m}\right)^{-\xi_x} y_x^m$$
$$\Rightarrow y_x^m = \frac{1}{(1-\gamma_x)} \left(\frac{q_f}{q_x^m}\right)^{\xi_x} m_f^x$$

$$y_{x,t}^m = \gamma_x^o \left(\frac{q_{x,t}^m}{q_{x,t}}\right)^{-\xi_x^o} y_{x,t} \tag{36}$$
$$\Rightarrow y_x^m = \gamma_x^o \left(\frac{q_x^m}{q_x}\right)^{-\xi_x^o} y_x$$
$$\Rightarrow \frac{1}{\gamma_x} \left(\frac{q_d}{q_x^m}\right)^{\xi_x} m_d^x = \gamma_x^o \left(\frac{q_x^m}{q_x}\right)^{-\xi_x^o} y_x$$
$$\Rightarrow \frac{m_d^x}{y_x} = \gamma_x^o \left(\frac{q_x^m}{q_x}\right)^{-\xi_x^o} \gamma_x \left(\frac{q_d}{q_x^m}\right)^{-\xi_x} \equiv \omega_x^d$$

$$y_{x,t}^m = \gamma_x^o \left(\frac{q_{x,t}^m}{q_{x,t}}\right)^{-\xi_x^o} y_{x,t} \tag{34, 36}$$
$$\Rightarrow y_x^m = \gamma_x^o \left(\frac{q_x^m}{q_x}\right)^{-\xi_x^o} y_x$$
$$\Rightarrow \frac{1}{(1-\gamma_x)} \left(\frac{q_f}{q_x^m}\right)^{\xi_x} m_f^x = \gamma_x^o \left(\frac{q_x^m}{q_x}\right)^{-\xi_x^o} y_x$$
$$\Rightarrow \frac{m_f^x}{y_x} = \gamma_x^o \left(\frac{q_x^m}{q_x}\right)^{-\xi_x^o} (1-\gamma_x) \left(\frac{q_f}{q_x^m}\right)^{-\xi_x} \equiv \omega_x^f$$

$$o_t^x = (1-\gamma_x^o) \left(\frac{q_{o,t}}{q_{x,t}}\right)^{-\xi_x^o} y_{x,t} \tag{37}$$
$$\Rightarrow o^x = (1-\gamma_x^o) \left(\frac{q_o}{q_x}\right)^{-\xi_x^o} y_x$$
$$\Rightarrow \frac{o^x}{y_x} = (1-\gamma_x^o) \left(\frac{q_o}{q_x}\right)^{-\xi_x^o} \equiv \omega_x^o$$

국내 중간재 시장 청산

$$
\begin{aligned}
y_{x,t} &= \Delta_{x,t} x_t \\
\Rightarrow y_x &= s_x x
\end{aligned}
\tag{73}
$$

$$
\begin{aligned}
m_{d,t} &= m^c_{d,t} + m^i_{d,t} + m^x_{d,t} \\
\Rightarrow m_d &= m^c_d + m^i_d + m^x_d \\
\Rightarrow \underbrace{y}_{=m_d} &= \underbrace{\omega^d_c y_c}_{=m^c_d} + \underbrace{\omega^d_i y_i}_{=m^i_d} + \underbrace{\omega^d_x y_x}_{=m^x_d} \\
\Rightarrow y &= \omega^d_c (c + g_c) + \omega^d_i (i + g_i) + \omega^d_x (s_x x) \\
\Rightarrow \omega^d_c c + \omega^d_x s_x x &= y - \omega^d_c g_c - \omega^d_i (i + g_i) \qquad (*)
\end{aligned}
\tag{115}
$$

국제수지 균형

$$
\begin{aligned}
m_{f,t} &= m^c_{f,t} + m^i_{f,t} + m^x_{f,t} \\
\Rightarrow m_f &= m^c_f + m^i_f + m^x_f \\
\Rightarrow m_f &= \omega^f_c y_c + \omega^f_i y_i + \omega^f_x y_x \\
\Rightarrow m_f &= \omega^f_c (c + g_c) + \omega^f_i (i + g_i) + \omega^f_x (s_x x)
\end{aligned}
\tag{116}
$$

$$
\begin{aligned}
o_t &= o^c_t + o^i_t + o^x_t \\
\Rightarrow o &= o^c + o^i + o^x \\
\Rightarrow o &= \omega^o_c y_c + \omega^o_i y_i + \omega^o_x y_x \\
\Rightarrow o &= \omega^o_c (c + g_c) + \omega^o_i (i + g_i) + \omega^o_x (s_x x)
\end{aligned}
\tag{117}
$$

$$
\begin{aligned}
b^f_t &= \tilde{e}_t \frac{R^*_{t-1} \Phi_{t-1}}{\pi_{c,t}} b^f_{t-1} + q^*_{x,t} x_t - (rer_t \Delta_{f,t} m_{f,t} + q_{o,t} o_t) \\
\Rightarrow b^f &= \tilde{e} \frac{R^* \Phi}{\pi_c} b^f + q^*_x x - (rer \Delta_f m_f + q_o o) \\
\Rightarrow & \left(1 - \tilde{e} \frac{R^* \Phi}{\pi_c}\right) b^f = q^*_x x - (rer \Delta_f m_f + q_o o) \\
\Rightarrow & q^*_x x - \left(1 - \tilde{e} \frac{R^* \Phi}{\pi_c}\right) b^f \\
& = rer \Delta_f \left[\omega^f_c (c + g_c) + \omega^f_i (i + g_i) + \omega^f_x (s_x x)\right] + q_o \left[\omega^o_c (c + g_c) + \omega^o_i (i + g_i) + \omega^o_x (s_x x)\right] \\
\Rightarrow & q^*_x x - \left(1 - \tilde{e} \frac{R^* \Phi}{\pi_c}\right) b^f
\end{aligned}
\tag{119}
$$

$$= (rer\Delta_f\omega_c^f + q_o\omega_c^o)c + s_x(rer\Delta_f\omega_x^f + q_o\omega_x^o)x + rer\Delta_f\left[\omega_c^f g_c + \omega_i^f(i+g_i)\right] + q_o\left[\omega_c^o g_c + \omega_i^o(i+g_i)\right]$$

$$\Rightarrow -(rer\Delta_f\omega_c^f + q_o\omega_c^o)c + \left[q_x^* - s_x(rer\Delta_f\omega_x^f + q_o\omega_x^o)\right]x$$

$$= rer\Delta_f\left[\omega_c^f g_c + \omega_i^f(i+g_i)\right] + q_o\left[\omega_c^o g_c + \omega_i^o(i+g_i)\right] + \left(1 - \tilde{e}\frac{R^*\Phi}{\pi_c}\right)b^f \qquad (**)$$

식 (*)와 (**)를 행렬식으로 다음과 같이 나타낼 수 있다.

$$\begin{pmatrix} \omega_c^d & s_x\omega_x^d \\ -(rer\Delta_f\omega_c^f + q_o\omega_c^o) & q_x^* - s_x(rer\Delta_f\omega_x^f + q_o\omega_x^o) \end{pmatrix}\begin{pmatrix} c \\ x \end{pmatrix}$$
$$= \begin{pmatrix} y - \omega_c^d g_c - \omega_i^d(i+g_i) \\ rer\Delta_f\left[\omega_c^f g_c + \omega_i^f(i+g_i)\right] + q_o\left[\omega_c^o g_c + \omega_i^o(i+g_i)\right] + \left(1 - \tilde{e}\frac{R^*\Phi}{\pi_c}\right)b^f \end{pmatrix}$$

위 행렬식의 우변은 상수 벡터이므로 c, x를 행렬식을 이용하여 계산할 수 있다.

c와 x가 주어지면 y_c와 y_x를 다음과 같이 얻을 수 있다. y_i는 앞에서 이미 구하였다.

$$\begin{aligned} y_{c,t} &= c_t + g_{c,t} + \Psi(u_t)\bar{k}_t \\ \Rightarrow y_c &= c + g_c + \underbrace{\Psi(u)}_{=0}\bar{k} \\ \Rightarrow y_c &= c + g_c \end{aligned} \qquad (113)$$

$$\begin{aligned} y_{x,t} &= \Delta_{x,t}x_t \\ \Rightarrow y_x &= s_x x \end{aligned} \qquad (73)$$

한편, x가 주어지면 $d_x, d_x^o, x_{x,n}, x_{x,d}$를 알 수 있다.

$$\begin{aligned} d_{x,t} &= \left(\frac{q_{x,t}^*}{q_{x,t}} - \Delta_{x,t}\right)x_t \\ \Rightarrow d_x &= \left(\frac{q_x^*}{q_x} - s_x\right)x \end{aligned} \qquad (77)$$

$$\begin{aligned} d_{x,t} &= (1-\omega)d_{x,t}^o \\ \Rightarrow d_x &= (1-\omega)d_x^o \\ \Rightarrow d_x^o &= \frac{d_x}{1-\omega} \end{aligned} \qquad (89)$$

$$x_{x,n} = \frac{x\frac{q_x}{q_x^*}}{1 - \theta_x Q_x\left(\pi_x^*\right)^{\varepsilon_x(1-\gamma_x^{idx})}}$$

$$x_{x,d} = \frac{x}{1 - \theta_x Q_x \left(\pi_x^*\right)^{(\varepsilon_x - 1)(1 - \gamma_x^{idx})}}$$

y_c, y_i, y_x가 주어지면 다음을 계산할 수 있다.

$$\begin{aligned}
m_d^c &= \omega_c^d y_c \\
m_f^c &= \omega_c^f y_c \\
o^c &= \omega_c^o y_c \\
y_c^m &= \frac{1}{\gamma_c} \left(\frac{q_d}{q_c^m} \right)^{\xi_c} m_d^c
\end{aligned}$$

$$\begin{aligned}
m_d^i &= \omega_i^d y_i \\
m_f^i &= \omega_i^f y_i \\
o^i &= \omega_i^o y_i \\
y_i^m &= \frac{1}{\gamma_i} \left(\frac{q_d}{q_i^m} \right)^{\xi_i} m_d^i
\end{aligned}$$

$$\begin{aligned}
m_d^x &= \omega_x^d y_x \\
m_f^x &= \omega_x^f y_x \\
o^x &= \omega_x^o y_x \\
y_x^m &= \frac{1}{\gamma_x} \left(\frac{q_d}{q_x^m} \right)^{\xi_x} m_d^x
\end{aligned}$$

$$\begin{aligned}
m_{f,t} &= m_{f,t}^c + m_{f,t}^i + m_{f,t}^x \\
\Rightarrow m_f &= m_f^c + m_f^i + m_f^x
\end{aligned} \tag{116}$$

$$\begin{aligned}
o_t &= o_t^c + o_t^i + o_t^x \\
\Rightarrow o &= o^c + o^i + o^x
\end{aligned} \tag{117}$$

$$\begin{aligned}
y_{f,t} &= \Delta_{f,t} m_{f,t} \\
\Rightarrow y_f &= \Delta_f m_f
\end{aligned} \tag{71}$$

$$d_{f,t} = m_{f,t} \left(1 - \frac{rer_t}{q_{f,t}} \Delta_{f,t} \right) \tag{76}$$

$$\Rightarrow d_f = m_f \left(1 - \frac{rer}{q_f}\Delta_f\right)$$

$$\begin{aligned} d_{f,t} &= (1-\omega) d^o_{f,t} \\ \Rightarrow d_f &= (1-\omega) d^o_f \\ \Rightarrow d^o_f &= \frac{d_f}{1-\omega} \end{aligned} \tag{88}$$

m_f를 알면 다음을 알 수 있다.

$$\begin{aligned} x_{f,n} &= \frac{m_f \left(\frac{rer}{q_f}\right)}{1 - \theta_f Q_f \pi_f^{\varepsilon_f(1-\gamma_f^{idx})}} \\ x_{f,d} &= \frac{m_f}{1 - \theta_f Q_f \pi_f^{(\varepsilon_f - 1)(1-\gamma_f^{idx})}} \end{aligned}$$

9.5 정액세/가계별 소비

본 모형의 동학(dynamics)에서는 정부부채(b_t)를 재정 잔차(residual)로 사용하나, 정상상태를 계산할 때에는 정액세(τ_t^L)를 재정 잔차로 사용한다.

$$\tau_t = \tau_{c,t} c_t + \tau_{n,t} w_t n_t^{eff} + \tau_{k,t} r_t^k k_t + \tau_{d,t} \left(q_{d,t} d_{d,t} + q_{f,t} d_{f,t} + q_{x,t} d_{x,t}\right) + \tau_{r,t} \left(\frac{R_{t-1} - 1}{\pi_{c,t}}\right) b_{t-1} + \tau_t^L \tag{45}$$

$$\Rightarrow \tau = \tau_c c + \tau_n w n^{eff} + \tau_k r^k k + \tau_d \left(q_d d_d + q_f d_f + q_x d_x\right) + \tau_r \left(\frac{R-1}{\pi_c}\right) b + \tau^L$$

$$\Rightarrow \tau^L = \tau - \left[\tau_c c + \tau_n w n^{eff} + \tau_k r^k k + \tau_d \left(q_d d_d + q_f d_f + q_x d_x\right) + \tau_r \left(\frac{R-1}{\pi_c}\right) b\right]$$

$$\begin{aligned} \omega \tau_t^{c,L} &= \alpha_t^L \tau_t^L \\ \Rightarrow \omega \tau^{c,L} &= \alpha^L \tau^L \\ \Rightarrow \tau^{c,L} &= \frac{\alpha^L \tau^L}{\omega} \end{aligned} \tag{93}$$

$$\begin{aligned} \tau_t^L &= \omega \tau_t^{c,L} + (1-\omega) \tau_t^{o,L} \\ \Rightarrow \tau^L &= \omega \tau^{c,L} + (1-\omega) \tau^{o,L} \\ \Rightarrow (1-\omega) \tau^{o,L} &= \tau^L - \omega \tau^{c,L} \\ \Rightarrow \tau^{o,L} &= \frac{\tau^L - \omega \tau^{c,L}}{1-\omega} \end{aligned} \tag{90}$$

$$
\begin{aligned}
c_t^c &= \frac{1}{1+\tau_{c,t}}\left[(1-\tau_{n,t})w_t n_t^{c,eff} + tr_t^c - \tau_t^{c,L}\right] \\
\Rightarrow c^c &= \frac{1}{1+\tau_c}\left[(1-\tau_n)w n^{c,eff} + tr^c - \tau^{c,L}\right]
\end{aligned} \tag{1}
$$

$$
\begin{aligned}
c_t &= \omega c_t^c + (1-\omega)c_t^o \\
\Rightarrow c &= \omega c^c + (1-\omega)c^o \\
\Rightarrow (1-\omega)c^o &= c - \omega c^c \\
\Rightarrow c^o &= \frac{c-\omega c^c}{1-\omega}
\end{aligned} \tag{80}
$$

$$
\begin{aligned}
\lambda_t(1+\tau_{c,t}) &= \zeta_{b,t}(c_t^o - \hbar c_{t-1}^o)^{-\sigma} - \hbar\beta\mathbb{E}_t\left[\zeta_{b,t+1}(c_{t+1}^o - \hbar c_t^o)^{-\sigma}\right] \\
\Rightarrow \lambda(1+\tau_c) &= \underbrace{\zeta_b}_{=1}(c^o - \hbar c^o)^{-\sigma} - \hbar\beta\underbrace{\zeta_b}_{=1}(c^o - \hbar c^o)^{-\sigma} \\
\Rightarrow \lambda(1+\tau_c) &= (1-\hbar\beta)(c^o - \hbar c^o)^{-\sigma} \\
\Rightarrow \lambda(1+\tau_c) &= (c^o)^{-\sigma}(1-\hbar\beta)(1-\hbar)^{-\sigma} \\
\Rightarrow \lambda &= \frac{(c^o)^{-\sigma}(1-\hbar\beta)(1-\hbar)^{-\sigma}}{1+\tau_c}
\end{aligned} \tag{6}
$$

λ가 주어지면 앞에서 살펴보았듯이 $\xi = \frac{\lambda q_i}{\mu_i}$로 정해진다.

9.6 y^*와 χ의 결정

$rer = 1$이 되도록 하는 y^*는 다음과 같다.

$$
\begin{aligned}
x_t &= \left(\frac{q_{x,t}^*}{rer_t}\right)^{-\eta_f} y_t^* \\
\Rightarrow x &= \left(\frac{q_x^*}{rer}\right)^{-\eta_f} y^* \\
\Rightarrow y^* &= x\left(\frac{q_x^*}{rer}\right)^{\eta_f}
\end{aligned} \tag{39}
$$

$n = 1/3$이 되도록 하는 χ는 다음과 같다.

$$
\begin{aligned}
x_{w,d,t} &= \lambda_t(1-\tau_{n,t})w_t n_t^{eff} + \beta\theta_w\mathbb{E}_t\left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w}\pi_c^{1-\iota_w})\gamma_w^{idx}}\right)^{\varepsilon_w - 1} x_{w,d,t+1}\right] \\
\Rightarrow x_{w,d} &= \lambda(1-\tau_n)w n^{eff} + \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})\gamma_w^{idx}}\right)^{\varepsilon_w - 1} x_{w,d}
\end{aligned} \tag{14}
$$

$$\Rightarrow x_{w,d} = \lambda(1-\tau_n) w n^{eff} + \beta\theta_w \pi_w^{(1-\gamma_w^{idx})(\varepsilon_w-1)} x_{w,d}$$
$$\Rightarrow \left[1 - \beta\theta_w \pi_w^{(1-\gamma_w^{idx})(\varepsilon_w-1)}\right] x_{w,d} = \lambda(1-\tau_n) w n^{eff}$$
$$\Rightarrow x_{w,d} = \frac{\lambda(1-\tau_n) w n^{eff}}{1 - \beta\theta_w \pi_w^{(1-\gamma_w^{idx})(\varepsilon_w-1)}}$$

$$(\tilde{w}_t^{opt})^{1+\varepsilon_w\phi} = \left(\frac{\varepsilon_w}{\varepsilon_w - 1}\mu_{w,t}\right)\frac{x_{w,n,t}}{x_{w,d,t}} \tag{12}$$
$$\Rightarrow (\tilde{w}^{opt})^{1+\varepsilon_w\phi} = \left(\frac{\varepsilon_w}{\varepsilon_w - 1}\underbrace{\mu_w}_{=1}\right)\frac{x_{w,n}}{x_{w,d}}$$
$$\Rightarrow x_{w,n} = \left(\frac{\varepsilon_w - 1}{\varepsilon_w}\right) x_{w,d} (\tilde{w}_t^{opt})^{1+\varepsilon_w\phi}$$

$$x_{w,n,t} = \zeta_{b,t}\chi\left(n_t^{eff}\right)^{1+\phi} + \beta\theta_w \mathbb{E}_t\left[\left(\frac{\pi_{w,t+1}}{(\pi_{w,t}^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w^{idx}}}\right)^{\varepsilon_w(1+\phi)} x_{w,n,t+1}\right] \tag{13}$$
$$\Rightarrow x_{w,n} = \underbrace{\zeta_b}_{=1}\chi\left(n^{eff}\right)^{1+\phi} + \beta\theta_w\left(\frac{\pi_w}{(\pi_w^{\iota_w}\pi_c^{1-\iota_w})^{\gamma_w^{idx}}}\right)^{\varepsilon_w(1+\phi)} x_{w,n}$$
$$\Rightarrow \chi\left(n^{eff}\right)^{1+\phi} = \left[1 - \beta\theta_w \pi_w^{(1-\gamma_w^{idx})\varepsilon_w(1+\phi)}\right] x_{w,n}$$
$$\Rightarrow \chi = \left[1 - \beta\theta_w \pi_w^{(1-\gamma_w^{idx})\varepsilon_w(1+\phi)}\right]\frac{x_{w,n}}{(n^{eff})^{1+\phi}}$$

10 제19장 추세를 포함한 중규모 모형과 베이지안 추정

제19장의 완성형 모형(이중 추세 + 수출수요 이동인자)의 정상상태는 제18장 절에서 유도한 캐스케이드를 그대로 쓰되, 성장 정상화가 요구하는 **앵커 몇 개만 교체**하면 얻어진다. 이 절은 그 앵커들을 유도하고, 두 개의 불변 명제와 수치 검증을 기록한다. 정상상태에서 외생 과정은 $g = \bar{g} = 1.0098805$(표본 평균 GDP 성장률), $g_v = \bar{g}_v = 1$, $\zeta_x = 1$이다.

10.1 성장 앵커의 유도

(i) 이자율 블록. 오일러 방정식(본책 (14))의 정상상태에서

$$1 = \beta\,\bar{g}^{-\sigma}\,\frac{R - \tau_r(R-1)}{\bar{\pi}} \;\Rightarrow\; R - \tau_r(R-1) = \frac{\bar{\pi}\,\bar{g}^{\sigma}}{\beta}$$

이므로 세후 실질이자율 앵커가 $1/\beta$에서 $\bar{g}^{\sigma}/\beta$로 바뀐다. 해외 금리의 정상상태도 같은 논리로 $R^* = \bar{\pi}\bar{g}^{\sigma}/\beta$($\Phi = 1$ 기준)이다.

(ii) 자본 블록. 그림자가격 식(본책 (18))의 정상상태에 투자1계 조건의 정상상태 $\xi = \lambda q_i (S = S' = 0, \mu_i = 1)$를 대입하면

$$\lambda q_i = \beta\,\frac{\bar{g}^{-\sigma}}{\bar{g}_v}\Big[\lambda\big((1-\tau_k)r^k - \Psi\big) + \lambda q_i(1-\delta)\Big] \;\Rightarrow\; (1-\tau_k)\,r^k = q_i\left[\frac{\bar{g}^{\sigma}\bar{g}_v}{\beta} - (1-\delta)\right]$$

($u=1, \Psi=0$). 자본 축적식(본책 (11))의 정상상태에서 유효 감가율은

$$i = \left[1 - \frac{1-\delta}{\bar{g}\,\bar{g}_v}\right]\bar{k}, \qquad k = \frac{\bar{k}}{\bar{g}\,\bar{g}_v}$$

로, 감가상각 보전에 자본을 두 추세만큼 키우는 몫이 더해진다. 정부자본(본책 (50))도 같은 형태다.

(iii) 칼보 재귀식 분모. 전방 계수가 $\beta\theta g^{1-\sigma}$ 이므로 모든 재귀 보조변수의 정상상태 분모가

$$1-\beta\theta \;\longrightarrow\; 1-\beta\theta\,\bar{g}^{1-\sigma}$$

로 바뀐다(네 부문 공통). $\sigma>1$ 이면 $\bar{g}^{1-\sigma}<1$ 이라 분모가 커지는데, 이는 성장으로 미래 이윤의 유효 할인이 강해지는 효과다.

(iv) 습관 한계효용. 본책 (13)의 정상상태는

$$\lambda(1+\tau_c) = \left(1-\hbar\beta\bar{g}^{-\sigma}\right)\left(1-\frac{\hbar}{\bar{g}}\right)^{-\sigma}(c^o)^{-\sigma}$$

로, 무추세판의 $(1-\hbar\beta)(1-\hbar)^{-\sigma}$ 에서 두 인자가 각각 $\bar{g}$ 조정을 받는다.

(v) 시차 명목잔고와 임금. 정부 예산제약·이자소득세·국제수지의 시차 잔고 항은 $Rb/(\bar{\pi}\bar{g})$ 꼴이 되고 (본책 (49)·(51)·(125)), 임금 인플레이션의 정상상태는 $\pi_w = \bar{\pi}\bar{g}$ 다(본책 (109)).

10.2 두 개의 불변 명제와 검증

명제 1 ($\bar{g}_v=1 \cdot \zeta_x=1$ 불변). $\bar{g}_v=1$ 이면 위 앵커에서 $\bar{g}_v$ 가 모두 소거되고, $\zeta_x=1$ 이면 수출수요식이 제18장 (39)로 돌아가므로, 이중 추세·이동인자 모형의 정상상태는 공통 추세 하나만 있는 성장판의 정상상태와 수치까지 완전히 같다. 제18장 개정에서 도입된 수정 UIP와 시변 물가목표도 정상상태에 흔적을 남기지 않으므로($\tilde{e}=1, \bar{\pi}_t=\bar{\pi}$) 이 명제에 포함된다. 이 명제는 구현 검증에 쓰인다: `nk_soe_medium_est2_ss.py`와 단일 추세판 스크립트의 출력이 전 변수에서 일치해야 하며, 실제로 기계 정밀도로 일치한다.

명제 2 ($\bar{g}\to 1$ 극한). $\bar{g}=1$을 넣으면 모든 앵커가 제18장의 값으로 돌아가므로 정상상태 전체가 제18장 캐스케이드의 해와 일치해야 한다. 이 극한 대조 역시 기계 정밀도로 확인된다(제16장에서 습관화한 퇴화 극한 검증).

수치 결과. $\bar{g}=1.0098805, \bar{g}_v=1$에서 캐스케이드는 7회 반복으로 수렴하며($k_g$ 잔차 0), 주요 값은

$$y=0.654769, \quad c=0.488599, \quad i=0.087890, \quad x=0.258602, \quad k_g=1.164734$$

이고 목표 변수 $n=1/3, rer=1$이 정확히 달성된다. 전체 154개 변수의 값은 `nk_soe_medium_est2_ss.py`가 `model_ss_values2.inc`로 출력하며, Dynare의 `resid` 검사에서 154개 방정식의 잔차가 모두 0임을 확인했다.